Master's Thesis

Focusing Time-Dependent Nonlinear Schrödinger Equation in $\mathbb{R}^2$

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Submitted in fulfillment of the requirements for the degree M.Sc. in Mathematics.

Declaration of Authorship

I hereby declare that I have written this master's thesis independently and that I have not used any sources or aids other than those explicitly indicated in the thesis. All ideas that are taken from the references, either verbatim or in substance, are marked as such.

I also declare that this thesis has not been submitted, in the same or in a similar form, to any other examination board, and that it has not been published elsewhere.

München, 5th of January 2026

Boyan Angelov Kugiyski

Signature

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0 Introduction

In, this section, we briefly explain the history and concepts that led to the nonlinear Schrödinger equation, and also explain why the focusing time-dependent model in $\mathbb{R}^2$ is mathematically interesting and relevant. The main goal is to motivate the variational and dynamical analysis carried out later in Sections 2, 3, and 4.

0.1 History and Development of the Schrödinger Equation

The Schrödinger equation stands as one of the most influential achievements of early twentieth-century physics. As the foundational equation of non-relativistic quantum mechanics, it describes the evolution of quantum states and provides a bridge between wave-like and particle-like descriptions of matter. Its development was not an isolated discovery, but rather the culmination of several decades of conceptual debates, experimental observations, and mathematical innovations.

0.2 From Classical Physics to Quantum Ideas

The late nineteenth century witnessed increasing tensions within classical physics. Classical electrodynamics and Newtonian mechanics struggled to explain phenomena such as black-body radiation, the photoelectric effect[8], and atomic spectra. Max Planck[16]'s quantization of energy in 1900 initiated a shift in physical understanding, suggesting that microscopic systems did not behave according to classical continuity. Shortly thereafter, Albert Einstein's interpretation of the photoelectric effect in terms of light quanta further supported the idea that energy exchanges occur in discrete packets.

Niels Bohr's atomic model (1913) deepened the conceptual departure from classical predictions. By postulating quantized electron orbits and introducing the correspondence principle, Bohr demonstrated that discrete spectral lines could be obtained from assumptions alien to classical mechanics. Nevertheless, Bohr's theory remained a hybrid construction without a general dynamical framework.

0.3 Wave-Particle Duality and de Broglie's Hypothesis

A decisive step toward a new mechanics was taken by Louis de Broglie[5] in 1924. Drawing on symmetry analogies between light and matter, de Broglie proposed that all particles exhibit wave-like behavior with wavelength

$$\lambda = \frac{h}{p},$$

where h is Planck's constant and p the momentum of the particle. This hypothesis, initially speculative, soon received experimental confirmation through electron diffraction experiments conducted by Davisson and Germer (1927) [7], and independently by Thomson.

De Broglie's thesis suggested the need for a wave equation governing the dynamics of matter waves. The problem was to identify a differential equation consistent with fundamental physical principles, especially energy conservation and the correspondence principle.

0.4 Schrödinger's Derivation and Interpretation

Erwin Schrödinger[21] succeeded in 1926 by proposing the now-famous wave equation. Motivated by Hamilton-Jacobi theory and de Broglie's relations, Schrödinger introduced a complex-valued wave function $\psi(t,x)$ satisfying

$$i\,\partial_t\psi = -\frac{1}{2m}\Delta\psi + V(x)\psi.$$

This formulation not only reproduced the energy levels of the hydrogen atom with remarkable accuracy, but also provided a coherent, mathematically sound framework for describing the evolution of quantum states.

Initially, Schrödinger interpreted $|\psi|^2$ as a distributed charge density, but Max Born[3] soon proposed (also in 1926) the probabilistic interpretation that became a cornerstone of quantum theory. This interpretation reconciled the wave formalism with discrete measurement outcomes, establishing quantum mechanics as a fundamentally statistical theory.

0.5 Extensions Toward Nonlinear and Time-Dependent Models

While Schrödinger's original equation is linear, nonlinear variants arise as effective models in several dispersive regimes, in particular, when a weak nonlinearity interacts with dispersion over long time scales. A standard example is the nonlinear Schrödinger equation (NLS), which appears in nonlinear optics and, also, as an effective description for Bose-Einstein condensates and related mean-field limits; see, for instance, [6].

In Section 4 of this thesis, we consider the focusing cubic NLS on $\mathbb{R}^2$,

$$i\,\partial_t u + \Delta u + |u|^2 u = 0, \qquad x \in \mathbb{R}^2, \quad t > 0,$$

with initial datum

$$u(0,\cdot) = u_0 \in H^1(\mathbb{R}^2).$$

A key feature of this equation is its invariance under the scaling

$$u(t,x) \;\longmapsto\; u_\lambda(t,x) \coloneqq \lambda\, u(\lambda^2 t, \lambda x), \qquad \lambda > 0,$$

and, in dimension 2, this scaling preserves the L^2-norm, which is why the equation is called L^2*-critical* (or *mass-critical*). The focusing sign is reflected in the conserved energy

$$E(u(t)) \coloneqq \frac{1}{2}\int_{\mathbb{R}^2} |\nabla u(t,x)|^2\,dx - \frac{1}{4}\int_{\mathbb{R}^2} |u(t,x)|^4\,dx,$$

and, together with the conserved mass

$$M(u(t)) \coloneqq \int_{\mathbb{R}^2} |u(t,x)|^2\,dx, \qquad t > 0$$

this places the analysis in a regime where concentration phenomena and finite-time blow-up can occur, see Subsection 4.1.3. See [6] for the general well-posedness framework and basic conserved quantities.

0.6 Mathematical Development and Analytical Framework

The L^2-critical focusing NLS is, in a precise sense, governed by a sharp functional inequality that balances the dispersive term $\int |\nabla u|^2$ against the nonlinear term $\int |u|^4$. More specifically, the sharp Gagliardo-Nirenberg inequality on $\mathbb{R}^2$ reads

$$\int_{\mathbb{R}^2} |u(x)|^4\,dx \le C_{\mathrm{GN}}\Big(\int_{\mathbb{R}^2} |u(x)|^2\,dx\Big)\Big(\int_{\mathbb{R}^2} |\nabla u(x)|^2\,dx\Big), \qquad \forall u \in H^1(\mathbb{R}^2),$$

where $C_{\text{GN}} > 0$ is the optimal constant; see Theorem 2.1.1. This sharp form, including the existence of optimizers and the relation to the stationary (ground state) equation, goes back to Weinstein's variational approach [25]. In, the present thesis, the proof strategy in Subsection 2.1 is variational: one considers the scale-invariant functional associated with the inequality, shows that a maximizing sequence can be chosen nonnegative and radially symmetric by rearrangement, and then proves compactness to obtain an optimizer. The corresponding Euler-Lagrange equation for an optimizer leads, after normalization, to the stationary equation

$$-\Delta Q + Q - Q^3 = 0 \qquad \text{on } \mathbb{R}^2,$$

where Q is the *ground state*, see Subsection 2.2.

Besides existence, a major structural input for the dynamical theory is uniqueness, namely, that there is a unique positive radial solution of this equation. In this thesis, this uniqueness statement is established in Section 3 by specializing the general uniqueness theory of McLeod and Serrin [15] to the case $n = 2$ and $f(u) = u^3 - u$ (see Theorem 3.3.1).

Finally, for the blow-up analysis in the mass-critical regime, the relevant mechanism is concentration-compactness at the level of the L^2-norm: sequences bounded in $H^1(\mathbb{R}^2)$ may fail to be compact only through translation and scaling effects. This principle is the backbone of the blow-up theory developed by Hmidi and Keraani [13], and, in Section 4, it is used to relate finite-time blow-up to concentration of mass on shrinking spatial scales and, in the minimal-mass case, to the ground-state profile. See Subsection 4.2, Theorem 4.2.3.

0.7 Outlook and Goal of The Thesis

The goal of this thesis is to connect, in a self-contained way, three central themes in the L^2-critical focusing NLS on $\mathbb{R}^2$: sharp functional inequalities, the structure of the ground state, and the mechanism of finite-time blow-up. The core variational object is the sharp Gagliardo-Nirenberg inequality, whose optimal constant C_{GN} and optimizers control the balance between dispersion and focusing [25].

Section 2 is divided into three subsections. In Subsection 2.1, we establish the sharp inequality, prove that the optimal constant is attained. In Subsection 2.2, we derive the stationary equation satisfied by the corresponding optimizer. In Subsection 2.3, we complement the existence theory of Subsection 2.1 and the Euler-Lagrange analysis of Subsection 2.2 by identifying the equality case and by expressing the sharp Gagliardo-Nirenberg constant in closed form via the normalized ground state $\mathcal{Q}$. Concretely, using Pohozaev-type identities for $\mathcal{Q}$ (Lemma 2.3.1), we obtain the explicit relation

$$C_{\text{GN}} = \frac{2}{||\mathcal{Q}||^2_{L^2(\mathbb{R}^2)}},$$

which links the optimal inequality constant to the mass of the stationary profile.

A second, independent, structural point is uniqueness: if the stationary equation admits a unique positive radial solution, then, up to the natural symmetries of the equation, the optimizer of the sharp Gagliardo-Nirenberg inequality is uniquely determined. In Section 3, we prove this uniqueness result by adapting the method of McLeod and Serrin [15] to our concrete nonlinearity and dimension.

With these variational and elliptic ingredients in place, Section 4 studies the blow-up dynamics of the time-dependent equation in the mass-critical regime. Following Hmidi and Keraani [13], we prove mass concentration at blow-up: if a solution blows up in finite time, then, as the blow-up time is approached, at least the ground state mass concentrates in a ball whose radius shrinks at the natural scaling rate. Moreover, in the minimal-mass case, we obtain a precise asymptotic description in terms of the ground-state profile, and we conclude with a classification of minimal-mass finite-time blow-up data, again, in the spirit of [13].

1 Preliminaries

In, this section, we collect the functional-analytic and measure-theoretic tools used throughout the thesis. The purpose is to fix notation and assumptions, so that later arguments, in Sections 2, 3, and 4, can focus on the core ideas rather than repeating background material.

Throughout this section, let $d \in \mathbb{N}$. Unless stated otherwise, all function spaces are over $\mathbb{R}^d$, equipped with the Lebesgue σ-algebra $\mathcal{L}(\mathbb{R}^d)$ and Lebesgue measure μ. In later sections, we are going to be working with $d = 2$. Standard references for the material recalled below are [19, 4, 20, 14, 1, 9, 6, 24, 12, 22].

1.1 Banach Spaces

Definition 1.1.1 (Normed and Banach spaces)**.** *Let $(X, ||\cdot||_X)$ be a normed vector space over $\mathbb{R}$ or $\mathbb{C}$. We say that X is a Banach space if it is complete with respect to the metric $d(x,y) := ||x-y||_X$, i.e., every Cauchy sequence $(x_n)_{n\in\mathbb{N}} \subseteq X$ converges in X.*

Definition 1.1.2 (Bounded linear maps and operator norm)**.** *Let X, Y be normed vector spaces. We define*

$$\mathcal{L}(X,Y) := \{T : X \to Y \text{ linear} \mid \exists C > 0 \text{ such that } ||Tx||_Y \le C\,||x||_X,\ \forall x \in X\}.$$

For $T \in \mathcal{L}(X,Y)$, the operator norm is

$$||T||_{\mathcal{L}(X,Y)} := \sup\left\{ \frac{||Tx||_Y}{||x||_X} \;\middle|\; x \in X,\ x \neq 0 \right\}.$$

Lemma 1.1.3 (Completeness of $\mathcal{L}(X,Y)$)**.** *Let X be a normed vector space and let Y be a Banach space. Then $\mathcal{L}(X,Y)$, equipped with the operator norm $||\cdot||_{\mathcal{L}(X,Y)}$, is a Banach space.*

Proof. Let $(T_n)_{n\in\mathbb{N}} \subseteq \mathcal{L}(X,Y)$ be a Cauchy sequence with respect to $||\cdot||_{\mathcal{L}(X,Y)}$. We must show that there exists $T \in \mathcal{L}(X,Y)$ such that $||T_n - T||_{\mathcal{L}(X,Y)} \to 0$.

Step 1: Pointwise convergence on X. Fix $x \in X$. Since (T_n) is Cauchy in $\mathcal{L}(X,Y)$, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that, for all $m, n \ge N$,

$$||T_n - T_m||_{\mathcal{L}(X,Y)} < \frac{\varepsilon}{\max\{1, ||x||_X\}}.$$

Hence, for all $m, n \ge N$,

$$||T_n x - T_m x||_Y \le ||T_n - T_m||_{\mathcal{L}(X,Y)}\,||x||_X < \frac{\varepsilon}{\max\{1, ||x||_X\}}\,||x||_X \le \varepsilon.$$

Thus $(T_n x)_{n\in\mathbb{N}}$ is a Cauchy sequence in Y. Since Y is complete, there exists an element $Tx \in Y$ such that

$$T_n x \to Tx \quad \text{in } Y, \text{ as } n \to \infty.$$

We have therefore defined a map $T : X \to Y$ by

$$Tx := \lim_{n\to\infty} T_n x.$$

Step 2: Linearity of T. Let $x_1, x_2 \in X$ and $\alpha, \beta \in \mathbb{C}$. For every n,

$$T_n(\alpha x_1 + \beta x_2) = \alpha T_n x_1 + \beta T_n x_2.$$

Passing to the limit in Y as $n \to \infty$ yields

$$T(\alpha x_1 + \beta x_2) = \alpha T x_1 + \beta T x_2,$$

so T is linear.

Step 3: Boundedness of T. Since (T_n) is Cauchy in $\mathcal{L}(X, Y)$, it is bounded: there exists $M > 0$ such that

$$||T_n||_{\mathcal{L}(X,Y)} \leq M, \qquad \forall n \in \mathbb{N}.$$

Fix $x \in X$. For every n,

$$||T_n x||_Y \leq ||T_n||_{\mathcal{L}(X,Y)} \, ||x||_X \leq M \, ||x||_X.$$

Taking the limit as $n \to \infty$ and using $T_n x \to Tx$ in Y gives

$$||Tx||_Y \leq M \, ||x||_X.$$

Hence $T \in \mathcal{L}(X, Y)$ and $||T||_{\mathcal{L}(X,Y)} \leq M$.

Step 4: Convergence in operator norm. Fix $\varepsilon > 0$. Since (T_n) is Cauchy in operator norm, there exists $N \in \mathbb{N}$ such that

$$||T_n - T_m||_{\mathcal{L}(X,Y)} < \varepsilon, \qquad \forall n, m \geq N.$$

Fix $n \geq N$ and $x \in X$. Letting $m \to \infty$ in

$$||T_n x - T_m x||_Y \leq ||T_n - T_m||_{\mathcal{L}(X,Y)} \, ||x||_X$$

and using $T_m x \to Tx$ yields

$$||T_n x - Tx||_Y \leq \varepsilon \, ||x||_X.$$

Taking the supremum over all $x \neq 0$ gives

$$||T_n - T||_{\mathcal{L}(X,Y)} \leq \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we conclude that $||T_n - T||_{\mathcal{L}(X,Y)} \to 0$.

Thus $\mathcal{L}(X, Y)$ is complete. ■

After Definition 1.1.2, we state several equivalent characterizations of the operator norm, which we will use interchangeably.

Lemma 1.1.4 (Equivalent formulas for the operator norm)**.** *Let X, Y be normed vector spaces and let $T \in \mathcal{L}(X, Y)$. Assume that $X \neq \{0\}$. Then the following quantities are equal. [19, 4]*

(i) The operator norm from Definition 1.1.2,

$$||T||_{\mathcal{L}(X,Y)} = \sup\left\{ \frac{||Tx||_Y}{||x||_X} \;\middle|\; x \in X, \; x \neq 0 \right\}.$$

(ii) The inf-characterization,

$$\inf\{c \geq 0 \mid ||Tx||_Y \leq c \, ||x||_X, \; \forall x \in X\}.$$

(iii) The supremum on the closed unit ball,

$$\sup\{||Tx||_Y \mid x \in X, \; ||x||_X \leq 1\}.$$

(iv) The supremum on the open unit ball,

$$\sup\{||Tx||_Y \mid x \in X,\ ||x||_X < 1\}.$$

(v) The supremum on the unit sphere,

$$\sup\{||Tx||_Y \mid x \in X,\ ||x||_X = 1\}.$$

If $X = \{0\}$, then $\mathcal{L}(X, Y) = \{0\}$ and the common value is interpreted as 0.

Proof. We prove that the quantities in (i)-(v) are equal by showing (i)=(ii)=(iii)=(v) and (iii)=(iv).

Step 1: Equality between (i) and (ii). Set

$$I := \inf\{c \geq 0 \mid ||Tx||_Y \leq c\,||x||_X,\ \forall x \in X\}.$$

We first show that (i)$\leq$ (ii), i.e., $||T||_{\mathcal{L}(X,Y)} \leq I$. Let $c \geq 0$ satisfy $||Tx||_Y \leq c\,||x||_X$ for all $x \in X$. Fix $x \in X$ with $x \neq 0$. Dividing by $||x||_X > 0$ yields

$$\frac{||Tx||_Y}{||x||_X} \leq c.$$

Taking the supremum over all $x \neq 0$ gives $||T||_{\mathcal{L}(X,Y)} \leq c$. Since this holds for every admissible c, we obtain $||T||_{\mathcal{L}(X,Y)} \leq I$.

We now show that (ii)$\leq$ (i), i.e., $I \leq ||T||_{\mathcal{L}(X,Y)}$. We claim that

$$||Tx||_Y \leq ||T||_{\mathcal{L}(X,Y)}\,||x||_X, \qquad \forall x \in X.$$

Indeed, if $x = 0$ then $||Tx||_Y = 0$. If $x \neq 0$, then, by the definition of the supremum in (i),

$$\frac{||Tx||_Y}{||x||_X} \leq ||T||_{\mathcal{L}(X,Y)},$$

which is equivalent to the desired inequality. Thus, the constant $c := ||T||_{\mathcal{L}(X,Y)}$ is admissible in the inf-set defining I, hence

$$I \leq ||T||_{\mathcal{L}(X,Y)}.$$

Combining the two inequalities yields (i)=(ii).

Step 2: Equality between (ii) and (iii). Let

$$A := \inf\{c \geq 0 \mid ||Tx||_Y \leq c\,||x||_X,\ \forall x \in X\}, \qquad B := \sup\{||Tx||_Y \mid ||x||_X \leq 1\}.$$

We first show $B \leq A$. Let $c \geq 0$ satisfy $||Tx||_Y \leq c\,||x||_X$ for all $x \in X$. Then, for every x with $||x||_X \leq 1$,

$$||Tx||_Y \leq c\,||x||_X \leq c.$$

Taking the supremum over all $||x||_X \leq 1$ yields $B \leq c$. Since this holds for every admissible c, we obtain $B \leq A$.

We now show $A \leq B$. Fix $x \in X$. If $x = 0$, then $||Tx||_Y = 0 \leq B\,||x||_X$. If $x \neq 0$, set $y := x/||x||_X$, so that $||y||_X = 1$, and hence

$$||Tx||_Y = ||T(||x||_X y)||_Y = ||x||_X\,||Ty||_Y \leq ||x||_X\,B.$$

Thus, for all $x \in X$,

$$||Tx||_Y \leq B\,||x||_X.$$

Therefore B is admissible in the inf-set defining A, hence $A \leq B$. Combining the inequalities yields (ii)=(iii).

Step 3: Equality between (iii) and (v). Since $\{x \mid ||x||_X = 1\} \subseteq \{x \mid ||x||_X \leq 1\}$, we have

$$\sup\{||Tx||_Y \mid ||x||_X = 1\} \leq \sup\{||Tx||_Y \mid ||x||_X \leq 1\}.$$

For the reverse inequality, let $x \in X$ with $||x||_X \leq 1$. If $x = 0$, then $||Tx||_Y = 0$, and the desired bound holds. If $x \neq 0$, set $y \coloneqq x/||x||_X$. Then $||y||_X = 1$ and

$$||Tx||_Y = ||x||_X\,||Ty||_Y \leq ||Ty||_Y \leq \sup\{||Tz||_Y \mid ||z||_X = 1\}.$$

Taking the supremum over all x with $||x||_X \leq 1$ yields

$$\sup\{||Tx||_Y \mid ||x||_X \leq 1\} \leq \sup\{||Tz||_Y \mid ||z||_X = 1\}.$$

Thus (iii)=(v).

Step 4: Equality between (iii) and (iv). Since $\{x \mid ||x||_X < 1\} \subseteq \{x \mid ||x||_X \leq 1\}$, we have

$$\sup\{||Tx||_Y \mid ||x||_X < 1\} \leq \sup\{||Tx||_Y \mid ||x||_X \leq 1\}.$$

Conversely, let $x \in X$ with $||x||_X \leq 1$. For each $n \in \mathbb{N}$, define $x_n \coloneqq \frac{n}{n+1}x$. Then $||x_n||_X < 1$ and

$$||Tx_n||_Y = \left\|T\Big(\frac{n}{n+1}x\Big)\right\|_Y = \frac{n}{n+1}\,||Tx||_Y.$$

Hence $||Tx_n||_Y \to ||Tx||_Y$ as $n \to \infty$, which implies

$$||Tx||_Y \leq \sup\{||Ty||_Y \mid ||y||_X < 1\}.$$

Taking the supremum over all $||x||_X \leq 1$ yields

$$\sup\{||Tx||_Y \mid ||x||_X \leq 1\} \leq \sup\{||Ty||_Y \mid ||y||_X < 1\}.$$

Thus (iii)=(iv).

By Step 1, Step 2, Step 3, and Step 4, the quantities in (i)-(v) are equal. ∎

Definition 1.1.5 (Dual space). *Let $(X, ||\cdot||_X)$ be a normed vector space. The dual space X^* is the space of all bounded linear functionals $\varphi : X \to \mathbb{C}$, equipped with the norm*

$$||\varphi||_{X^*} \coloneqq \sup\left\{\frac{|\varphi(x)|}{||x||_X} \;\middle|\; x \in X,\ x \neq 0\right\}.$$

Definition 1.1.6 (Weak and weak-$*$ convergence). *Let X be a Banach space.*

(i) A sequence $(x_n)_{n\in\mathbb{N}} \subseteq X$ converges weakly to $x \in X$, written $x_n \rightharpoonup x$ in X, if

$$\varphi(x_n) \to \varphi(x), \qquad \forall \varphi \in X^*.$$

(ii) A sequence $(\psi_n)_{n\in\mathbb{N}} \subseteq X^*$ converges weak-$*$ to $\psi \in X^*$, written $\psi_n \rightharpoonup^* \psi$ in X^*, if

$$\psi_n(x) \to \psi(x), \qquad \forall x \in X.$$

Theorem 1.1.7 (Banach-Alaoglu)**.** *Let X be a Banach space. Then the closed unit ball of X^* is compact in the weak-$*$ topology.*

Proof. See [19, 4]. ■

Theorem 1.1.8 (Banach-Steinhaus (uniform boundedness principle))**.** *Let X be a Banach space and let Y be a normed vector space. Let $(T_n)_{n\in\mathbb{N}} \subseteq \mathcal{L}(X, Y)$. Assume that, for every $x \in X$, the set $\{||T_n x||_Y \mid n \in \mathbb{N}\}$ is bounded in $\mathbb{R}$. Then*

$$\sup_{n\in\mathbb{N}} ||T_n||_{\mathcal{L}(X,Y)} < \infty.$$

Proof. See [19, 4]. ■

1.2 Hilbert Spaces

Definition 1.2.1 (Hilbert spaces)**.** *A Hilbert space is a complex vector space H equipped with an inner product $\langle\cdot,\cdot\rangle_H$ such that*

(i) $\langle\cdot,\cdot\rangle_H$ is conjugate-linear in the first argument and linear in the second argument,

(ii) $\langle u, v\rangle_H = \overline{\langle v, u\rangle_H}$, for all $u, v \in H$,

(iii) $\langle u, u\rangle_H \geq 0$, for all $u \in H$, and $\langle u, u\rangle_H = 0$ implies $u = 0$,

(iv) H is complete with respect to the induced norm $||u||_H \coloneqq \sqrt{\langle u, u\rangle_H}$.

Remark 1.2.2 ($L^2(\mathbb{R}^d)$ as a Hilbert space)**.** *The space $L^2(\mathbb{R}^d)$ is a Hilbert space with inner product*

$$\langle f, g\rangle_{L^2(\mathbb{R}^d)} \coloneqq \int_{\mathbb{R}^d} \overline{f(x)}\, g(x)\, dx, \qquad f, g \in L^2(\mathbb{R}^d).$$

Proof. See [19, 4]. ■

Theorem 1.2.3 (Riesz representation theorem)**.** *Let H be a Hilbert space. Then, for every $\ell \in H^*$, there exists a unique $v \in H$ such that*

$$\ell(u) = \langle u, v\rangle_H, \qquad \forall u \in H,$$

and $||\ell||_{H^*} = ||v||_H$.

Proof. See [19, 4]. ■

Theorem 1.2.4 (Banach-Steinhaus)**.** *Let H be a Hilbert space and let $(u_n)_{n\in\mathbb{N}} \subseteq H$. Assume that $u_n \rightharpoonup u$ in H, in the sense of Definition 1.1.6. Then,*

$$\sup_{n\in\mathbb{N}} ||u_n||_H < \infty.$$

Proof. Define, for each $n \in \mathbb{N}$, the bounded linear functional $\ell_n \in H^*$ by

$$\ell_n(v) \coloneqq \langle v, u_n\rangle_H, \qquad v \in H.$$

By the Riesz representation theorem (Theorem 1.2.3),

$$||\ell_n||_{H^*} = ||u_n||_H, \qquad \forall n \in \mathbb{N}.$$

Fix $v \in H$. Since $u_n \rightharpoonup u$ in H, we have

$$\ell_n(v) = \langle v, u_n \rangle_H \to \langle v, u \rangle_H, \qquad \text{as } n \to \infty.$$

In particular, the sequence $(\ell_n(v))_{n\in\mathbb{N}}$ is bounded in $\mathbb{C}$, for every fixed $v \in H$. Applying Theorem 1.1.8 with $X \coloneqq H$, $Y \coloneqq \mathbb{C}$, and $T_n \coloneqq \ell_n$, yields

$$\sup_{n\in\mathbb{N}} ||\ell_n||_{H^*} < \infty.$$

Using $||\ell_n||_{H^*} = ||u_n||_H$, we get

$$\sup_{n\in\mathbb{N}} ||u_n||_H < \infty$$

and this concludes the proof. ■

Theorem 1.2.5 (Banach-Alaoglu). *Let H be a separable Hilbert space and let $(u_n)_{n\in\mathbb{N}} \subseteq H$. Assume that*

$$\sup_{n\in\mathbb{N}} ||u_n||_H < \infty.$$

Then there exist a subsequence $(u_{n_k})_{k\in\mathbb{N}}$ and an element $u \in H$ such that

$$u_{n_k} \rightharpoonup u \quad \textit{in } H, \textit{ as } k \to \infty.$$

Proof. Fix an orthonormal basis $(e_j)_{j\in\mathbb{N}}$ of H (which exists since H is separable). For each $j \in \mathbb{N}$ define the scalar sequence

$$a_n^{(j)} \coloneqq \langle e_j, u_n \rangle_H.$$

By Cauchy-Schwarz,

$$|a_n^{(j)}| = |\langle e_j, u_n \rangle_H| \le ||e_j||_H \, ||u_n||_H = ||u_n||_H,$$

hence, since (u_n) is bounded in H, for every fixed j the sequence $(a_n^{(j)})_{n\in\mathbb{N}}$ is bounded in $\mathbb{C}$. Therefore, for $j = 1$, there exists a subsequence $(u_{n_k^{(1)}})_{k\in\mathbb{N}}$ such that $(a^{(1)}_{n_k^{(1)}})_{k\in\mathbb{N}}$ converges in $\mathbb{C}$.

Inductively, assume that we have constructed a subsequence $(u_{n_k^{(j)}})_{k\in\mathbb{N}}$ such that, for every $m \in \{1, \dots, j\}$, the scalar sequence

$$\big(\langle e_m, u_{n_k^{(j)}} \rangle_H\big)_{k\in\mathbb{N}}$$

converges in $\mathbb{C}$. Since $\big(\langle e_{j+1}, u_{n_k^{(j)}} \rangle_H\big)_{k\in\mathbb{N}}$ is bounded in $\mathbb{C}$, it has a convergent subsequence; denote by $(u_{n_k^{(j+1)}})_{k\in\mathbb{N}}$ a subsequence of $(u_{n_k^{(j)}})_{k\in\mathbb{N}}$ along which $\langle e_{j+1}, u_{n_k^{(j+1)}} \rangle_H$ converges. Then, for every $m \in \{1, \dots, j\}$, the sequence $\langle e_m, u_{n_k^{(j+1)}} \rangle_H$ still converges, because it is a subsequence of a convergent sequence. This completes the induction.

Define the diagonal subsequence by $v_k \coloneqq u_{n_k^{(k)}}$. Then, for every fixed $m \in \mathbb{N}$, the scalar sequence $\langle e_m, v_k \rangle_H$ converges in $\mathbb{C}$ as $k \to \infty$, because for $k \ge m$, v_k belongs to the subsequence constructed at stage m.

For each $m \in \mathbb{N}$ let

$$b_m \coloneqq \lim_{k\to\infty} \langle e_m, v_k \rangle_H \in \mathbb{C}.$$

We claim that $(b_m)_{m\in\mathbb{N}} \in \ell^2(\mathbb{N})$. Indeed, let $M \in \mathbb{N}$. By Bessel's inequality, for each k,

$$\sum_{m=1}^{M} |\langle e_m, v_k \rangle_H|^2 \le ||v_k||_H^2.$$

Taking $\liminf_{k\to\infty}$ on both sides and using convergence of each term on the left yields

$$\sum_{m=1}^{M} |b_m|^2 = \sum_{m=1}^{M} \lim_{k\to\infty} |\langle e_m, v_k\rangle_H|^2 = \lim_{k\to\infty} \sum_{m=1}^{M} |\langle e_m, v_k\rangle_H|^2 \le \liminf_{k\to\infty} ||v_k||_H^2.$$

Since (v_k) is a subsequence of (u_n) and (u_n) is bounded, the right-hand side is finite. Thus the partial sums $\sum_{m=1}^{M} |b_m|^2$ are bounded uniformly in M, which implies $(b_m) \in \ell^2$.

Since $(b_m) \in \ell^2$ and (e_m) is an orthonormal basis, there exists a unique $u \in H$ such that

$$u = \sum_{m=1}^{\infty} b_m e_m, \qquad \text{with convergence in } H,$$

and, moreover,

$$\langle e_m, u\rangle_H = b_m, \qquad \forall m \in \mathbb{N}.$$

We now show that $v_k \rightharpoonup u$ in H. Let $w \in H$. Since (e_m) is an orthonormal basis, there exist coefficients $(c_m)_{m\in\mathbb{N}} \in \ell^2$ such that

$$w = \sum_{m=1}^{\infty} c_m e_m \quad \text{in } H, \qquad c_m = \langle e_m, w\rangle_H.$$

Fix $\varepsilon > 0$. Choose $M \in \mathbb{N}$ such that

$$\Big\| w - \sum_{m=1}^{M} c_m e_m \Big\|_H < \varepsilon.$$

Then, for every k,

$$\begin{aligned} |\langle w, v_k - u\rangle_H| &\le \Big| \Big\langle w - \sum_{m=1}^{M} c_m e_m,\, v_k - u \Big\rangle_H \Big| + \Big| \Big\langle \sum_{m=1}^{M} c_m e_m,\, v_k - u \Big\rangle_H \Big| \\ &\le \Big\| w - \sum_{m=1}^{M} c_m e_m \Big\|_H ||v_k - u||_H + \Big| \sum_{m=1}^{M} c_m \langle e_m, v_k - u\rangle_H \Big|. \end{aligned}$$

Since (v_k) is bounded in H and $u \in H$, there exists $K_0 > 0$ such that $||v_k - u||_H \le K_0$ for all k. Hence the first term is bounded by εK_0.

For the second term, note that, for each fixed $m \in \{1, \ldots, M\}$,

$$\langle e_m, v_k - u\rangle_H = \langle e_m, v_k\rangle_H - \langle e_m, u\rangle_H \to b_m - b_m = 0, \qquad \text{as } k \to \infty.$$

Therefore, since the sum is finite, we have

$$\sum_{m=1}^{M} c_m \langle e_m, v_k - u\rangle_H \to 0, \qquad \text{as } k \to \infty.$$

Thus,

$$\limsup_{k\to\infty} |\langle w, v_k - u\rangle_H| \le \varepsilon K_0.$$

Since $\varepsilon > 0$ was arbitrary, we conclude that $\langle w, v_k\rangle_H \to \langle w, u\rangle_H$ for every $w \in H$, i.e., $v_k \rightharpoonup u$ in H.

This concludes the proof. ■

Remark 1.2.6. *Theorems 1.2.4 and 1.2.5 can be viewed as the Hilbert-space versions of Banach-Steinhaus and Banach-Alaoglu combined with separability, respectively. We note that the separability of H in Theorem 1.2.5 is essential. Otherwise, we could not obtain such a characterization for sequences and we would have to, instead, work with nets. See [19, 4].*

1.3 L^p-Spaces and L^p_{loc}-Spaces

Definition 1.3.1 (L^p spaces)**.** *Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. Let $1 \le p < \infty$. We define*

$$L^p(\Omega) \coloneqq \left\{ f : \Omega \to \mathbb{C} \text{ measurable} \;\middle|\; \int_\Omega |f(x)|^p \, d\mu(x) < \infty \right\},$$

equipped with the norm

$$||f||_{L^p(\Omega)} \coloneqq \Big(\int_\Omega |f(x)|^p \, d\mu(x) \Big)^{1/p}.$$

Moreover, we define

$$L^\infty(\Omega) \coloneqq \{ f : \Omega \to \mathbb{C} \text{ measurable} \mid \exists C \ge 0 \text{ such that } |f(x)| \le C, \text{ for } \mu\text{-a.e. } x \in \Omega \},$$

equipped with the norm

$$||f||_{L^\infty(\Omega)} \coloneqq \inf\{ C \ge 0 \mid |f(x)| \le C, \text{ for } \mu\text{-a.e. } x \in \Omega \}.$$

Theorem 1.3.2 (Hölder)**.** *Let $1 \le p, q \le \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then*

$$||fg||_{L^1(\Omega)} = \int_\Omega |f(x)g(x)| \, d\mu(x) \le ||f||_{L^p(\Omega)} \, ||g||_{L^q(\Omega)}, \qquad \forall f \in L^p(\Omega), \ \forall g \in L^q(\Omega).$$

Theorem 1.3.3 (Minkowski)**.** *Let $1 \le p \le \infty$. Then*

$$||f + g||_{L^p(\Omega)} \le ||f||_{L^p(\Omega)} + ||g||_{L^p(\Omega)}, \qquad \forall f, g \in L^p(\Omega).$$

Remark 1.3.4. *Theorems 1.3.2 and 1.3.3 are standard, see [20, 14] for their proofs.*

Definition 1.3.5 (L^p_{loc} spaces)**.** *Moreover, we define*

$$L^p_{\mathrm{loc}}(\Omega) \coloneqq \big\{ f : \Omega \to \mathbb{C} \text{ measurable} \;\big|\; \forall K \subseteq \Omega \text{ compact}, \ f\big|_K \in L^p(K) \big\},$$

where $L^p(K)$ is understood with respect to the restricted measure space $(K, \mathcal{A} \cap K, \mu|_K)$. For $p = \infty$, we define

$$L^\infty_{\mathrm{loc}}(\Omega) \coloneqq \big\{ f : \Omega \to \mathbb{C} \text{ measurable} \;\big|\; \forall K \subseteq \Omega \text{ compact}, \ f\big|_K \in L^\infty(K) \big\}.$$

1.4 Sobolev Spaces

Definition 1.4.1 (Weak derivatives)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open and let $u \in L^1_{\mathrm{loc}}(\Omega)$. Let $\alpha \in \mathbb{N}_0^d$. We say that $v \in L^1_{\mathrm{loc}}(\Omega)$ is the weak derivative $D^\alpha u$ if*

$$\int_\Omega u(x) \, D^\alpha \varphi(x) \, dx = (-1)^{|\alpha|} \int_\Omega v(x) \, \varphi(x) \, dx, \qquad \forall \varphi \in C_c^\infty(\Omega).$$

In this case, we write $v = D^\alpha u$.

Definition 1.4.2 (Sobolev spaces $W^{k,p}(\Omega)$)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open, let $1 \leq p \leq \infty$, and let $k \in \mathbb{N}$. We define*

$$W^{k,p}(\Omega) := \Big\{ u \in L^p(\Omega) \,\Big|\, \forall \alpha \in \mathbb{N}_0^d \text{ with } |\alpha| \leq k,\ \exists D^\alpha u \in L^p(\Omega) \text{ in the sense of Definition 1.4.1} \Big\}.$$

Definition 1.4.3 (Sobolev norms)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open, let $k \in \mathbb{N}$, and let $1 \leq p \leq \infty$. For $u \in W^{k,p}(\Omega)$ we define*

$$||u||_{W^{k,p}(\Omega)} := \begin{cases} \Big(\sum\limits_{|\alpha| \leq k} ||D^\alpha u||_{L^p(\Omega)}^p \Big)^{1/p}, & 1 \leq p < \infty, \\ \max\limits_{|\alpha| \leq k} ||D^\alpha u||_{L^\infty(\Omega)}, & p = \infty. \end{cases}$$

Theorem 1.4.4 (Completeness of Sobolev spaces)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open, let $k \in \mathbb{N}$, and let $1 \leq p \leq \infty$. Then $(W^{k,p}(\Omega), ||\cdot||_{W^{k,p}(\Omega)})$ is a Banach space. Moreover, if $p = 2$, then $W^{k,2}(\Omega)$ is a Hilbert space with inner product*

$$\langle u, v \rangle_{W^{k,2}(\Omega)} := \sum_{|\alpha| \leq k} \int_\Omega \overline{D^\alpha u(x)}\, D^\alpha v(x)\, dx, \qquad u, v \in W^{k,2}(\Omega).$$

Proof. See [1, 4, 9]. ■

Definition 1.4.5 (The space $H^k(\Omega)$)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open and let $k \in \mathbb{N}$. We define*

$$H^k(\Omega) := W^{k,2}(\Omega).$$

Definition 1.4.6 (The space $H^1(\mathbb{R}^d)$)**.** *We define*

$$H^1(\mathbb{R}^d) := \Big\{ u \in L^2(\mathbb{R}^d) \,\Big|\, \nabla u \in L^2(\mathbb{R}^d; \mathbb{C}^d) \Big\},$$

where ∇u is understood in the sense of weak derivatives. The norm is

$$||u||_{H^1(\mathbb{R}^d)} := \sqrt{||u||_{L^2(\mathbb{R}^d)}^2 + ||\nabla u||_{L^2(\mathbb{R}^d)}^2},$$

or, after squaring on both sides,

$$||u||_{H^1(\mathbb{R}^d)}^2 := ||u||_{L^2(\mathbb{R}^d)}^2 + ||\nabla u||_{L^2(\mathbb{R}^d)}^2.$$

Theorem 1.4.7 (Density of smooth functions in Sobolev spaces)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open, let $k \in \mathbb{N}_0$, and let $1 \leq p < \infty$. Then*

$$W^{k,p}(\Omega) \cap C^\infty(\Omega) \text{ is dense in } W^{k,p}(\Omega).$$

In particular, $H^k(\Omega) \cap C^\infty(\Omega)$ is dense in $H^k(\Omega) = W^{k,2}(\Omega)$.
Moreover, if $\Omega = \mathbb{R}^d$, then $C_c^\infty(\mathbb{R}^d)$ is dense in $W^{k,p}(\mathbb{R}^d)$ (and hence in $H^k(\mathbb{R}^d)$). [1, 4, 9]

Theorem 1.4.8 (Rellich-Kondrachov compactness on bounded domains)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be bounded and open. Then the embedding*

$$H^1(\Omega) \hookrightarrow L^2(\Omega)$$

is compact. [1, 4, 9]

1.5 Sobolev Inequalities

Theorem 1.5.1 (Sobolev embedding for $1 \le p < d$)**.** *Let $d \ge 2$, let $1 \le p < d$, and set $p^* := \frac{dp}{d-p}$. Then there exists $C > 0$ such that*

$$||u||_{L^{p^*}(\mathbb{R}^d)} \le C\,||u||_{W^{1,p}(\mathbb{R}^d)}, \qquad \forall u \in W^{1,p}(\mathbb{R}^d).$$

Theorem 1.5.2 (The H^1-Sobolev range)**.** *Let $d \ge 1$.*

If $d \ge 3$, set $2^* := \frac{2d}{d-2}$. Then, $\forall p \in [2, 2^*]$, $\exists C > 0$ such that

$$||u||_{L^p(\mathbb{R}^d)} \le C\,||u||_{H^1(\mathbb{R}^d)}, \qquad \forall u \in H^1(\mathbb{R}^d).$$

Moreover, $\exists C > 0$ such that

$$||u||_{L^{2^*}(\mathbb{R}^d)} \le C\,||\nabla u||_{L^2(\mathbb{R}^d)}, \qquad \forall u \in H^1(\mathbb{R}^d).$$

If $d = 2$, then, $\forall q \in [2, \infty)$, $\exists C > 0$ such that

$$||u||_{L^q(\mathbb{R}^2)} \le C\,||u||_{H^1(\mathbb{R}^2)}, \qquad \forall u \in H^1(\mathbb{R}^2).$$

If $d = 1$, then, $\forall q \in [2, \infty]$, $\exists C > 0$ such that

$$||u||_{L^q(\mathbb{R})} \le C\,||u||_{H^1(\mathbb{R})}, \qquad \forall u \in H^1(\mathbb{R}).$$

In particular,

$$H^1(\mathbb{R}) \hookrightarrow C(\mathbb{R}) \cap L^\infty(\mathbb{R}).$$

Theorem 1.5.3 (A non-sharp mass-critical Gagliardo-Nirenberg inequality in $\mathbb{R}^d$)**.** *There exists a constant $C_0 > 0$, depending only on d, such that*

$$||u||_{L^{2+\frac{4}{d}}(\mathbb{R}^d)} \le C_0\,||u||_{L^2(\mathbb{R}^d)}^{\frac{2}{d+2}}\,||\nabla u||_{L^2(\mathbb{R}^d)}^{\frac{d}{d+2}}, \qquad \forall u \in H^1(\mathbb{R}^d).$$

Remark 1.5.4. *Theorems 1.5.1, 1.5.2, and 1.5.3 are standard. For their proofs, see [1, 4, 9, 14]. In later sections, we will be working with $d = 2$.*

1.6 Time-Dependent Banach Space-Valued Functions

In this subsection, we define continuity and differentiability for maps with values in Banach spaces, as well as the Bochner spaces used for time-dependent PDE solutions. The goal is to state the regularity classes for solutions of the nonlinear Schrödinger equation precisely, which will be used in Section 4 when formulating conservation laws and blow-up criteria.

Since solutions, in Section 4, are viewed as maps $t \mapsto u(t)$ with values in Sobolev spaces, we recall basic function spaces for Banach space-valued maps. [4, 6]

Definition 1.6.1 (Continuity and differentiability with values in a Banach space)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $(X, ||\cdot||_X)$ be a Banach space.*

(i) We write $u \in C(I; X)$ if $u : I \to X$ is continuous with respect to the norm topology of X.

(ii) We write $u \in C^1(I; X)$ if there exists a map $\dot{u} : I \to X$ such that

$$\lim_{h \to 0} \Big\| \frac{u(t+h) - u(t)}{h} - \dot{u}(t) \Big\|_X = 0, \qquad \forall t \in I,$$

and such that $t \mapsto \dot{u}(t)$ is continuous from I into X.

Definition 1.6.2 (Bochner spaces)**.** *Let $1 \leq p < \infty$, let $I \subseteq \mathbb{R}$ be an interval, and let $(X, ||\cdot||_X)$ be a Banach space. We define*

$$L^p(I;X) \coloneqq \left\{ u : I \to X \text{ strongly measurable} \;\middle|\; \int_I ||u(t)||_X^p \, dt < \infty \right\},$$

equipped with the norm

$$||u||_{L^p(I;X)} \coloneqq \Big(\int_I ||u(t)||_X^p \, dt \Big)^{1/p}.$$

1.7 Rearrangements

In, this subsection, we introduce symmetric decreasing rearrangements and the basic equimeasurability and inequalities associated with them. The goal is to provide the rearrangement tools needed later, since they are of main importance in the variational analysis of Section 2. With the help of rearrangements, we reduce variational problems to radial, nonnegative, and more compact classes of functions. See [14].

Throughout this subsection, $|E|$ denotes the d-dimensional Lebesgue measure of a (Lebesgue) measurable set $E \subseteq \mathbb{R}^d$. Let $u : \mathbb{R}^d \to \mathbb{R}$ be measurable. In applications below, we will only use $u \geq 0$.

Definition 1.7.1 (Distribution function and decreasing rearrangement)**.** *For $t \geq 0$ define the distribution function*

$$\mu_u(t) \coloneqq \big|\{x \in \mathbb{R}^d : \; u(x) > t\}\big|.$$

The (right-continuous) decreasing rearrangement $u^{\#} : (0, \infty) \to [0, \infty]$ is defined by

$$u^{\#}(s) \coloneqq \inf\{t \geq 0 \mid \mu_u(t) \leq s\}.$$

Lemma 1.7.2 (Basic properties of distribution functions)**.** *Let $u : \mathbb{R}^d \to [0, \infty]$ be measurable.*

(i) The map $t \mapsto \mu_u(t)$ is nonincreasing on $[0, \infty)$.

(ii) The map $t \mapsto \mu_u(t)$ is right-continuous on $[0, \infty)$.

(iii) Let $p \in [1, \infty)$. If $u \in L^p(\mathbb{R}^d)$, then $\mu_u(t) < \infty$, $\forall t > 0$.

Proof. (i) Let $0 \leq t_1 < t_2$. Then $\{u > t_2\} \subseteq \{u > t_1\}$, hence, by monotonicity of Lebesgue measure,

$$\mu_u(t_2) = |\{u > t_2\}| \leq |\{u > t_1\}| = \mu_u(t_1).$$

(ii) Let $t \geq 0$ and let $(t_n)_{n\in\mathbb{N}} \subseteq (0, \infty)$ satisfy $t_n \downarrow t$ as $n \to \infty$. Set $E_n \coloneqq \{u > t_n\}$ and $E \coloneqq \{u > t\}$. Then $(E_n)_{n\in\mathbb{N}}$ is an increasing sequence of measurable sets, and

$$\bigcup_{n=1}^{\infty} E_n = E.$$

Indeed, if $x \in \bigcup_{n\in\mathbb{N}} E_n$, then $\exists n$ such that $u(x) > t_n \geq t$, hence $u(x) > t$, so $x \in E$. Conversely, if $x \in E$, then $u(x) > t$. Since $t_n \downarrow t$, $\exists n$ such that $t_n < u(x)$, hence $u(x) > t_n$, so $x \in E_n \subseteq \bigcup_{k\in\mathbb{N}} E_k$. Therefore, by continuity from below of Lebesgue measure,

$$\mu_u(t) = |E| = \lim_{n\to\infty} |E_n| = \lim_{n\to\infty} \mu_u(t_n),$$

which is the right-continuity of μ_u at t.

(iii) Let $p \in [1, \infty)$ and assume $u \in L^p(\mathbb{R}^d)$. Fix $t > 0$ and set $E := \{u > t\}$. Then $u(x)^p \geq t^p$ holds for all $x \in E$, hence

$$t^p |E| \leq \int_E u(x)^p \, dx \leq \int_{\mathbb{R}^d} u(x)^p \, dx < \infty.$$

Therefore, $|E| = \mu_u(t) \leq t^{-p} \int_{\mathbb{R}^d} u(x)^p \, dx < \infty$.

■

Definition 1.7.3 (Symmetric rearrangement of measurable sets)**.** *Let $\omega_d := |\{x \in \mathbb{R}^d : \ |x| < 1\}|$ be the volume of the unit ball in $\mathbb{R}^d$. Let $E \subseteq \mathbb{R}^d$ be measurable. If $|E| \in [0, \infty)$, we define E^* by*

$$E^* := \Big\{ x \in \mathbb{R}^d \ \Big| \ |x| < r_E \Big\}, \qquad r_E := \Big(\frac{|E|}{\omega_d} \Big)^{1/d},$$

so that $|E^*| = |E|$. If $|E| = \infty$, we set $E^* := \mathbb{R}^d$.

Definition 1.7.4 (Symmetric decreasing rearrangement)**.** *The symmetric decreasing rearrangement $u^* : \mathbb{R}^d \to [0, \infty]$ is defined by*

$$u^*(x) := u^{\#}(\omega_d |x|^d).$$

Then, u^* is radial and nonincreasing as a function of $|x|$.

Theorem 1.7.5 (Layer-cake representation)**.** *Let $d \in \mathbb{N}$, and let $u : \mathbb{R}^d \to [0, \infty]$ be measurable. Then:*

(i) One has the pointwise identity

$$u(x) = \int_0^\infty \mathbf{1}_{\{u(x) > t\}} \, dt, \qquad \forall x \in \mathbb{R}^d.$$

(ii) For every $p \in (0, \infty)$, one has

$$\int_{\mathbb{R}^d} |u(x)|^p \, dx = p \int_0^\infty t^{p-1} \, \mu_u(t) \, dt,$$

where $\mu_u(t) := \Big| \{x \in \mathbb{R}^d : \ u(x) > t\} \Big|$.

Theorem 1.7.6 (Equimeasurability and L^p-invariance)**.** *Let $d \in \mathbb{N}$, and let $u : \mathbb{R}^d \to [0, \infty]$ be measurable. Let u^* be the symmetric decreasing rearrangement from Definition 1.7.4.*

(i) (Equimeasurability) One has

$$\mu_{u^*}(t) = \mu_u(t), \qquad \forall t > 0.$$

(ii) (L^p-invariance, with convention) Let $p \in [1, \infty]$. Then

$$||u^*||_{L^p(\mathbb{R}^d)} = ||u||_{L^p(\mathbb{R}^d)},$$

with the convention that the value is $+\infty$ if $u \notin L^p(\mathbb{R}^d)$.

Proof. **Proof of (i).** Fix $t > 0$. By Definition 1.7.4, we have, by construction,

$$\{x \in \mathbb{R}^d : \ u^*(x) > t\} = B_{r(t)}(0) \quad \text{with} \quad |B_{r(t)}(0)| = \mu_u(t).$$

Taking Lebesgue measures yields

$$\mu_{u^*}(t) = |\{u^* > t\}| = |B_{r(t)}(0)| = \mu_u(t),$$

which proves (i).

Proof of (ii) for $p \in [1, \infty)$. Fix $p \in [1, \infty)$. We use Theorem 1.7.5 (ii) and (i). If $u \notin L^p(\mathbb{R}^d)$, then

$$\int_{\mathbb{R}^d} |u(x)|^p \, dx = +\infty,$$

and Theorem 1.7.5 (ii) gives

$$p \int_0^\infty t^{p-1} \mu_u(t) \, dt = +\infty.$$

Using $\mu_{u^*} = \mu_u$ from (i) and applying Theorem 1.7.5 (ii) again yields

$$\int_{\mathbb{R}^d} |u^*(x)|^p \, dx = +\infty,$$

so both norms are $+\infty$ and the claimed equality holds under the convention.

Assume now that $u \in L^p(\mathbb{R}^d)$, i.e.,

$$\int_{\mathbb{R}^d} |u(x)|^p \, dx < \infty.$$

Then Theorem 1.7.5 (ii) gives

$$\int_{\mathbb{R}^d} |u(x)|^p \, dx = p \int_0^\infty t^{p-1} \mu_u(t) \, dt,$$

and, again, by (i),

$$p \int_0^\infty t^{p-1} \mu_u(t) \, dt = p \int_0^\infty t^{p-1} \mu_{u^*}(t) \, dt.$$

Applying Theorem 1.7.5 (ii) to u^* yields

$$p \int_0^\infty t^{p-1} \mu_{u^*}(t) \, dt = \int_{\mathbb{R}^d} |u^*(x)|^p \, dx.$$

Thus,

$$\int_{\mathbb{R}^d} |u(x)|^p \, dx = \int_{\mathbb{R}^d} |u^*(x)|^p \, dx,$$

and taking p-th roots proves $||u||_{L^p(\mathbb{R}^d)} = ||u^*||_{L^p(\mathbb{R}^d)}$.

Proof of (ii) for $p = \infty$. Set

$$M := ||u||_{L^\infty(\mathbb{R}^d)} = \inf\Big\{ C \geq 0 \;\Big|\; |u(x)| \leq C, \text{ for a.e. } x \in \mathbb{R}^d \Big\}.$$

We show that $||u^*||_{L^\infty(\mathbb{R}^d)} = M$.

First we prove $||u^*||_{L^\infty(\mathbb{R}^d)} \leq M$. Fix $\varepsilon > 0$. By definition of the infimum, there exists $C_\varepsilon \geq 0$ such that

$$C_\varepsilon \leq M + \varepsilon \quad \text{and} \quad |u(x)| \leq C_\varepsilon, \text{ for a.e. } x \in \mathbb{R}^d.$$

Hence $\mu_u(C_\varepsilon) = |\{u > C_\varepsilon\}| = 0$. By (i), $\mu_{u^*}(C_\varepsilon) = \mu_u(C_\varepsilon) = 0$, i.e.,

$$|\{u^* > C_\varepsilon\}| = 0.$$

Thus $u^*(x) \le C_\varepsilon$ for a.e. x, hence

$$||u^*||_{L^\infty(\mathbb{R}^d)} \le C_\varepsilon \le M + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we obtain $||u^*||_{L^\infty(\mathbb{R}^d)} \le M$.

Second we prove $M \le ||u^*||_{L^\infty(\mathbb{R}^d)}$. Let $C \ge 0$ satisfy $u^*(x) \le C$ for a.e. x. Then $|\{u^* > C\}| = 0$, so $\mu_{u^*}(C) = 0$, and by (i), also $\mu_u(C) = 0$. Thus $|u(x)| \le C$ for a.e. x, and by definition of M we get $M \le C$. Taking the infimum over such C yields $M \le ||u^*||_{L^\infty(\mathbb{R}^d)}$. This completes the proof. ■

Before stating the coarea formula, we briefly clarify the notation $d\mathcal{H}^{d-1}$. Here, $\mathcal{H}^{d-1}$ denotes the $(d-1)$-dimensional Hausdorff measure on $\mathbb{R}^d$. In particular, if $S \subseteq \mathbb{R}^d$ is a C^1 hypersurface, then $\mathcal{H}^{d-1}|_S$ coincides with the usual surface measure on S. We also note that, for $u \in W^{1,1}(\mathbb{R}^d)$, the level sets $\{u = t\}$ are $\mathcal{H}^{d-1}$-measurable for a.e. $t \in \mathbb{R}$. For background on Hausdorff measures, surface measure, and the coarea formula, see [10, 11].

Theorem 1.7.7 (Coarea formula)**.** *Let $d \in \mathbb{N}$, let $u \in W^{1,1}(\mathbb{R}^d)$, and let $g : \mathbb{R}^d \to [0,\infty]$ be measurable. Then*

$$\int_{\mathbb{R}^d} g(x)\,|\nabla u(x)|\,dx = \int_{-\infty}^{\infty} \left(\int_{\{u=t\}} g(x)\,d\mathcal{H}^{d-1}(x) \right) dt.$$

We now state an isoperimetric inequality in the setting of sets of finite perimeter. If $E \subseteq \mathbb{R}^d$ is measurable, its perimeter $P(E)$ is understood in the sense of geometric measure theory (equivalently, as the total variation of the distributional gradient of $\mathbf{1}_E$). In this framework, if $\partial^* E$ denotes the reduced boundary of E, then $P(E) = \mathcal{H}^{d-1}(\partial^* E)$. For definitions and basic properties of sets of finite perimeter, reduced boundaries, and the isoperimetric inequality in this generality, see [10, 11].

Theorem 1.7.8 (Isoperimetric inequality in $\mathbb{R}^d$)**.** *Let $d \in \mathbb{N}$ and let $E \subseteq \mathbb{R}^d$ be a measurable set of finite perimeter. Let E^* be the open ball centered at 0 such that $|E^*| = |E|$. Then*

$$P(E) \ge P(E^*),$$

where $P(\cdot)$ denotes the perimeter (in the sense of sets of finite perimeter).

We now prove (ii). We will integrate over level sets $\{u = t\}$ using the $(d-1)$-dimensional Hausdorff measure. The symbol $\mathcal{H}^{d-1}$ denotes the $(d-1)$-dimensional Hausdorff measure on $\mathbb{R}^d$. It is characterized by the property that if $\Sigma \subseteq \mathbb{R}^d$ is a C^1 hypersurface, then $\mathcal{H}^{d-1}(\Sigma)$ coincides with the usual surface area of Σ. We write $d\mathcal{H}^{d-1}$ for integration with respect to this measure; see [18, 11, 10].

Theorem 1.7.9 (Pólya-Szegő inequality)**.** *Let $u \in H^1(\mathbb{R}^d)$ with $u \ge 0$ a.e., and let u^* be its symmetric decreasing rearrangement. Then $u^* \in H^1(\mathbb{R}^d)$ and*

$$||\nabla u^*||_{L^2(\mathbb{R}^d)} \le ||\nabla u||_{L^2(\mathbb{R}^d)}.$$

Proof. The proof is divided into two parts.

Part 1: Proof for $u \in C_c^\infty(\mathbb{R}^d)$, $u \ge 0$.

Step 1: Level sets and distribution function. For $t \ge 0$ define the superlevel set

$$E_t := \left\{ x \in \mathbb{R}^d \;\middle|\; u(x) > t \right\},$$

and define the distribution function

$$\mu_u(t) := |E_t|.$$

Since u is smooth and compactly supported, $\mu_u(t) < \infty$ for all $t > 0$ and $\mu_u(t) = 0$ for all $t > ||u||_\infty$.

Step 2: Formulas from the coarea theorem. We apply Theorem 1.7.7 twice. First take $g(x) \coloneqq |\nabla u(x)|$. Since u is smooth, $|\nabla u|$ is measurable and nonnegative. We obtain

$$\int_{\mathbb{R}^d} |\nabla u(x)|^2 \, dx = \int_{-\infty}^{\infty} \left(\int_{\{u=t\}} |\nabla u(x)| \, d\mathcal{H}^{d-1}(x) \right) dt.$$

Because $u \geq 0$, the level sets $\{u = t\}$ are empty for $t < 0$, hence

$$\int_{\mathbb{R}^d} |\nabla u|^2 \, dx = \int_{0}^{\infty} \left(\int_{\{u=t\}} |\nabla u| \, d\mathcal{H}^{d-1} \right) dt. \tag{1}$$

Second, take $g(x) \coloneqq 1/|\nabla u(x)|$ on the set where $\nabla u(x) \neq 0$ and $g(x) \coloneqq 0$ where $\nabla u(x) = 0$. Then g is measurable and nonnegative. The coarea formula gives

$$\int_{\mathbb{R}^d} \mathbf{1}_{\{u>t\}}(x) \, dx = \int_{t}^{\infty} \left(\int_{\{u=s\}} \frac{1}{|\nabla u(x)|} \, d\mathcal{H}^{d-1}(x) \right) ds \quad \text{for every } t \geq 0.$$

The left-hand side equals $\mu_u(t)$. Therefore

$$\mu_u(t) = \int_{t}^{\infty} \left(\int_{\{u=s\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^{d-1} \right) ds \quad \text{for every } t \geq 0. \tag{2}$$

Step 3: Differentiate (2) to obtain $-\mu_u'(t)$. The function $t \mapsto \mu_u(t)$ is monotone nonincreasing, hence differentiable for a.e. t. Fix a value t at which μ_u is differentiable. From (2), for $h > 0$,

$$\mu_u(t+h) - \mu_u(t) = -\int_{t}^{t+h} \left(\int_{\{u=s\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^{d-1} \right) ds.$$

Divide by h:

$$\frac{\mu_u(t+h) - \mu_u(t)}{h} = -\frac{1}{h} \int_{t}^{t+h} \left(\int_{\{u=s\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^{d-1} \right) ds.$$

As $h \downarrow 0$, the right-hand side converges to the integrand at $s = t$ for a.e. t by the Lebesgue differentiation theorem, applied to the locally integrable function

$$s \mapsto \int_{\{u=s\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^{d-1}.$$

Hence for a.e. t,

$$-\mu_u'(t) = \int_{\{u=t\}} \frac{1}{|\nabla u|} \, d\mathcal{H}^{d-1}. \tag{3}$$

Step 4: A Cauchy-Schwarz estimate on the level set $\{u = t\}$. For a.e. $t > 0$ the set $\{u = t\}$ is a $(d-1)$-dimensional C^1 hypersurface (up to finitely many connected components), hence it has finite $\mathcal{H}^{d-1}$ measure. Define $P(E_t) \coloneqq \mathcal{H}^{d-1}(\partial E_t)$; for smooth u and regular t this equals $\mathcal{H}^{d-1}(\{u = t\})$. Then

$$P(E_t) = \int_{\{u=t\}} 1 \, d\mathcal{H}^{d-1}.$$

Apply Cauchy-Schwarz on $\{u=t\}$ with respect to the measure $\mathcal{H}^{d-1}$:

$$\left(\int_{\{u=t\}} 1\, d\mathcal{H}^{d-1}\right)^2 = \left(\int_{\{u=t\}} |\nabla u|^{1/2} \cdot |\nabla u|^{-1/2}\, d\mathcal{H}^{d-1}\right)^2 \leq$$

$$\leq \left(\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{d-1}\right)\left(\int_{\{u=t\}} \frac{1}{|\nabla u|}\, d\mathcal{H}^{d-1}\right).$$

Using (3), for a.e. t this becomes

$$\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{d-1} \geq \frac{P(E_t)^2}{-\mu_u'(t)}. \tag{4}$$

Step 5: Compare with the rearranged level sets using isoperimetry. Let u^* be the symmetric decreasing rearrangement. For each t, the set

$$E_t^* := \{x \mid u^*(x) > t\}$$

is a ball centered at the origin with $|E_t^*| = |E_t| = \mu_u(t)$, hence $\mu_{u^*}(t) = \mu_u(t)$. By Theorem 1.7.8, for a.e. t,

$$P(E_t) \geq P(E_t^*).$$

Applying (4) and using $\mu_{u^*} = \mu_u$ yields, for a.e. t,

$$\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{d-1} \geq \frac{P(E_t)^2}{-\mu_u'(t)} \geq \frac{P(E_t^*)^2}{-\mu_u'(t)} = \int_{\{u^*=t\}} |\nabla u^*|\, d\mathcal{H}^{d-1}.$$

Step 6: integrate in t using (1). Integrate the previous inequality over $t \in (0,\infty)$ and use (1) for both u and u^*:

$$\int_{\mathbb{R}^d} |\nabla u|^2\, dx = \int_0^\infty \left(\int_{\{u=t\}} |\nabla u|\, d\mathcal{H}^{d-1}\right) dt \geq \int_0^\infty \left(\int_{\{u^*=t\}} |\nabla u^*|\, d\mathcal{H}^{d-1}\right) dt = \int_{\mathbb{R}^d} |\nabla u^*|^2\, dx.$$

Thus $||\nabla u^*||_2 \leq ||\nabla u||_2$ for smooth compactly supported u.

Part 2: Extension from C_c^∞ to H^1. Let $u \in H^1(\mathbb{R}^d)$, $u \geq 0$ a.e. Choose $u_n \in C_c^\infty(\mathbb{R}^d)$ with $u_n \geq 0$ and $u_n \to u$ in $H^1(\mathbb{R}^d)$. Then $||u_n - u||_2 \to 0$ and $||\nabla u_n - \nabla u||_2 \to 0$. For each n, Part 1 gives $||\nabla u_n^*||_2 \leq ||\nabla u_n||_2$. Because $||u_n^*||_2 = ||u_n||_2$ by (i) and $||\nabla u_n^*||_2 \leq ||\nabla u_n||_2$, the sequence (u_n^*) is bounded in $H^1(\mathbb{R}^d)$.

Hence there exists a subsequence (still denoted u_n^*) and $v \in H^1(\mathbb{R}^d)$ such that $u_n^* \rightharpoonup v$ weakly in $H^1(\mathbb{R}^d)$. We now identify v with u^*.

Step 1: Almost everywhere convergence of a subsequence of u_n^*. Since (u_n^*) is bounded in $H^1(\mathbb{R}^d)$, for every $m \in \mathbb{N}$ the sequence (u_n^*) is bounded in $H^1(B_m)$. By the Rellich-Kondrachov theorem on the bounded domain B_m, there exists a subsequence (depending on m) which converges strongly in $L^2(B_m)$. Using a diagonal extraction over $m \in \mathbb{N}$, we may assume, after passing to a subsequence (not relabeled), that

$$u_n^* \to v \quad \text{strongly in } L^2(B_m) \quad \text{for every } m \in \mathbb{N}.$$

Fix m. Strong convergence in $L^2(B_m)$ implies that there exists a further subsequence (depending on m) which converges pointwise a.e. on B_m. Applying, again, a diagonal extraction over $m \in \mathbb{N}$, we may assume, in addition, that

$$u_n^*(x) \to v(x) \quad \text{for a.e. } x \in \mathbb{R}^d.$$

Step 2: Almost everywhere convergence of a subsequence of u_n. Since $u_n \to u$ in $L^2(\mathbb{R}^d)$, there exists (after passing to a subsequence, not relabeled) a pointwise a.e. convergence

$$u_n(x) \to u(x) \quad \text{for a.e. } x \in \mathbb{R}^d.$$

Step 3: Convergence of distribution functions. Define, for $t \geq 0$,

$$\mu_f(t) := \left|\left\{x \in \mathbb{R}^d \;\middle|\; f(x) > t\right\}\right|.$$

Because $f \in L^2(\mathbb{R}^d)$ implies $\mu_f(t) \leq t^{-2}||f||_2^2 < \infty$ for every $t > 0$ (Chebyshev), all quantities below are finite for $t > 0$. We claim that the set of $t > 0$ such that $|\{u = t\}| > 0$ is at most countable. Indeed, write $\mathbb{R}^d = \bigcup_{m\in\mathbb{N}} B_m$ and, for $m, k \in \mathbb{N}$, set

$$A_{m,k} := \left\{t > 0 \;\middle|\; |\{u = t\} \cap B_m| \geq \frac{1}{k}\right\}.$$

For fixed (m, k), the sets $\{u = t\} \cap B_m$ are pairwise disjoint as t varies, and each has measure at least $1/k$. Since $|B_m| < \infty$, the set $A_{m,k}$ must be finite. Therefore

$$\{t > 0 \mid |\{u = t\}| > 0\} \subseteq \bigcup_{m\in\mathbb{N}} \bigcup_{k\in\mathbb{N}} A_{m,k}$$

is a countable union of finite sets, hence countable. The same argument applies to v.

Fix $t > 0$ such that $|\{u = t\}| = 0$ and $|\{v = t\}| = 0$. For such a t, the pointwise a.e. convergences imply

$$\mathbf{1}_{\{u_n>t\}}(x) \to \mathbf{1}_{\{u>t\}}(x) \quad \text{for a.e. } x, \qquad \mathbf{1}_{\{u_n^*>t\}}(x) \to \mathbf{1}_{\{v>t\}}(x) \quad \text{for a.e. } x.$$

Because $0 \leq \mathbf{1}_{\{u_n>t\}} \leq 1$ and $0 \leq \mathbf{1}_{\{u_n^*>t\}} \leq 1$, dominated convergence yields

$$\mu_{u_n}(t) = \int_{\mathbb{R}^d} \mathbf{1}_{\{u_n>t\}}\, dx \to \int_{\mathbb{R}^d} \mathbf{1}_{\{u>t\}}\, dx = \mu_u(t),$$

and

$$\mu_{u_n^*}(t) = \int_{\mathbb{R}^d} \mathbf{1}_{\{u_n^*>t\}}\, dx \to \int_{\mathbb{R}^d} \mathbf{1}_{\{v>t\}}\, dx = \mu_v(t).$$

On the other hand, by equimeasurability of rearrangement, $\mu_{u_n^*}(t) = \mu_{u_n}(t)$ for all n and all $t \geq 0$. Hence, for a.e. $t > 0$,

$$\mu_v(t) = \mu_u(t).$$

Step 4: Uniqueness of the symmetric decreasing rearrangement. Since each u_n^* is radial and nonincreasing, the pointwise a.e. limit v is radial and nonincreasing as well (up to modification on a null set). The symmetric decreasing rearrangement u^* is, by definition, radial, nonincreasing, and satisfies $\mu_{u^*}(t) = \mu_u(t)$ for every $t \geq 0$. We now show that any radial nonincreasing function with $\mu_f = \mu_u$ must coincide with u^* a.e.

Let $\omega_d \coloneqq |B_1(0)|$. For a radial nonincreasing function f, define

$$\rho_f(t) \coloneqq \Big(\frac{\mu_f(t)}{\omega_d}\Big)^{1/d} \quad \text{for } t > 0.$$

Then $\{f > t\}$ is the ball of radius $\rho_f(t)$, hence $f(r) > t$ iff $r < \rho_f(t)$ for a.e. $t > 0$. Define the radial profile

$$\widetilde{f}(r) \coloneqq \sup\{t > 0 \mid r < \rho_f(t)\}.$$

Then $\widetilde{f}(|x|) = f(x)$ for a.e. $x \in \mathbb{R}^d$. Applying this construction to v and to u^*, and using $\mu_v(t) = \mu_u(t) = \mu_{u^*}(t)$ for a.e. $t > 0$, we obtain $\rho_v(t) = \rho_{u^*}(t)$ for a.e. $t > 0$, hence $\widetilde{v}(r) = \widetilde{u^*}(r)$ for a.e. $r > 0$. Therefore $v = u^*$ a.e.

Finally, weak lower semicontinuity of the L^2 norm gives

$$||\nabla u^*||_2 = ||\nabla v||_2 \leq \liminf_{n\to\infty} ||\nabla u_n^*||_2 \leq \liminf_{n\to\infty} ||\nabla u_n||_2 = ||\nabla u||_2.$$

■

1.8 Arzelà-Ascoli

Lemma 1.8.1 (Arzelà-Ascoli)**.** *Let $m \in \mathbb{N}$ and let $K \subseteq \mathbb{R}^m$ be compact and let $(f_n) \subseteq C(K)$. Assume:*

(i) $\sup_n \|f_n\|_{L^\infty(K)} < \infty$,

(ii) for every $\varepsilon > 0$ there exists $\delta(\varepsilon) = \delta > 0$ such that $\forall n \in \mathbb{N}$ and $\forall x, y \in K$, $|x - y| < \delta$ implies $|f_n(x) - f_n(y)| < \varepsilon$ (equicontinuity).

Then there exists a subsequence (f_{n_k}) and $f \in C(K)$ such that $f_{n_k} \to f$ uniformly on K.

Proof. We give a complete proof of the classical Arzelà-Ascoli theorem; compare, e.g., [20].

Step 1: Fix a countable dense subset of K. Because $K \subsetneq \mathbb{R}^m$ is compact, it is separable. Therefore there exists a countable dense subset

$$D \coloneqq \{x_1, x_2, x_3, \dots\} \subseteq K$$

such that the closure of D in K equals K.

Step 2: Extract a subsequence that converges at each point of D (diagonal extraction). By assumption (i), there exists $M < \infty$ such that for all n,

$$\|f_n\|_{L^\infty(K)} \leq M.$$

In particular, for each fixed $j \in \mathbb{N}$, the scalar sequence $(f_n(x_j))_{n\in\mathbb{N}}$ is bounded in $\mathbb{R}$:

$$|f_n(x_j)| \leq M \qquad \text{for all } n.$$

A bounded sequence in $\mathbb{R}$ has a convergent subsequence (Bolzano-Weierstrass). We now perform an iterative construction.

Step 3: Construction of nested subsequences.

(a) For $j = 1$, choose a subsequence $(f_n^{(1)})_{n\in\mathbb{N}}$ of (f_n) such that $(f_n^{(1)}(x_1))_{n\in\mathbb{N}}$ converges.

(b) Suppose that for some $j \geq 2$ we have already chosen a subsequence $(f_n^{(j-1)})$ of (f_n) such that $(f_n^{(j-1)}(x_i))$ converges for each $i \in \{1, \dots, j-1\}$. Since $(f_n^{(j-1)}(x_j))$ is still bounded in $\mathbb{R}$, choose a further subsequence $(f_n^{(j)})$ of $(f_n^{(j-1)})$ such that $(f_n^{(j)}(x_j))$ converges. Then $(f_n^{(j)}(x_i))$ converges for each $i \leq j$, because $(f_n^{(j)})$ is a subsequence of $(f_n^{(j-1)})$.

This produces a family of subsequences

$$(f_n^{(1)}) \supset (f_n^{(2)}) \supset (f_n^{(3)}) \supset \cdots$$

such that for each j, the sequence $(f_n^{(j)}(x_i))$ converges for all $i \le j$.

Step 4: Define the diagonal subsequence. Set

$$g_k := f_k^{(k)} \qquad (k \in \mathbb{N}).$$

Because $(f_n^{(k)})$ is a subsequence of $(f_n^{(j)})$ whenever $k \ge j$, it follows that for each fixed j, the tail $(g_k)_{k \ge j}$ is a subsequence of $(f_n^{(j)})$. Therefore, for each fixed j, the real sequence $(g_k(x_j))_{k\in\mathbb{N}}$ converges.

Define

$$\ell_j := \lim_{k\to\infty} g_k(x_j) \in \mathbb{R}.$$

Step 5: Show (g_k) is Cauchy in $L^\infty(K)$. We prove that for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $k, \ell \ge N$,

$$\|g_k - g_\ell\|_{L^\infty(K)} < \varepsilon.$$

Fix $\varepsilon > 0$.

Step 5(a): Choose δ from equicontinuity. By assumption (ii) (equicontinuity), there exists $\delta > 0$ such that for all $n \in \mathbb{N}$ and all $x, y \in K$,

$$|x - y| < \delta \implies |g_n(x) - g_n(y)| < \varepsilon/3.$$

(We may apply (ii) to the sequence (g_n) because (g_n) is a subsequence of (f_n) and inherits the same equicontinuity modulus.)

Step 5(b): Cover K by finitely many δ-balls centered at points of D. Because D is dense in K, for every $x \in K$ there exists some $x_j \in D$ with $|x - x_j| < \delta/2$. Therefore the family of open balls

$$\{B(x_j, \delta/2) \cap K\}_{j\in\mathbb{N}}$$

is an open cover of K (relative to K). Since K is compact, there exists a finite subcover: there exist indices $j_1, \dots, j_J$ such that

$$K \subseteq \bigcup_{r=1}^{J} \big(B(x_{j_r}, \delta/2) \cap K\big).$$

In particular, for every $x \in K$ there exists $r \in \{1, \dots, J\}$ such that

$$|x - x_{j_r}| < \frac{\delta}{2} < \delta. \tag{5}$$

Step 5(c): Use pointwise convergence at finitely many centers to obtain uniform Cauchy. For each fixed $r \in \{1, \dots, J\}$, the sequence $(g_k(x_{j_r}))_{k\in\mathbb{N}}$ converges (Step 2), hence it is Cauchy. Therefore there exists $N_r \in \mathbb{N}$ such that for all $k, \ell \ge N_r$,

$$|g_k(x_{j_r}) - g_\ell(x_{j_r})| < \varepsilon/3.$$

Let

$$N := \max\{N_1, \dots, N_J\}.$$

Then for all $k, \ell \geq N$ and all $r \in \{1, \ldots, J\}$,

$$|g_k(x_{j_r}) - g_\ell(x_{j_r})| < \varepsilon/3. \tag{6}$$

Now fix any $x \in K$. Choose r such that (6) applies and (6) holds at x_{j_r}, and also (6) holds for all $k, \ell \geq N$. Then for $k, \ell \geq N$, using the triangle inequality,

$$|g_k(x) - g_\ell(x)| \leq |g_k(x) - g_k(x_{j_r})| + |g_k(x_{j_r}) - g_\ell(x_{j_r})| + |g_\ell(x_{j_r}) - g_\ell(x)|.$$

The middle term is $< \varepsilon/3$ by (6). For the first and third terms, since $|x - x_{j_r}| < \delta$ by (5), equicontinuity gives

$$|g_k(x) - g_k(x_{j_r})| < \varepsilon/3, \qquad |g_\ell(x_{j_r}) - g_\ell(x)| < \varepsilon/3.$$

Therefore

$$|g_k(x) - g_\ell(x)| < \varepsilon.$$

Since this holds for all $x \in K$, we obtain

$$\|g_k - g_\ell\|_{L^\infty(K)} < \varepsilon \quad \text{for all } k, \ell \geq N.$$

Thus (g_k) is Cauchy in $(C(K), \|\cdot\|_{L^\infty(K)})$.

Step 6: Existence of a uniform limit f, and continuity of f. The space $(C(K), \|\cdot\|_{L^\infty(K)})$ is complete, because K is compact. We justify completeness explicitly:

Let (h_n) be Cauchy in $L^\infty(K)$ with each $h_n \in C(K)$. For each $x \in K$, the real sequence $(h_n(x))$ is Cauchy because

$$|h_n(x) - h_m(x)| \leq \|h_n - h_m\|_{L^\infty(K)}.$$

Hence $(h_n(x))$ converges in $\mathbb{R}$. Define

$$h(x) := \lim_{n \to \infty} h_n(x).$$

We show $h_n \to h$ uniformly. Fix $\varepsilon > 0$. Choose N such that for all $n, m \geq N$, $\|h_n - h_m\|_\infty < \varepsilon$. Fix $n \geq N$ and let $m \to \infty$. For each x,

$$|h_n(x) - h(x)| = \lim_{m \to \infty} |h_n(x) - h_m(x)| \leq \limsup_{m \to \infty} \|h_n - h_m\|_\infty \leq \varepsilon.$$

Taking the supremum over x gives $\|h_n - h\|_\infty \leq \varepsilon$ for all $n \geq N$. Therefore $h_n \to h$ uniformly.

Finally, h is continuous because it is a uniform limit of continuous functions: fix $x \in K$ and $\varepsilon > 0$. Choose N such that $\|h - h_N\|_\infty < \varepsilon/3$. Since h_N is continuous at x, there exists $\delta > 0$ such that $|x - y| < \delta$ implies $|h_N(x) - h_N(y)| < \varepsilon/3$. Then for such y,

$$|h(x) - h(y)| \leq |h(x) - h_N(x)| + |h_N(x) - h_N(y)| + |h_N(y) - h(y)| < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon.$$

So h is continuous.

This proves completeness.

Apply completeness to the Cauchy sequence (g_k): there exists $f \in C(K)$ such that $\|g_k - f\|_{L^\infty(K)} \to 0$ as $k \to \infty$. Setting $f_{n_k} := g_k$ gives the desired subsequence. ∎

1.9 Brezis-Lieb Lemma

Lemma 1.9.1 (Brezis-Lieb Lemma)**.** *Let (Ω, μ) be a measure space. Let $p \in (1, \infty)$. Let $(f_n)_{n\in\mathbb{N}} \subseteq L^p(\Omega)$ be a bounded sequence, and assume that $f_n \to f$ almost everywhere in Ω, for some $f \in L^p(\Omega)$. Then*

$$||f_n||^p_{L^p(\Omega)} - ||f_n - f||^p_{L^p(\Omega)} \to ||f||^p_{L^p(\Omega)}. \tag{7}$$

Proof. We prove (7) by dominated convergence.

Fix $p \in (1, \infty)$. Define, for $a, b \in \mathbb{C}$,

$$G(a, b) \coloneqq |a|^p - |a - b|^p - |b|^p.$$

We will show that $G(f_n, f) \to 0$ in $L^1(\Omega)$.

Step 1: Pointwise convergence. Since $f_n \to f$ almost everywhere, we have $f_n - f \to 0$ almost everywhere. Therefore, for almost every point $\omega \in \Omega$,

$$G(f_n(\omega), f(\omega)) = |f_n(\omega)|^p - |f_n(\omega) - f(\omega)|^p - |f(\omega)|^p \to |f(\omega)|^p - 0 - |f(\omega)|^p = 0.$$

Step 2: A pointwise upper bound by an L^1 function. We claim that there exists $C_p \in (0, \infty)$ such that

$$|G(a, b)| \le C_p\big(|a|^{p-1}|b| + |b|^p\big), \qquad \forall a, b \in \mathbb{C}. \tag{8}$$

To prove (8), we use the mean value theorem on the map $\Phi(z) \coloneqq |z|^p$ viewed as a C^1 map $\mathbb{R}^2 \to \mathbb{R}$. For $a, b \in \mathbb{C}$, write $a = a_1 + ia_2$, $b = b_1 + ib_2$ and identify them with vectors in $\mathbb{R}^2$. Then

$$\Phi(a) - \Phi(a - b) = \nabla\Phi(a - \theta b) \cdot b,$$

for some $\theta \in (0, 1)$, by the mean value theorem along the segment $a - sb$, $s \in [0, 1]$. A direct computation of the gradient yields

$$|\nabla\Phi(z)| \le p|z|^{p-1}, \qquad \forall z \in \mathbb{R}^2.$$

Therefore,

$$|\Phi(a) - \Phi(a - b)| \le p|a - \theta b|^{p-1}|b| \le p\big(|a| + |b|\big)^{p-1}|b|.$$

Using the elementary inequality $(\alpha + \beta)^{p-1} \le 2^{p-2}(\alpha^{p-1} + \beta^{p-1})$ for $\alpha, \beta \ge 0$, we obtain

$$|\Phi(a) - \Phi(a - b)| \le p2^{p-2}\big(|a|^{p-1}|b| + |b|^p\big).$$

Since $G(a, b) = \Phi(a) - \Phi(a - b) - |b|^p$, we get

$$|G(a, b)| \le |\Phi(a) - \Phi(a - b)| + |b|^p \le (p2^{p-2})|a|^{p-1}|b| + (p2^{p-2} + 1)|b|^p.$$

Thus (8) holds with $C_p \coloneqq \max\{p2^{p-2}, \, p2^{p-2} + 1\}$.

We now apply (8) with $a = f_n(\omega)$ and $b = f(\omega)$. We obtain, for almost every $\omega \in \Omega$,

$$|G(f_n(\omega), f(\omega))| \le C_p\Big(|f_n(\omega)|^{p-1}|f(\omega)| + |f(\omega)|^p\Big).$$

We now verify that the right-hand side is integrable uniformly in n.

Since (f_n) is bounded in $L^p(\Omega)$, there exists $M \in (0,\infty)$ such that $||f_n||_p \leq M$ for all n. By Hölder's inequality, for each n,

$$\int_\Omega |f_n|^{p-1}|f| \leq \Big(\int_\Omega |f_n|^p\Big)^{(p-1)/p}\Big(\int_\Omega |f|^p\Big)^{1/p} \leq M^{p-1}||f||_p.$$

Also $\int_\Omega |f|^p = ||f||_p^p < \infty$. Hence,

$$\sup_{n\in\mathbb{N}} \int_\Omega \Big(|f_n|^{p-1}|f| + |f|^p\Big) < \infty,$$

so the dominating functions are in $L^1(\Omega)$.

Step 3: Dominated convergence. By Step 1, $G(f_n, f) \to 0$ almost everywhere. By Step 2, $|G(f_n, f)|$ is dominated by an L^1 function uniformly in n. Therefore, by the dominated convergence theorem,

$$\int_\Omega |G(f_n, f)|\, d\mu \to 0.$$

Expanding $G(f_n, f)$, this is exactly (7). ∎

1.10 Fourier Transform

In several places, in particular, when interpreting elliptic operators and Sobolev spaces with possibly negative exponents, we use the Fourier transform on $\mathbb{R}^d$. [24, 12]

Definition 1.10.1 (Schwartz space). *The Schwartz space is*

$$\mathcal{S}(\mathbb{R}^d) := \left\{ f \in C^\infty(\mathbb{R}^d;\mathbb{C}) \;\middle|\; \sup_{x\in\mathbb{R}^d} \big|x^\alpha\, \partial^\beta f(x)\big| < \infty,\ \forall \alpha,\beta \in \mathbb{N}_0^d \right\}.$$

For $\alpha, \beta \in \mathbb{N}_0^d$, we set the seminorm

$$p_{\alpha,\beta}(f) := \sup_{x\in\mathbb{R}^d} \big|x^\alpha\, \partial^\beta f(x)\big|.$$

We always endow $\mathcal{S}(\mathbb{R}^d)$ with the locally convex topology generated by the family $(p_{\alpha,\beta})_{\alpha,\beta\in\mathbb{N}_0^d}$. [22, 24, 12]

Definition 1.10.2 (Tempered distributions). *The space of tempered distributions is the continuous dual of $\mathcal{S}(\mathbb{R}^d)$,*

$$\mathcal{S}'(\mathbb{R}^d) := \Big\{ T : \mathcal{S}(\mathbb{R}^d) \to \mathbb{C} \;\Big|\; T \text{ is linear and continuous} \Big\},$$

where continuity is meant with respect to the topology of Definition 1.10.1. [22, 24, 12]

Definition 1.10.3 (Fourier transform on $\mathcal{S}(\mathbb{R}^d)$). *For $f \in \mathcal{S}(\mathbb{R}^d)$, we define $\widehat{f} : \mathbb{R}^d \to \mathbb{C}$ by*

$$\widehat{f}(\xi) := (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix\cdot\xi} f(x)\, dx, \qquad \xi \in \mathbb{R}^d.$$

Theorem 1.10.4 (Inversion and basic identities on $\mathcal{S}(\mathbb{R}^d)$). *Let $f \in \mathcal{S}(\mathbb{R}^d)$. Then $\widehat{f} \in \mathcal{S}(\mathbb{R}^d)$, and one has the inversion formula*

$$f(x) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{ix\cdot\xi}\, \widehat{f}(\xi)\, d\xi, \qquad \forall x \in \mathbb{R}^d.$$

Moreover, $\forall \alpha \in \mathbb{N}_0^d$,

$$\widehat{(\partial^\alpha f)}(\xi) = (i\xi)^\alpha \widehat{f}(\xi), \qquad \forall \xi \in \mathbb{R}^d.$$

[24, 12]

Definition 1.10.5 (Fourier transform on $\mathcal{S}'(\mathbb{R}^d)$)**.** *Let* $T \in \mathcal{S}'(\mathbb{R}^d)$*. We define* $\widehat{T} \in \mathcal{S}'(\mathbb{R}^d)$ *by*

$$\widehat{T}(\varphi) := T(\widehat{\varphi}), \qquad \forall \varphi \in \mathcal{S}(\mathbb{R}^d),$$

where $\widehat{\varphi}$ *is the Fourier transform from Definition 1.10.3. [22, 24, 12]*

Theorem 1.10.6 (Plancherel)**.** *The Fourier transform extends uniquely to a unitary operator on* $L^2(\mathbb{R}^d)$*, and one has*

$$||\widehat{f}||_{L^2(\mathbb{R}^d)} = ||f||_{L^2(\mathbb{R}^d)}, \qquad \forall f \in L^2(\mathbb{R}^d).$$

[24, 12]

Definition 1.10.7 (Sobolev spaces via Fourier transform)**.** *Let* $s \in \mathbb{R}$*. We define*

$$H^s(\mathbb{R}^d) := \left\{ u \in \mathcal{S}'(\mathbb{R}^d) \;\middle|\; (1 + |\xi|^2)^{s/2}\, \widehat{u}(\xi) \in L^2(\mathbb{R}^d) \right\},$$

equipped with the norm

$$||u||_{H^s(\mathbb{R}^d)} := ||(1 + |\xi|^2)^{s/2} \widehat{u}||_{L^2(\mathbb{R}^d)}.$$

Remark 1.10.8. *For integer* $s \geq 0$*, Definition 1.10.7 is equivalent to the weak-derivative definition of* $H^s(\mathbb{R}^d) = W^{s,2}(\mathbb{R}^d)$*. [12, 9]*

1.11 Distributions

In this subsection, we recall distributions, test functions, and weak derivatives in the sense of distributions. This provides the rigorous meaning of differential operators acting on low-regularity functions, which is essential in Sections 2 and 4 when solutions are only known to lie in Sobolev spaces.

When we write identities such as $-\Delta u + u = f$ for functions u that are not classically twice differentiable, the correct interpretation is in the sense of distributions. [22, 4, 9]

Definition 1.11.1 (Test functions and distributions)**.** *We define the space of test functions by*

$$C_c^\infty(\mathbb{R}^d; \mathbb{C}) := \left\{ \varphi \in C^\infty(\mathbb{R}^d; \mathbb{C}) \;\middle|\; \mathrm{supp}(\varphi) \textit{ is compact} \right\}.$$

A distribution on $\mathbb{R}^d$ *is a continuous linear functional*

$$T : C_c^\infty(\mathbb{R}^d; \mathbb{C}) \to \mathbb{C},$$

where continuity is with respect to the usual test-function topology. The space of distributions is denoted by $\mathcal{D}'(\mathbb{R}^d)$*. [22, 4, 9]*

Definition 1.11.2 (Locally integrable functions as distributions)**.** *Let* $u \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$*. We associate to* u *the distribution* $T_u \in \mathcal{D}'(\mathbb{R}^d)$ *defined by*

$$T_u(\varphi) := \int_{\mathbb{R}^d} u(x)\, \varphi(x)\, dx, \qquad \forall \varphi \in C_c^\infty(\mathbb{R}^d; \mathbb{C}).$$

[4, 9]

Definition 1.11.3 (Distributional derivatives)**.** *Let $T \in \mathcal{D}'(\mathbb{R}^d)$ and let $\alpha \in \mathbb{N}_0^d$. The distributional derivative $D^\alpha T \in \mathcal{D}'(\mathbb{R}^d)$ is defined by*

$$(D^\alpha T)(\varphi) := (-1)^{|\alpha|} T(D^\alpha \varphi), \qquad \forall \varphi \in C_c^\infty(\mathbb{R}^d; \mathbb{C}).$$

Definition 1.11.4 (The space $H^{-1}(\mathbb{R}^d)$)**.** *We define*

$$H^{-1}(\mathbb{R}^d) := (H^1(\mathbb{R}^d))^*,$$

the space of continuous linear functionals on $H^1(\mathbb{R}^d)$. [4, 6]

Lemma 1.11.5 (The Laplacian maps H^1 into H^{-1})**.** *Let $d \in \mathbb{N}$. Then $\forall u \in H^1(\mathbb{R}^d)$, the distribution Δu defines an element of $H^{-1}(\mathbb{R}^d)$. Moreover, the map*

$$\Delta : H^1(\mathbb{R}^d) \to H^{-1}(\mathbb{R}^d)$$

is bounded. [6, 4]

Proof. **Step 1: Well-definedness of L_u.** Let $u \in H^1(\mathbb{R}^d)$. We define a functional L_u initially on $C_c^\infty(\mathbb{R}^d; \mathbb{C})$ by

$$L_u(\varphi) := -\int_{\mathbb{R}^d} \nabla u(x) \cdot \nabla \varphi(x)\, dx, \qquad \varphi \in C_c^\infty(\mathbb{R}^d; \mathbb{C}).$$

We first check that L_u is well-defined. Indeed, $\nabla u \in L^2(\mathbb{R}^d; \mathbb{C}^d)$ by the definition of $H^1(\mathbb{R}^d)$, and $\nabla \varphi \in C_c^\infty(\mathbb{R}^d; \mathbb{C}^d) \subseteq L^2(\mathbb{R}^d; \mathbb{C}^d)$, hence the pointwise product $\nabla u \cdot \nabla \varphi$ belongs to $L^1(\mathbb{R}^d)$ by Cauchy-Schwarz, and the integral is finite.

Step 2: Boundedness of L_u. Next, we show that L_u is bounded with respect to the H^1-norm on test functions. Let $\varphi \in C_c^\infty(\mathbb{R}^d; \mathbb{C})$. Then

$$|L_u(\varphi)| = \Big| \int_{\mathbb{R}^d} \nabla u(x) \cdot \nabla \varphi(x)\, dx \Big| \le \int_{\mathbb{R}^d} |\nabla u(x)|\, |\nabla \varphi(x)|\, dx \le ||\nabla u||_{L^2(\mathbb{R}^d)}\, ||\nabla \varphi||_{L^2(\mathbb{R}^d)}.$$

Since $||\nabla \varphi||_{L^2(\mathbb{R}^d)} \le ||\varphi||_{H^1(\mathbb{R}^d)}$, we obtain

$$|L_u(\varphi)| \le ||\nabla u||_{L^2(\mathbb{R}^d)}\, ||\varphi||_{H^1(\mathbb{R}^d)} \le ||u||_{H^1(\mathbb{R}^d)}\, ||\varphi||_{H^1(\mathbb{R}^d)}.$$

Therefore, L_u extends uniquely by density of $C_c^\infty(\mathbb{R}^d)$ in $H^1(\mathbb{R}^d)$ to a continuous linear functional on $H^1(\mathbb{R}^d)$, which we still denote by L_u. By Definition 1.11.4, we have $L_u \in H^{-1}(\mathbb{R}^d)$ and

$$||L_u||_{H^{-1}(\mathbb{R}^d)} = \sup_{\psi \in H^1(\mathbb{R}^d) \setminus \{0\}} \frac{|L_u(\psi)|}{||\psi||_{H^1(\mathbb{R}^d)}} \le ||u||_{H^1(\mathbb{R}^d)}.$$

Step 3: Identify L_u with Δu. Finally, we identify L_u with Δu in the sense of distributions. Let $\varphi \in C_c^\infty(\mathbb{R}^d; \mathbb{C})$. Since $u \in H^1(\mathbb{R}^d)$, the weak gradient ∇u exists in L^2. By the definition of distributional derivatives applied componentwise, the distribution Δu satisfies

$$(\Delta u)(\varphi) = u(\Delta \varphi) = \sum_{j=1}^d u(\partial_{jj} \varphi) = -\sum_{j=1}^d (\partial_j u)(\partial_j \varphi) = -\int_{\mathbb{R}^d} \nabla u(x) \cdot \nabla \varphi(x)\, dx = L_u(\varphi),$$

where we used Definition 1.11.3 and Definition 1.11.2 in the intermediate equalities.

Hence, Δu coincides with L_u on test functions, and therefore defines an element of $H^{-1}(\mathbb{R}^d)$ with the stated bound. ∎

Remark 1.11.6. *Lemma 1.11.5 is the basic reason why, in Section 4, the Schrödinger equation is naturally formulated as an identity in $H^{-1}(\mathbb{R}^d)$ when the solution is only known to lie in $H^1(\mathbb{R}^d)$. [6]*

2 Sobolev Inequality and Minimizer

In this section, we develop the sharp Gagliardo-Nirenberg inequality in the relevant setting and prove the existence of an optimizer. The main goal is to identify the variational inequality, optimizer, and sharp constant that produce the ground-state profile and to derive the stationary Euler-Lagrange equation, which becomes the key ingredient for uniqueness in Section 3, and for the blow-up analysis in Section 4.

2.1 Existence of an Optimizer for the Sharp Gagliardo-Nirenberg Inequality

In, this subsection, we set up the sharp variational problem and show that the best constant is attained by a maximizer. The purpose is to construct a function in $H^1(\mathbb{R}^2)$ that later serves as the ground-state profile for both the stationary equation in Subsection 2.2 and the time-dependent one in Section 4.

2.1.1 Scale-invariant Sobolev (Gagliardo-Nirenberg) inequality in $\mathbb{R}^2$

In, this subsubsection, we state the scale-invariant form of the inequality and define the variational functional whose maximizer yields the sharp constant. The goal is to isolate the precise inequality that governs the critical nonlinear term, which later controls the threshold for the blow-up phenomenon in Section 4.

The classical Sobolev embedding in two dimensions states that $H^1(\mathbb{R}^2) \hookrightarrow L^p(\mathbb{R}^2)$ for every $p \in [2, \infty)$, with a bound of the form $||u||_{L^p} \leq C_p ||u||_{H^1}$. For the analysis of the cubic NLS, it is convenient to isolate the *scale-invariant* version at $p = 4$, which interpolates between the L^2-norm (mass) and the homogeneous H^1-seminorm (kinetic energy).

Theorem 2.1.1 (Scale-invariant Sobolev (Gagliardo-Nirenberg) inequality in $\mathbb{R}^2$)**.** *Let $u \in H^1(\mathbb{R}^2)$. Then, there exists a constant $C > 0$, depending only on the dimension, such that*

$$||u||_{L^4(\mathbb{R}^2)} \leq C\, ||u||_{L^2(\mathbb{R}^2)}^{\frac{1}{2}}\, ||\nabla u||_{L^2(\mathbb{R}^2)}^{\frac{1}{2}}.$$

Proof. We start from the (non scale-invariant) Sobolev embedding in $\mathbb{R}^2$:

$$||u||_{L^4(\mathbb{R}^2)} \leq C||u||_{H^1(\mathbb{R}^2)} = C\Big(||u||_{L^2(\mathbb{R}^2)}^2 + ||\nabla u||_{L^2(\mathbb{R}^2)}^2\Big)^{\frac{1}{2}}. \tag{9}$$

Fix $\ell > 0$ and define the scaled function

$$u_\ell(x) := \ell\, u(\ell x).$$

We compute its norms explicitly.

Step 1: The L^2 norm. Using the change of variables $y := \ell x$ (so $dy = \ell^2\, dx$),

$$||u_\ell||_{L^2(\mathbb{R}^2)}^2 = \int_{\mathbb{R}^2} |\ell u(\ell x)|^2\, dx = \int_{\mathbb{R}^2} \ell^2 |u(\ell x)|^2\, dx = \int_{\mathbb{R}^2} |u(y)|^2\, dy = ||u||_{L^2(\mathbb{R}^2)}^2.$$

Step 2: The gradient norm. For $j \in \{1, 2\}$ we have, by the chain rule,

$$\partial_j u_\ell(x) = \partial_j\big(\ell u(\ell x)\big) = \ell^2 (\partial_j u)(\ell x).$$

Hence $|\nabla u_\ell(x)|^2 = \ell^4 |\nabla u(\ell x)|^2$, and with $y = \ell x$,

$$||\nabla u_\ell||^2_{L^2(\mathbb{R}^2)} = \int_{\mathbb{R}^2} \ell^4 |\nabla u(\ell x)|^2 \, dx = \ell^2 \int_{\mathbb{R}^2} |\nabla u(y)|^2 \, dy = \ell^2 ||\nabla u||^2_{L^2(\mathbb{R}^2)}.$$

Step 3: The L^4 norm. Again with $y = \ell x$,

$$||u_\ell||^4_{L^4(\mathbb{R}^2)} = \int_{\mathbb{R}^2} |\ell u(\ell x)|^4 \, dx = \int_{\mathbb{R}^2} \ell^4 |u(\ell x)|^4 \, dx = \ell^2 \int_{\mathbb{R}^2} |u(y)|^4 \, dy = \ell^2 ||u||^4_{L^4(\mathbb{R}^2)},$$

so

$$||u_\ell||_{L^4(\mathbb{R}^2)} = \ell^{1/2} ||u||_{L^4(\mathbb{R}^2)}.$$

Step 4: Apply (9) **to u_ℓ and simplify.** By (9),

$$||u_\ell||_{L^4(\mathbb{R}^2)} \le C \Big(||u_\ell||^2_{L^2(\mathbb{R}^2)} + ||\nabla u_\ell||^2_{L^2(\mathbb{R}^2)} \Big)^{1/2}.$$

Insert the identities from Steps 1-3:

$$\ell^{1/2} ||u||_{L^4(\mathbb{R}^2)} \le C \Big(||u||^2_{L^2(\mathbb{R}^2)} + \ell^2 ||\nabla u||^2_{L^2(\mathbb{R}^2)} \Big)^{1/2}.$$

Square both sides (both sides are nonnegative):

$$\ell \, ||u||^2_{L^4(\mathbb{R}^2)} \le C^2 \Big(||u||^2_{L^2(\mathbb{R}^2)} + \ell^2 ||\nabla u||^2_{L^2(\mathbb{R}^2)} \Big).$$

Divide by $\ell > 0$:

$$||u||^2_{L^4(\mathbb{R}^2)} \le C^2 \Big(\frac{1}{\ell} ||u||^2_{L^2(\mathbb{R}^2)} + \ell ||\nabla u||^2_{L^2(\mathbb{R}^2)} \Big).$$

Step 5: Optimize over $\ell > 0$. Set $A := ||u||_{L^2(\mathbb{R}^2)}$ and $B := ||\nabla u||_{L^2(\mathbb{R}^2)}$. Then, the bound reads

$$||u||^2_{L^4(\mathbb{R}^2)} \le C^2 \Big(\frac{A^2}{\ell} + \ell B^2 \Big).$$

Consider the function $F : (0, \infty) \to (0, \infty)$ given by

$$F(\ell) := \frac{A^2}{\ell} + \ell B^2.$$

Its derivative is

$$F'(\ell) = -\frac{A^2}{\ell^2} + B^2.$$

The unique critical point satisfies $F'(\ell) = 0$, i.e.,

$$-\frac{A^2}{\ell^2} + B^2 = 0 \quad \iff \quad \ell^2 = \frac{A^2}{B^2} \quad \iff \quad \ell = \frac{A}{B}.$$

Since

$$F''(\ell) = \frac{2A^2}{\ell^3} > 0 \quad \text{for all } \ell > 0,$$

this critical point is the global minimizer. Therefore,

$$\inf_{\ell>0} F(\ell) = F(A/B) = \frac{A^2}{A/B} + \frac{A}{B}B^2 = AB + AB = 2AB.$$

Thus,

$$||u||^2_{L^4(\mathbb{R}^2)} \leq 2C^2 \, ||u||_{L^2(\mathbb{R}^2)} \, ||\nabla u||_{L^2(\mathbb{R}^2)}.$$

Taking square roots on both sides yields

$$||u||_{L^4(\mathbb{R}^2)} \leq \sqrt{2}\, C \, ||u||^{1/2}_{L^2(\mathbb{R}^2)} \, ||\nabla u||^{1/2}_{L^2(\mathbb{R}^2)},$$

which is the desired estimate (after renaming the constant). ■

We assume the sharp Gagliardo-Nirenberg inequality has been established, i.e., there exists a constant $C_0 > 0$ such that for all $u \in H^1(\mathbb{R}^2)$,

$$\int_{\mathbb{R}^2} |u(x)|^4 \, dx \leq C_0 \left(\int_{\mathbb{R}^2} |u(x)|^2 \, dx \right) \left(\int_{\mathbb{R}^2} |\nabla u(x)|^2 \, dx \right). \tag{10}$$

Define the functional $\mathcal{J} : H^1(\mathbb{R}^2) \setminus \{0\} \to (0, \infty)$ by

$$\mathcal{J}(u) := \frac{\|u\|^4_{L^4(\mathbb{R}^2)}}{\|u\|^2_{L^2(\mathbb{R}^2)} \, \|\nabla u\|^2_{L^2(\mathbb{R}^2)}},$$

and define the best constant by

$$C_{\mathrm{GN}} := \sup_{u \in H^1(\mathbb{R}^2) \setminus \{0\}} \mathcal{J}(u). \tag{11}$$

First, we prove the following simple result.

Lemma 2.1.2 (Positivity of the denominator of $\mathcal{J}$). *Let $u \in H^1(\mathbb{R}^2) \setminus \{0\}$. Then,*

$$||u||^2_{L^2(\mathbb{R}^2)} \, ||\nabla u||^2_{L^2(\mathbb{R}^2)} > 0,$$

so $\mathcal{J}(u) \in (0, \infty)$ is well-defined.

Proof. Since $u \neq 0$ in $L^2(\mathbb{R}^2)$, we have $||u||^2_{L^2(\mathbb{R}^2)} > 0$.

Assume for contradiction that $||\nabla u||^2_{L^2(\mathbb{R}^2)} = 0$. Then $|\nabla u| = 0$ almost everywhere, hence each weak partial derivative $\partial_{x_j} u$ vanishes almost everywhere. It follows that u is almost everywhere equal to a constant on each connected component of $\mathbb{R}^2$, hence (since $\mathbb{R}^2$ is connected) $u = c$ almost everywhere for some constant $c \in \mathbb{C}$. If $c \neq 0$, then $u \notin L^2(\mathbb{R}^2)$, because

$$\int_{\mathbb{R}^2} |u(x)|^2 \, dx = \int_{\mathbb{R}^2} |c|^2 \, dx = +\infty,$$

contradicting $u \in H^1(\mathbb{R}^2) \subseteq L^2(\mathbb{R}^2)$. Hence $c = 0$, which implies $u = 0$ almost everywhere, contradicting $u \neq 0$. Therefore $||\nabla u||^2_{L^2(\mathbb{R}^2)} > 0$. ■

Our goal now is to prove that the supremum in (11) is attained.

Remark 2.1.3 (Maximizers of $\mathcal{J}$ and minimizers of I)**.** *Define, for every $u \in H^1(\mathbb{R}^2) \setminus \{0\}$,*

$$\mathcal{J}(u) := \frac{||u||^4_{L^4(\mathbb{R}^2)}}{||u||^2_{L^2(\mathbb{R}^2)}\,||\nabla u||^2_{L^2(\mathbb{R}^2)}}, \qquad I(u) := \frac{||u||^2_{L^2(\mathbb{R}^2)}\,||\nabla u||^2_{L^2(\mathbb{R}^2)}}{||u||^4_{L^4(\mathbb{R}^2)}}.$$

Then for every such u one has

$$I(u) = \frac{1}{\mathcal{J}(u)}.$$

Consequently, the following statements are equivalent:

(i) There exists $Q \in H^1(\mathbb{R}^2) \setminus \{0\}$ such that

$$\mathcal{J}(Q) = \sup_{u \in H^1(\mathbb{R}^2)\setminus\{0\}} \mathcal{J}(u).$$

(ii) There exists $Q \in H^1(\mathbb{R}^2) \setminus \{0\}$ such that

$$I(Q) = \inf_{u \in H^1(\mathbb{R}^2)\setminus\{0\}} I(u).$$

Proof. Throughout, we work in the extended real line $[0, \infty]$ with the conventions

$$\frac{1}{0} := +\infty, \qquad \frac{1}{+\infty} := 0.$$

For every $u \neq 0$ in $H^1(\mathbb{R}^2)$, the definition of I gives $I(u) = 1/\mathcal{J}(u)$.

Step 1: An order-reversing property of $x \mapsto 1/x$ on $[0, \infty]$. Let $a, b \in [0, \infty]$. We claim that

$$a \leq b \quad \Longrightarrow \quad \frac{1}{b} \leq \frac{1}{a}. \tag{12}$$

If $a = 0$, then (12) reads $1/b \leq 1/0 = +\infty$, which is true. If $b = +\infty$, then (12) reads $1/+\infty = 0 \leq 1/a$, which is true. If $a, b \in (0, \infty)$, then $a \leq b$ implies $1/b \leq 1/a$ by multiplying the inequality $a \leq b$ by the positive number $(ab)^{-1}$.

Step 2: $\inf(1/A) = 1/(\sup A)$ for sets $A \subseteq [0, \infty]$. Let $A \subseteq [0, \infty]$ be any nonempty set and define

$$S := \sup A \in [0, \infty], \qquad B := \left\{\frac{1}{a} : a \in A\right\} \subseteq [0, \infty].$$

We claim that

$$\inf B = \frac{1}{S}. \tag{13}$$

First, for every $a \in A$ we have $a \leq S$, hence by (12),

$$\frac{1}{S} \leq \frac{1}{a}.$$

Since $1/a \in B$ for each $a \in A$, it follows that $1/S$ is a lower bound for B, hence

$$\frac{1}{S} \leq \inf B. \tag{14}$$

Conversely, by definition of supremum: for every $\varepsilon > 0$, there exists $a_\varepsilon \in A$ such that

$$S - \varepsilon < a_\varepsilon \le S.$$

If $S = 0$, then $A \subseteq \{0\}$ so $B \subseteq \{+\infty\}$ and $\inf B = +\infty = 1/0 = 1/S$, so (13) holds. Assume now $S \in (0, \infty]$. If $S < \infty$, choose $\varepsilon \in (0, S)$ so that $S - \varepsilon > 0$. Then $a_\varepsilon > 0$ and by (12),

$$\frac{1}{a_\varepsilon} < \frac{1}{S - \varepsilon}.$$

Since $1/a_\varepsilon \in B$, we obtain

$$\inf B \le \frac{1}{a_\varepsilon} < \frac{1}{S - \varepsilon}.$$

Letting $\varepsilon \downarrow 0$ yields $\inf B \le 1/S$. If $S = +\infty$, then for every $M > 0$ there exists $a_M \in A$ with $a_M > M$, hence $1/a_M < 1/M$. Therefore $\inf B \le 1/M$ for all $M > 0$, which implies $\inf B = 0 = 1/(+\infty) = 1/S$. In all cases we have shown

$$\inf B \le \frac{1}{S}. \tag{15}$$

Combining (14) and (15) gives (13).

Step 3: Equivalence of optimizing $\mathcal{J}$ and optimizing I. Let

$$A := \{\mathcal{J}(u) : u \in H^1(\mathbb{R}^2) \setminus \{0\}\} \subseteq [0, \infty].$$

Then

$$\sup_{u \neq 0} \mathcal{J}(u) = \sup A.$$

Also, because $I(u) = 1/\mathcal{J}(u)$ for $u \neq 0$,

$$\{I(u) : u \in H^1(\mathbb{R}^2) \setminus \{0\}\} = \left\{\frac{1}{a} : a \in A\right\}.$$

By (13),

$$\inf_{u \neq 0} I(u) = \inf\left\{\frac{1}{a} : a \in A\right\} = \frac{1}{\sup A} = \frac{1}{\sup_{u \neq 0} \mathcal{J}(u)}. \tag{16}$$

Now assume (i). Then there exists $Q \neq 0$ with $\mathcal{J}(Q) = \sup_{u \neq 0} \mathcal{J}(u)$. Taking reciprocals and using $I(Q) = 1/\mathcal{J}(Q)$ gives

$$I(Q) = \frac{1}{\mathcal{J}(Q)} = \frac{1}{\sup_{u \neq 0} \mathcal{J}(u)}.$$

By (16), the right-hand side equals $\inf_{u \neq 0} I(u)$, hence Q is a minimizer of I and (ii) holds.

Conversely, assume (ii). Then there exists $Q \neq 0$ with $I(Q) = \inf_{u \neq 0} I(u)$. Again taking reciprocals,

$$\mathcal{J}(Q) = \frac{1}{I(Q)} = \frac{1}{\inf_{u \neq 0} I(u)}.$$

Using (16) rewritten as $\inf_{u \neq 0} I(u) = 1/\sup_{u \neq 0} \mathcal{J}(u)$ yields

$$\mathcal{J}(Q) = \sup_{u \neq 0} \mathcal{J}(u),$$

so (i) holds. This completes the proof. ■

2.1.2 Scaling invariance

In, this subsubsection, we compute how the functional behaves under the natural scaling symmetry and use this to normalize maximizing sequences. This normalization is essential, as it prevents loss of compactness due to scaling and prepares the compactness step carried out later in Subsubsection 2.1.4.

Lemma 2.1.4 (Critical scaling in $d = 2$)**.** *For $\lambda > 0$ and $u : \mathbb{R}^2 \to \mathbb{C}$ define*

$$u_\lambda(x) := \lambda\, u(\lambda x).$$

If $u \in H^1(\mathbb{R}^2)$ then $u_\lambda \in H^1(\mathbb{R}^2)$ and

$$\|u_\lambda\|_{L^2(\mathbb{R}^2)} = \|u\|_{L^2(\mathbb{R}^2)}, \qquad \|\nabla u_\lambda\|_{L^2(\mathbb{R}^2)} = \lambda\, \|\nabla u\|_{L^2(\mathbb{R}^2)},$$

$$\|u_\lambda\|^4_{L^4(\mathbb{R}^2)} = \lambda^2\, \|u\|^4_{L^4(\mathbb{R}^2)}, \qquad \textit{and therefore} \qquad \mathcal{J}(u_\lambda) = \mathcal{J}(u).$$

Proof. We compute each norm explicitly.

Step 1: L^2-norm. Using the change of variables $y = \lambda x$ (so $dy = \lambda^2\, dx$),

$$\|u_\lambda\|_2^2 = \int_{\mathbb{R}^2} |\lambda u(\lambda x)|^2\, dx = \int_{\mathbb{R}^2} \lambda^2 |u(\lambda x)|^2\, dx = \int_{\mathbb{R}^2} |u(y)|^2\, dy = \|u\|_2^2.$$

Step 2: Gradient and $\dot{H}^1$-norm. For each $j \in \{1, 2\}$, by the chain rule for smooth functions applied pointwise,

$$\partial_j u_\lambda(x) = \partial_j\big(\lambda u(\lambda x)\big) = \lambda \cdot \sum_{k=1}^{2} (\partial_k u)(\lambda x)\, \partial_j(\lambda x_k) = \lambda \cdot \sum_{k=1}^{2} (\partial_k u)(\lambda x)\, (\lambda \delta_{jk}) = \lambda^2 (\partial_j u)(\lambda x).$$

Hence

$$|\nabla u_\lambda(x)|^2 = \sum_{j=1}^{2} |\partial_j u_\lambda(x)|^2 = \sum_{j=1}^{2} \lambda^4 |(\partial_j u)(\lambda x)|^2 = \lambda^4 |\nabla u(\lambda x)|^2.$$

Therefore, again using $y = \lambda x$,

$$\|\nabla u_\lambda\|_2^2 = \int_{\mathbb{R}^2} \lambda^4 |\nabla u(\lambda x)|^2\, dx = \lambda^2 \int_{\mathbb{R}^2} |\nabla u(y)|^2\, dy = \lambda^2 \|\nabla u\|_2^2,$$

so $\|\nabla u_\lambda\|_2 = \lambda \|\nabla u\|_2$.

Step 3: L^4-norm. Using $y = \lambda x$,

$$\|u_\lambda\|_4^4 = \int_{\mathbb{R}^2} |\lambda u(\lambda x)|^4\, dx = \int_{\mathbb{R}^2} \lambda^4 |u(\lambda x)|^4\, dx = \lambda^2 \int_{\mathbb{R}^2} |u(y)|^4\, dy = \lambda^2 \|u\|_4^4.$$

Step 4: Invariance of $\mathcal{J}$. Insert the identities into the definition of $\mathcal{J}$:

$$\mathcal{J}(u_\lambda) = \frac{\|u_\lambda\|_4^4}{\|u_\lambda\|_2^2 \|\nabla u_\lambda\|_2^2} = \frac{\lambda^2 \|u\|_4^4}{\|u\|_2^2 (\lambda^2 \|\nabla u\|_2^2)} = \mathcal{J}(u).$$

∎

2.1.3 Reduction to nonnegative real-valued functions

In this subsubsection, we reduce the variational problem to nonnegative real-valued functions without decreasing the values that the functional attains. The goal is to remove the dependency on the phase, and to place the problem into a setting where rearrangement inequalities can later, in Section 1.7, be applied.

Lemma 2.1.5 (A Lipschitz approximation of the modulus)**.** *For $\varepsilon > 0$ define $\psi_\varepsilon : \mathbb{C} \simeq \mathbb{R}^2 \to \mathbb{R}$ by*

$$\psi_\varepsilon(z) := \sqrt{|z|^2 + \varepsilon^2} - \varepsilon.$$

Then for all $z \in \mathbb{C}$,

(i) $0 \le \psi_\varepsilon(z) \le |z|$,

(ii) $\psi_\varepsilon(z) \to |z|$ *as* $\varepsilon \to 0$,

(iii) ψ_ε *is globally Lipschitz with Lipschitz constant* 1.

Proof. Fix $z \in \mathbb{C}$.

Proof of (i). Since $\sqrt{|z|^2 + \varepsilon^2} \ge \varepsilon$, we have $\psi_\varepsilon(z) \ge 0$. Also,

$$\psi_\varepsilon(z) \le |z| \iff \sqrt{|z|^2 + \varepsilon^2} - \varepsilon \le |z| \iff \sqrt{|z|^2 + \varepsilon^2} \le |z| + \varepsilon.$$

Squaring both sides (both are nonnegative) gives

$$|z|^2 + \varepsilon^2 \le |z|^2 + 2\varepsilon|z| + \varepsilon^2,$$

which holds because $2\varepsilon|z| \ge 0$. Hence $\psi_\varepsilon(z) \le |z|$.

Proof of (ii). For fixed z, the map $\varepsilon \mapsto \sqrt{|z|^2 + \varepsilon^2} - \varepsilon$ is continuous and

$$\lim_{\varepsilon \to 0} \psi_\varepsilon(z) = \sqrt{|z|^2 + 0} - 0 = |z|.$$

Proof of (iii). We compute the derivative explicitly. Identify $\mathbb{C}$ with $\mathbb{R}^2$ and write $z = (z_1, z_2)$. For $z \neq 0$, ψ_ε is differentiable and

$$\nabla\psi_\varepsilon(z) = \nabla\big((|z|^2 + \varepsilon^2)^{1/2}\big) = \frac{1}{2}(|z|^2 + \varepsilon^2)^{-1/2}\,\nabla(|z|^2) = \frac{1}{2}(|z|^2 + \varepsilon^2)^{-1/2}\,(2z) = \frac{z}{\sqrt{|z|^2 + \varepsilon^2}}.$$

Hence

$$|\nabla\psi_\varepsilon(z)| = \frac{|z|}{\sqrt{|z|^2 + \varepsilon^2}} \le 1.$$

At $z = 0$, the same formula gives $|\nabla\psi_\varepsilon(0)| = 0$. Since $|\nabla\psi_\varepsilon| \le 1$ everywhere and ψ_ε is C^1, the mean value theorem implies

$$|\psi_\varepsilon(z) - \psi_\varepsilon(w)| \le |z - w| \quad \text{for all } z, w \in \mathbb{C},$$

so ψ_ε is globally 1-Lipschitz. ∎

Lemma 2.1.6 (Reduction to nonnegative real-valued maximizing sequences)**.** *For every $u \in H^1(\mathbb{R}^2; \mathbb{C})$ there exists $v \in H^1(\mathbb{R}^2)$ with $v \ge 0$ such that*

$$\|v\|_2 = \|u\|_2, \qquad \|v\|_4 = \|u\|_4, \qquad \|\nabla v\|_2 \le \|\nabla u\|_2.$$

Consequently, for such u and v,

$$\mathcal{J}(u) \le \mathcal{J}(v).$$

Proof. Define $v := |u|$ pointwise.

Step 1: L^p-norms. For $p \in \{2,4\}$,

$$\|v\|_p^p = \int_{\mathbb{R}^2} |v(x)|^p\,dx = \int_{\mathbb{R}^2} ||u(x)||^p\,dx = \int_{\mathbb{R}^2} |u(x)|^p\,dx = \|u\|_p^p.$$

Step 2: $v \in H^1$ and the gradient bound. We use the Lipschitz approximations ψ_ε from Lemma 2.1.5 and set

$$v_\varepsilon(x) := \psi_\varepsilon(u(x)).$$

We prove:

(a) $v_\varepsilon \in H^1(\mathbb{R}^2)$ and $\|\nabla v_\varepsilon\|_2 \le \|\nabla u\|_2$ for each $\varepsilon > 0$,
(b) $v_\varepsilon \to v$ in $L^2(\mathbb{R}^2)$ as $\varepsilon \to 0$,
(c) (v_ε) is bounded in $H^1(\mathbb{R}^2)$, hence has a weakly convergent subsequence in H^1,
(d) the weak limit must equal v, and then weak lower semi-continuity gives $\|\nabla v\|_2 \le \|\nabla u\|_2$.

Step 2(a): $v_\varepsilon \in H^1$ and $\|\nabla v_\varepsilon\|_2 \le \|\nabla u\|_2$. Because ψ_ε is globally Lipschitz and $\psi_\varepsilon(0) = 0$, the composition $v_\varepsilon = \psi_\varepsilon \circ u$ belongs to $H^1(\mathbb{R}^2)$. Moreover, by the chain rule for Lipschitz functions in Sobolev spaces, there exists a measurable vector field ∇v_ε satisfying

$$|\nabla v_\varepsilon(x)| \le \mathrm{Lip}(\psi_\varepsilon)\,|\nabla u(x)| \quad \text{for a.e. } x \in \mathbb{R}^2.$$

Since $\mathrm{Lip}(\psi_\varepsilon) = 1$ (Lemma 2.1.5(iii)), we have

$$|\nabla v_\varepsilon(x)| \le |\nabla u(x)| \quad \text{for a.e. } x.$$

Integrating the square and using monotonicity of the integral gives

$$\|\nabla v_\varepsilon\|_2^2 = \int_{\mathbb{R}^2} |\nabla v_\varepsilon(x)|^2\,dx \le \int_{\mathbb{R}^2} |\nabla u(x)|^2\,dx = \|\nabla u\|_2^2.$$

Step 2(b): $v_\varepsilon \to v$ in L^2. Fix $x \in \mathbb{R}^2$. By Lemma 2.1.5(ii), $v_\varepsilon(x) = \psi_\varepsilon(u(x)) \to |u(x)| = v(x)$ as $\varepsilon \to 0$. Thus $v_\varepsilon \to v$ pointwise.

Also, by Lemma 2.1.5(i),

$$0 \le v_\varepsilon(x) \le |u(x)| \quad \text{for all } x,$$

and therefore

$$|v_\varepsilon(x) - v(x)|^2 \le (|u(x)| + |u(x)|)^2 = 4|u(x)|^2.$$

Since $u \in L^2(\mathbb{R}^2)$, the function $4|u|^2$ is integrable. Therefore the dominated convergence theorem applies to the sequence $|v_\varepsilon - v|^2$, and we obtain

$$\lim_{\varepsilon\to 0} \int_{\mathbb{R}^2} |v_\varepsilon(x) - v(x)|^2\,dx = 0,$$

which is exactly $v_\varepsilon \to v$ in L^2.

Step 2(c): Weak compactness in H^1. From Step 2(a) and Step 2(b) we have

$$\sup_{\varepsilon\in(0,1]} \|v_\varepsilon\|_2 \le \|u\|_2 \quad \text{and} \quad \sup_{\varepsilon\in(0,1]} \|\nabla v_\varepsilon\|_2 \le \|\nabla u\|_2,$$

so $(v_\varepsilon)_{\varepsilon\in(0,1]}$ is bounded in the Hilbert space $H^1(\mathbb{R}^2)$. Hence there exists a sequence $\varepsilon_k \downarrow 0$ and $w \in H^1$ such that $v_{\varepsilon_k} \rightharpoonup w$ weakly in H^1.

Step 2(d): Identification of the weak limit and gradient inequality. The weak convergence implies in particular weak convergence in L^2. But we also have strong convergence $v_{\varepsilon_k} \to v$ in L^2 (Step 2(b)). A sequence in a Hilbert space cannot converge strongly to one limit and weakly to a different limit. Therefore $w = v$. Finally, weak lower semi-continuity of the L^2-norm implies

$$\|\nabla v\|_2 \leq \liminf_{k\to\infty} \|\nabla v_{\varepsilon_k}\|_2 \leq \|\nabla u\|_2.$$

Combining Step 1 and Step 2 gives the lemma.

Consequently,

$$\mathcal{J}(v) = \frac{\|v\|^4_{L^4(\mathbb{R}^2)}}{\|v\|^2_{L^2(\mathbb{R}^2)}\,\|\nabla v\|^2_{L^2(\mathbb{R}^2)}} = \frac{\|u\|^4_{L^4(\mathbb{R}^2)}}{\|u\|^2_{L^2(\mathbb{R}^2)}\,\|\nabla v\|^2_{L^2(\mathbb{R}^2)}} \geq \frac{\|u\|^4_{L^4(\mathbb{R}^2)}}{\|u\|^2_{L^2(\mathbb{R}^2)}\,\|\nabla u\|^2_{L^2(\mathbb{R}^2)}} = \mathcal{J}(u),$$

because the numerator and the L^2 factor in the denominator are unchanged, while $\|\nabla v\|_2 \leq \|\nabla u\|_2$ makes the denominator smaller. ■

Corollary 2.1.7 (Monotonicity of the Gagliardo-Nirenberg functional under rearrangement)**.** *Let $d \in \mathbb{N}$, and let $u \in H^1(\mathbb{R}^d)$ with $u \geq 0$ a.e. Then,*

$$\mathcal{J}(u^*) \geq \mathcal{J}(u).$$

Proof. By Theorem 1.7.6 (ii), with $p = 2$ and $p = 4$, we have

$$||u^*||_2 = ||u||_2, \qquad ||u^*||_4 = ||u||_4.$$

By Theorem 1.7.9,

$$||\nabla u^*||_2 \leq ||\nabla u||_2.$$

Therefore,

$$\mathcal{J}(u^*) = \frac{||u^*||_4^4}{||u^*||_2^2\,||\nabla u^*||_2^2} \geq \frac{||u||_4^4}{||u||_2^2\,||\nabla u||_2^2} = \mathcal{J}(u).$$

■

2.1.4 A local compactness lemma on balls (proof for bounded sequences in $H^1(\mathbb{R}^2)$)

In this subsubsection, we prove a compactness mechanism on bounded regions that converts boundedness in H^1 into strong convergence in L^2 locally. The purpose is to get the compactness in order to extract a non-zero limit from a maximizing sequence, which is then upgraded to a maximizer later, in Theorem 2.1.11.

Lemma 2.1.8 (A quantitative mollification estimate in $L^2(\mathbb{R}^2)$)**.** *Let $\rho \in C_c^\infty(\mathbb{R}^2)$ satisfy $\rho \geq 0$ and $\int_{\mathbb{R}^2} \rho = 1$. For $\varepsilon > 0$ set $\rho_\varepsilon(x) := \varepsilon^{-2}\rho(x/\varepsilon)$ and define $u_\varepsilon := u * \rho_\varepsilon$. Then for every $u \in H^1(\mathbb{R}^2)$,*

$$\|u - u_\varepsilon\|_{L^2(\mathbb{R}^2)} \leq \varepsilon \left(\int_{\mathbb{R}^2} |z|\rho(z)\,dz \right) \|\nabla u\|_{L^2(\mathbb{R}^2)}. \tag{17}$$

Proof. Fix $u \in H^1(\mathbb{R}^2)$ and $\varepsilon > 0$.

Step 1: Write the difference as an average of translations. For a.e. x,

$$u(x) - u_\varepsilon(x) = u(x) - \int_{\mathbb{R}^2} u(x-y)\rho_\varepsilon(y)\,dy = \int_{\mathbb{R}^2} \rho_\varepsilon(y)\big(u(x) - u(x-y)\big)\,dy.$$

Step 2: Estimate $u(x) - u(x-y)$ using the fundamental theorem of calculus. For each fixed $y \in \mathbb{R}^2$ and a.e. x, the map $t \mapsto u(x-ty)$ is absolutely continuous on $[0,1]$ (because $u \in H^1$ implies u has representatives with ACL property on lines). For such x,

$$u(x) - u(x-y) = \int_0^1 \frac{d}{dt} u(x-ty)\,dt = \int_0^1 (-y)\cdot \nabla u(x-ty)\,dt.$$

Taking absolute values and using Cauchy-Schwarz in $\mathbb{R}^2$ gives

$$|u(x) - u(x-y)| \le \int_0^1 |y|\,|\nabla u(x-ty)|\,dt = |y| \int_0^1 |\nabla u(x-ty)|\,dt.$$

Step 3: Take L^2-norms and apply Minkowski. Combine Step 1 and Step 2:

$$|u(x) - u_\varepsilon(x)| \le \int_{\mathbb{R}^2} \rho_\varepsilon(y)\,|y| \int_0^1 |\nabla u(x-ty)|\,dt\,dy.$$

Take the L^2-norm in x and use Minkowski's integral inequality twice (first in y, then in t):

$$\|u - u_\varepsilon\|_2 \le \int_{\mathbb{R}^2} \rho_\varepsilon(y)\,|y| \int_0^1 \|\nabla u(\cdot - ty)\|_2\,dt\,dy.$$

For each fixed t and y, translation invariance of Lebesgue measure implies $\|\nabla u(\cdot - ty)\|_2 = \|\nabla u\|_2$. Thus

$$\|u - u_\varepsilon\|_2 \le \int_{\mathbb{R}^2} \rho_\varepsilon(y)\,|y| \int_0^1 \|\nabla u\|_2\,dt\,dy = \Big(\int_{\mathbb{R}^2} \rho_\varepsilon(y)\,|y|\,dy\Big)\,\|\nabla u\|_2.$$

Step 4: Compute $\int \rho_\varepsilon(y)|y|\,dy$. Substitute $y = \varepsilon z$ (so $dy = \varepsilon^2 dz$):

$$\int_{\mathbb{R}^2} \rho_\varepsilon(y)|y|\,dy = \int_{\mathbb{R}^2} \varepsilon^{-2}\rho(y/\varepsilon)|y|\,dy = \int_{\mathbb{R}^2} \varepsilon^{-2}\rho(z)\,|\varepsilon z|\,\varepsilon^2\,dz = \varepsilon \int_{\mathbb{R}^2} |z|\rho(z)\,dz.$$

Insert this into Step 3 to obtain (17). ■

Lemma 2.1.9 (Local L^2-compactness of bounded $H^1(\mathbb{R}^2)$ sequences)**.** *Fix $R > 0$ and let (u_n) be bounded in $H^1(\mathbb{R}^2)$. Then there exist a subsequence (u_{n_k}) and $u \in L^2(B_R)$ such that*

$$u_{n_k} \to u \quad \textit{strongly in } L^2(B_R).$$

Proof. Let $M_0 := \sup_n \|u_n\|_2$ and $M_1 := \sup_n \|\nabla u_n\|_2$; both are finite by hypothesis.

Fix $\varepsilon \in (0,1]$ and define $v_{n,\varepsilon} := u_n * \rho_\varepsilon$.

Step 1: $\|u_n - v_{n,\varepsilon}\|_{L^2(\mathbb{R}^2)}$ is uniformly small for small ε. Lemma 2.1.8 gives

$$\|u_n - v_{n,\varepsilon}\|_2 \le \varepsilon\Big(\int |z|\rho(z)\,dz\Big)\,\|\nabla u_n\|_2 \le \varepsilon\Big(\int |z|\rho(z)\,dz\Big)\,M_1.$$

Define the constant $C_\rho := \int_{\mathbb{R}^2} |z|\rho(z)\,dz$. Then,

$$\sup_{n\in\mathbb{N}} \|u_n - v_{n,\varepsilon}\|_2 \le \varepsilon\,C_\rho\,M_1. \tag{18}$$

Step 2: For fixed ε, the family $(v_{n,\varepsilon})_n$ is uniformly bounded and equicontinuous on $\overline{B_R}$. Fix $x \in \mathbb{R}^2$. Using Cauchy-Schwarz,

$$|v_{n,\varepsilon}(x)| = \left| \int_{\mathbb{R}^2} u_n(x-y)\rho_\varepsilon(y)\, dy \right| \le \left(\int_{\mathbb{R}^2} |u_n(x-y)|^2\, dy \right)^{1/2} \left(\int_{\mathbb{R}^2} |\rho_\varepsilon(y)|^2\, dy \right)^{1/2}.$$

Translation invariance implies $\int |u_n(x-y)|^2 dy = \int |u_n(y)|^2 dy = \|u_n\|_2^2 \le M_0^2$. Also,

$$\|\rho_\varepsilon\|_2^2 = \int \varepsilon^{-4} \rho(y/\varepsilon)^2\, dy = \varepsilon^{-2} \int \rho(z)^2\, dz,$$

so $\|\rho_\varepsilon\|_2 = \varepsilon^{-1}\|\rho\|_2$. Therefore

$$\sup_{x\in\mathbb{R}^2} |v_{n,\varepsilon}(x)| \le M_0\, \varepsilon^{-1} \|\rho\|_2,$$

which is a uniform bound in n (for fixed ε).

Next, compute the gradient:

$$\nabla v_{n,\varepsilon} = \nabla(u_n * \rho_\varepsilon) = u_n * (\nabla \rho_\varepsilon).$$

The same Cauchy-Schwarz argument yields

$$\|\nabla v_{n,\varepsilon}\|_\infty \le \|u_n\|_2\, \|\nabla \rho_\varepsilon\|_2.$$

We compute $\|\nabla\rho_\varepsilon\|_2$ explicitly: since $\nabla\rho_\varepsilon(y) = \varepsilon^{-3}(\nabla\rho)(y/\varepsilon)$ in $d = 2$,

$$\|\nabla\rho_\varepsilon\|_2^2 = \int \varepsilon^{-6} |\nabla\rho(y/\varepsilon)|^2\, dy = \varepsilon^{-4} \int |\nabla\rho(z)|^2\, dz,$$

so $\|\nabla\rho_\varepsilon\|_2 = \varepsilon^{-2}\|\nabla\rho\|_2$. Hence

$$\sup_n \|\nabla v_{n,\varepsilon}\|_\infty \le M_0\, \varepsilon^{-2} \|\nabla\rho\|_2 < \infty.$$

Now for $x, y \in \overline{B_R}$, the mean value theorem gives

$$|v_{n,\varepsilon}(x) - v_{n,\varepsilon}(y)| \le |x-y|\, \|\nabla v_{n,\varepsilon}\|_\infty \le |x-y|\, M_0\, \varepsilon^{-2} \|\nabla\rho\|_2.$$

This is an equicontinuity estimate uniform in n.

Step 3: Apply Arzelà-Ascoli to $(v_{n,\varepsilon})$ on $\overline{B_R}$. By Step 2 and Lemma 1.8.1, there exists a subsequence (still denoted $v_{n,\varepsilon}$) and a continuous function v_ε on $\overline{B_R}$ such that $v_{n,\varepsilon} \to v_\varepsilon$ uniformly on $\overline{B_R}$. Uniform convergence implies convergence in $L^2(B_R)$:

$$\|v_{n,\varepsilon} - v_\varepsilon\|_{L^2(B_R)}^2 = \int_{B_R} |v_{n,\varepsilon}(x) - v_\varepsilon(x)|^2\, dx \le |B_R|\, \|v_{n,\varepsilon} - v_\varepsilon\|_{L^\infty(\overline{B_R})}^2 \to 0.$$

Step 4: Remove mollification using 18. Let $\delta > 0$. Choose $\varepsilon > 0$ so small that $\varepsilon C_\rho M_1 < \delta/3$. Then choose N such that for $n, m \ge N$,

$$\|v_{n,\varepsilon} - v_{m,\varepsilon}\|_{L^2(B_R)} < \delta/3.$$

For $n, m \ge N$,

$$\|u_n - u_m\|_{L^2(B_R)} \le \|u_n - v_{n,\varepsilon}\|_{L^2(B_R)} + \|v_{n,\varepsilon} - v_{m,\varepsilon}\|_{L^2(B_R)} + \|v_{m,\varepsilon} - u_m\|_{L^2(B_R)} \le$$

$$\le \|u_n - v_{n,\varepsilon}\|_2 + \frac{\delta}{3} + \|u_m - v_{m,\varepsilon}\|_2 < \delta,$$

where the last inequality uses 18. Thus (u_n) is Cauchy in $L^2(B_R)$ along a subsequence, hence converges strongly in $L^2(B_R)$ to some u. ■

2.1.5 Radial tail estimate

In this subsubsection, we control the mass of radial functions away from the origin in terms of their H^1-norms. The goal is to prevent the escape of the mass to spatial infinity, so that local compactness can be combined with control of the tails to yield global strong convergence for suitable normalized sequences.

Lemma 2.1.10 (Radial L^4 tail bound)**.** *Let $u \in H^1_{\mathrm{rad}}(\mathbb{R}^2)$. Then for every $R > 0$,*

$$\int_{|x|\geq R} |u(x)|^4 \, dx \leq \frac{1}{\pi R} \, \|u\|_2^3 \, \|\nabla u\|_2.$$

Proof. Write $u(x) = f(r)$ with $r = |x|$.

Step 1: Pointwise bound for $f(r)$. For a.e. $r > 0$,

$$|f(r)|^2 = -\int_r^\infty \frac{d}{ds}|f(s)|^2 \, ds = -2\Re \int_r^\infty f(s)\overline{f'(s)} \, ds \leq 2\int_r^\infty |f(s)| \, |f'(s)| \, ds.$$

Insert weights and use Cauchy-Schwarz:

$$\int_r^\infty |f||f'| \, ds = \int_r^\infty (\sqrt{s}|f'(s)|) \, \frac{|f(s)|}{\sqrt{s}} \, ds \leq \left(\int_r^\infty s|f'(s)|^2 \, ds\right)^{1/2} \left(\int_r^\infty \frac{|f(s)|^2}{s} \, ds\right)^{1/2}.$$

For $s \geq r$ we have $1/s^2 \leq 1/r^2$, hence

$$\int_r^\infty \frac{|f(s)|^2}{s} \, ds = \int_r^\infty |f(s)|^2 s \, \frac{1}{s^2} \, ds \leq \frac{1}{r^2} \int_0^\infty |f(s)|^2 s \, ds.$$

Therefore

$$|f(r)|^2 \leq \frac{2}{r} \left(\int_0^\infty s|f'(s)|^2 \, ds\right)^{1/2} \left(\int_0^\infty |f(s)|^2 s \, ds\right)^{1/2}.$$

Convert the integrals to $\|u\|_2$ and $\|\nabla u\|_2$ using polar coordinates:

$$\|u\|_2^2 = 2\pi \int_0^\infty |f(s)|^2 s \, ds, \qquad \|\nabla u\|_2^2 = 2\pi \int_0^\infty |f'(s)|^2 s \, ds.$$

Thus

$$|f(r)|^2 \leq \frac{2}{r} \cdot \frac{1}{2\pi} \, \|u\|_2 \, \|\nabla u\|_2 = \frac{1}{\pi r} \, \|u\|_2 \, \|\nabla u\|_2.$$

Step 2: Tail estimate. For $|x| \geq R$ we have $|u(x)|^2 \leq \frac{1}{\pi R}\|u\|_2\|\nabla u\|_2$. Hence

$$\int_{|x|\geq R} |u|^4 = \int_{|x|\geq R} |u|^2 \, |u|^2 \leq \left(\sup_{|x|\geq R} |u(x)|^2\right) \int_{|x|\geq R} |u(x)|^2 \, dx \leq \frac{1}{\pi R} \, \|u\|_2 \, \|\nabla u\|_2 \cdot \|u\|_2^2,$$

which is the claimed inequality. ∎

2.1.6 Existence of a maximizer

In this subsubsection, we assemble the previous results and compactness tools to show that the sharp constant is attained by a maximizer. This is most important variational result for the rest of the thesis, since it produces the ground-state profile that appears in the stationary equation in Subsection 2.2, and the minimal-mass dynamics in Section 4.

Theorem 2.1.11 (Existence of an optimizer for (10))**.** *There exists $Q \in H^1(\mathbb{R}^2)$ such that*

(i) $Q \geq 0$ almost everywhere,

(ii) Q is radial and nonincreasing in $|x|$,

(iii) $\mathcal{J}(Q) = C_{\mathrm{GN}}$.

Proof. **Step 1: Choose a maximizing sequence.** By definition of the supremum (11), there exists $(u_n) \subseteq H^1(\mathbb{R}^2) \setminus \{0\}$ such that

$$\lim_{n\to\infty} \mathcal{J}(u_n) = C_{\mathrm{GN}}. \tag{19}$$

Step 2: Reduce to nonnegative functions. By Lemma 2.1.6, for each n there exists $v_n \in H^1(\mathbb{R}^2)$ with $v_n \geq 0$ and $\mathcal{J}(v_n) \geq \mathcal{J}(u_n)$. Replacing u_n by v_n preserves (11) and (10), and preserves (19) with $\geq$. Thus we may assume $u_n \geq 0$ for all n.

Step 3: Normalize $\|u_n\|_2 = 1$ and $\|\nabla u_n\|_2 = 1$ by scaling. Define

$$a_n \coloneqq \|u_n\|_2, \qquad \tilde{u}_n \coloneqq \frac{u_n}{a_n}.$$

Then $\|\tilde{u}_n\|_2 = 1$ and $\mathcal{J}(\tilde{u}_n) = \mathcal{J}(u_n)$. Next define $\lambda_n \coloneqq \|\nabla \tilde{u}_n\|_2^{-1}$ and set

$$w_n \coloneqq (\tilde{u}_n)_{\lambda_n}.$$

Lemma 2.1.4 implies

$$\|w_n\|_2 = \|\tilde{u}_n\|_2 = 1, \qquad \|\nabla w_n\|_2 = \lambda_n \|\nabla \tilde{u}_n\|_2 = 1, \qquad \mathcal{J}(w_n) = \mathcal{J}(\tilde{u}_n) = \mathcal{J}(u_n).$$

In particular, from (19),

$$\lim_{n\to\infty} \|w_n\|_4^4 = \lim_{n\to\infty} \mathcal{J}(w_n) = C_{\mathrm{GN}}. \tag{20}$$

Step 4: Rearrange to radial decreasing functions. Let w_n^* be the symmetric decreasing rearrangement. Corollary 2.1.7 implies $\mathcal{J}(w_n^*) \geq \mathcal{J}(w_n)$.

Moreover, Theorem 1.7.6 (ii), with $p = 2$ and $p = 4$, implies $\|w_n^*\|_2 = \|w_n\|_2 = 1$ and $\|w_n^*\|_4 = \|w_n\|_4$. Theorem 1.7.9 implies $\|\nabla w_n^*\|_2 \leq \|\nabla w_n\|_2 = 1$.

If $\|\nabla w_n^*\|_2 = 1$, set $\widetilde{w}_n \coloneqq w_n^*$. If $\|\nabla w_n^*\|_2 < 1$, define $\mu_n \coloneqq \|\nabla w_n^*\|_2^{-1} > 1$ and set $\widetilde{w}_n \coloneqq (w_n^*)_{\mu_n}$. Then Lemma 2.1.4 gives $\|\widetilde{w}_n\|_2 = 1$, $\|\nabla \widetilde{w}_n\|_2 = 1$ and $\mathcal{J}(\widetilde{w}_n) = \mathcal{J}(w_n^*) \geq \mathcal{J}(w_n)$.

Thus, after replacing w_n by $\widetilde{w}_n$, we may assume:

$$w_n \geq 0, \quad w_n \in H^1_{\mathrm{rad}}(\mathbb{R}^2) \text{ and } r \mapsto w_n(r) \text{ is nonincreasing}, \quad \|w_n\|_2 = \|\nabla w_n\|_2 = 1, \tag{21}$$

and $\|w_n\|_4^4$ still converges to C_{GN} by (20) and the monotonicity of $\mathcal{J}$ under rearrangement.

Step 5: Extract a subsequence converging strongly in $L^2(B_R)$ for every R. Fix $R \in \mathbb{N}$. Since (w_n) is bounded in $H^1(\mathbb{R}^2)$, Lemma 2.1.9 gives a subsequence converging strongly in $L^2(B_R)$. Perform this extraction for $R = 1, 2, 3, \ldots$ and use a diagonal argument: there exists a single subsequence (still denoted w_n) and a function Q such that for every $R \in \mathbb{N}$,

$$w_n \to Q \quad \text{strongly in } L^2(B_R). \tag{22}$$

Step 6: Identify $Q \in H^1$ and preserve radial monotonicity. Because $\|w_n\|_2$ and $\|\nabla w_n\|_2$ are bounded, the Banach-Alaoglu theorem yields (after passing to a further subsequence) a function $Q \in H^1(\mathbb{R}^2)$ such that $w_n \rightharpoonup Q$ weakly in H^1. The strong local L^2 convergence (22) forces that weak limit to coincide with the local L^2 limit.

Each w_n is radial and satisfies $w_n(x) = w_n(Ux)$ for every rotation $U \in SO(2)$. Let $T_U f(x) := f(Ux)$. Then T_U is an isometry on L^2 and on H^1. Since $T_U w_n = w_n$ for all n, we have $T_U w_n \rightharpoonup T_U Q$ weakly in H^1. On the other hand $w_n \rightharpoonup Q$ weakly in H^1. Weak limits are unique, so $T_U Q = Q$ for every rotation U, meaning Q is radial.

To prove radial monotonicity, write $w_n(x) = f_n(|x|)$ and $Q(x) = f(|x|)$. For almost every pair of radii $0 < r_1 < r_2$, we can choose points x_1, x_2 with $|x_j| = r_j$ such that $w_n(x_j) \to Q(x_j)$ along a further subsequence (use pointwise a.e. convergence extracted from $L^2(B_R)$ convergence). Since w_n is nonincreasing in r, we have $w_n(x_1) \geq w_n(x_2)$ for all n. Passing to the limit gives $Q(x_1) \geq Q(x_2)$, hence $f(r_1) \geq f(r_2)$ for almost every $r_1 < r_2$. Therefore Q is radial and nonincreasing (in the almost everywhere sense).

Also $Q \geq 0$ a.e. because each $w_n \geq 0$ and $w_n \to Q$ in L^2_{loc} implies convergence in measure on each ball.

Step 7: Strong L^4 convergence on balls, using a cutoff and (10)**.** Fix $R > 0$. Choose $\eta_R \in C_c^\infty(\mathbb{R}^2)$ such that $0 \leq \eta_R \leq 1$, $\eta_R \equiv 1$ on B_R, and $\eta_R \equiv 0$ outside B_{2R}. Define

$$g_{n,R} := \eta_R\,(w_n - Q).$$

Then $g_{n,R} \in H^1(\mathbb{R}^2)$. We show $\|g_{n,R}\|_2 \to 0$ and $\sup_n \|\nabla g_{n,R}\|_2 < \infty$.

First,

$$\|g_{n,R}\|_2^2 = \int |\eta_R|^2 |w_n - Q|^2 \leq \int_{B_{2R}} |w_n - Q|^2 = \|w_n - Q\|_{L^2(B_{2R})}^2 \to 0$$

by (22) with radius $2R$.

Next, compute the gradient:

$$\nabla g_{n,R} = (\nabla \eta_R)(w_n - Q) + \eta_R \nabla (w_n - Q).$$

Taking L^2 norms and using $(a+b)^2 \leq 2a^2 + 2b^2$ gives

$$\|\nabla g_{n,R}\|_2^2 \leq 2\|(\nabla \eta_R)(w_n - Q)\|_2^2 + 2\|\eta_R \nabla (w_n - Q)\|_2^2.$$

The first term satisfies

$$\|(\nabla \eta_R)(w_n - Q)\|_2^2 \leq \|\nabla \eta_R\|_\infty^2 \int_{B_{2R}} |w_n - Q|^2 = \|\nabla \eta_R\|_\infty^2 \, \|w_n - Q\|_{L^2(B_{2R})}^2,$$

which is bounded uniformly in n because $\|w_n - Q\|_{L^2(B_{2R})} \to 0$ in particular is bounded.

For the second term,

$$\|\eta_R \nabla (w_n - Q)\|_2 \leq \|\nabla w_n\|_2 + \|\nabla Q\|_2,$$

since $|\eta_R| \leq 1$. Using (21), $\|\nabla w_n\|_2 = 1$ for all n, so the right-hand side is bounded. Therefore, $\sup_n \|\nabla g_{n,R}\|_2 < \infty$ and $\|g_{n,R}\|_2 \to 0$. Apply (10) to $g_{n,R}$:

$$\|g_{n,R}\|_4^4 \leq C_{\text{GN}} \, \|g_{n,R}\|_2^2 \, \|\nabla g_{n,R}\|_2^2 \to 0.$$

Because $\eta_R \equiv 1$ on B_R, we have

$$\int_{B_R} |w_n - Q|^4 \, dx \le \int_{\mathbb{R}^2} |g_{n,R}|^4 \, dx = \|g_{n,R}\|_4^4 \to 0.$$

Thus

$$w_n \to Q \quad \text{strongly in } L^4(B_R) \quad \text{for every } R > 0. \tag{23}$$

Step 8: cCntrol of the L^4 tail and global L^4 convergence. By Lemma 2.1.10 and (21), for every $R > 0$ and every n,

$$\int_{|x|\ge R} |w_n(x)|^4 \, dx \le \frac{1}{\pi R} \|w_n\|_2^3 \|\nabla w_n\|_2 = \frac{1}{\pi R}. \tag{24}$$

Also, $Q \in H^1(\mathbb{R}^2)$ implies $Q \in L^4(\mathbb{R}^2)$ by (10), since $\|Q\|_4^4 \le C_{\mathrm{GN}} \|Q\|_2^2 \|\nabla Q\|_2^2 < \infty$. Therefore the sets $E_R := \{|x| \ge R\}$ decrease to the empty set, and monotone convergence gives

$$\lim_{R\to\infty} \int_{|x|\ge R} |Q(x)|^4 \, dx = 0. \tag{25}$$

Now, fix $\varepsilon > 0$. Choose R large enough so that

$$\frac{1}{\pi R} \le \varepsilon \quad \text{and} \quad \int_{|x|\ge R} |Q|^4 \le \varepsilon,$$

which is possible by (24) and (25). Then, by (23) with radius R, there exists N such that for all $n \ge N$,

$$\int_{|x|\le R} |w_n - Q|^4 \, dx \le \varepsilon.$$

For the exterior region, use the inequality $|a-b|^4 \le 8(|a|^4 + |b|^4)$ for all $a, b \in \mathbb{C}$, which follows from $|a-b|^2 \le 2(|a|^2 + |b|^2)$ and squaring. Thus for $n \ge N$,

$$\int_{|x|\ge R} |w_n - Q|^4 \le 8 \int_{|x|\ge R} |w_n|^4 + 8 \int_{|x|\ge R} |Q|^4 \le 8\varepsilon + 8\varepsilon = 16\varepsilon.$$

Hence for $n \ge N$,

$$\|w_n - Q\|_4^4 = \int_{|x|\le R} |w_n - Q|^4 + \int_{|x|\ge R} |w_n - Q|^4 \le \varepsilon + 16\varepsilon = 17\varepsilon.$$

Since ε is arbitrary, $\|w_n - Q\|_4 \to 0$, i.e.

$$w_n \to Q \quad \text{strongly in } L^4(\mathbb{R}^2). \tag{26}$$

Step 9: Q attains the supremum and satisfies the normalizations. From (26) and (20),

$$\|Q\|_4^4 = \lim_{n\to\infty} \|w_n\|_4^4 = C_{\mathrm{GN}},$$

so $Q \not\equiv 0$.

Weak lower semi-continuity of the norm gives

$$\|Q\|_2 \le \liminf_{n\to\infty} \|w_n\|_2 = 1, \qquad \|\nabla Q\|_2 \le \liminf_{n\to\infty} \|\nabla w_n\|_2 = 1. \tag{27}$$

Therefore, $\|Q\|_2^2\|\nabla Q\|_2^2 \le 1$. By definition of C_{GN} as a supremum, $\mathcal{J}(Q) \le C_{\mathrm{GN}}$. Since $\|Q\|_4^4 = C_{\mathrm{GN}}$, this inequality reads

$$\frac{C_{\mathrm{GN}}}{\|Q\|_2^2\|\nabla Q\|_2^2} \le C_{\mathrm{GN}}.$$

Because $C_{\mathrm{GN}} > 0$, we can divide by it and obtain

$$\|Q\|_2^2\|\nabla Q\|_2^2 \ge 1.$$

Combining with ≤ 1 yields

$$\|Q\|_2 = 1, \qquad \|\nabla Q\|_2 = 1.$$

Finally,

$$\mathcal{J}(Q) = \frac{\|Q\|_4^4}{\|Q\|_2^2\|\nabla Q\|_2^2} = \|Q\|_4^4 = C_{\mathrm{GN}}.$$

This proves that Q attains the supremum and has the stated properties. ■

2.2 The Ground State Equation

In this subsection, we derive the Euler-Lagrange equation satisfied by an optimizer and identify it with the stationary nonlinear Schrödinger equation. The goal is to connect the variational maximizer to a PDE, which later allows us to use the uniqueness theory from Section 3, and to identify blow-up profiles in Section 4.

In Subsection 2.1 we constructed a maximizer $Q \in H^1(\mathbb{R}^2)$ for the sharp Gagliardo-Nirenberg functional, with the following properties:

$$Q \ge 0 \text{ a.e.}, \qquad Q \text{ is radial and nonincreasing in } |x|,$$

and (after the normalizations carried out there)

$$\|Q\|_{L^2(\mathbb{R}^2)} = 1, \qquad \|\nabla Q\|_{L^2(\mathbb{R}^2)} = 1, \qquad \|Q\|_{L^4(\mathbb{R}^2)}^4 = C_{\mathrm{GN}}. \tag{28}$$

The goal of this subsection is to derive an elliptic equation satisfied by Q, and then to rescale it into the standard "ground state" $\mathcal{Q}$ form

$$-\Delta\mathcal{Q} + \mathcal{Q} - \mathcal{Q}^3 = 0 \qquad \text{in } \mathbb{R}^2. \tag{29}$$

2.2.1 Preparatory calculus: first-order expansions

Lemma 2.2.1 (A one-dimensional expansion at $\varepsilon = 0$)**.** *Let $a, b \in \mathbb{R}$ and define $F : (-\varepsilon_0, \varepsilon_0) \to (0, \infty)$ by*

$$F(\varepsilon) := \sqrt{1 + 2a\varepsilon + b\varepsilon^2},$$

where $\varepsilon_0 > 0$ is chosen so that the expression under the square root is positive. Then F is differentiable at $\varepsilon = 0$ and

$$F(0) = 1, \qquad F'(0) = a.$$

Proof. We compute the difference quotient explicitly:

$$\frac{F(\varepsilon)-F(0)}{\varepsilon}=\frac{\sqrt{1+2a\varepsilon+b\varepsilon^2}-1}{\varepsilon}.$$

Multiply numerator and denominator by the conjugate $\sqrt{1+2a\varepsilon+b\varepsilon^2}+1$:

$$\frac{\sqrt{1+2a\varepsilon+b\varepsilon^2}-1}{\varepsilon}=\frac{(1+2a\varepsilon+b\varepsilon^2)-1}{\varepsilon\big(\sqrt{1+2a\varepsilon+b\varepsilon^2}+1\big)}=\frac{2a+b\varepsilon}{\sqrt{1+2a\varepsilon+b\varepsilon^2}+1}.$$

As $\varepsilon\to 0$, the denominator converges to $1+1=2$. Therefore the difference quotient converges to $(2a)/2=a$. ■

Lemma 2.2.2 (First variations of the L^2, $\dot{H}^1$, and L^4 quantities)**.** *Let $Q\in H^1(\mathbb{R}^2)$ be real-valued and let $\varphi\in C_c^\infty(\mathbb{R}^2)$ be real-valued. For $\varepsilon\in\mathbb{R}$ define $w_\varepsilon:=Q+\varepsilon\varphi$. Then:*

(i) The map $\varepsilon\mapsto\|w_\varepsilon\|_2$ is differentiable at $\varepsilon=0$ and

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\|w_\varepsilon\|_2=\int_{\mathbb{R}^2}Q(x)\varphi(x)\,dx.$$

(ii) The map $\varepsilon\mapsto\|\nabla w_\varepsilon\|_2$ is differentiable at $\varepsilon=0$ and

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\|\nabla w_\varepsilon\|_2=\int_{\mathbb{R}^2}\nabla Q(x)\cdot\nabla\varphi(x)\,dx.$$

(iii) The map $\varepsilon\mapsto\|w_\varepsilon\|_4^4$ is differentiable at $\varepsilon=0$ and

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\|w_\varepsilon\|_4^4=4\int_{\mathbb{R}^2}Q(x)^3\,\varphi(x)\,dx.$$

Proof. We prove each statement by reducing to an explicit polynomial identity and then applying Lemma 2.2.1.

Proof of (i). First compute $\|w_\varepsilon\|_2^2$:

$$\|w_\varepsilon\|_2^2=\int_{\mathbb{R}^2}(Q+\varepsilon\varphi)^2=\int_{\mathbb{R}^2}\big(Q^2+2\varepsilon Q\varphi+\varepsilon^2\varphi^2\big)=\|Q\|_2^2+2\varepsilon\int Q\varphi+\varepsilon^2\|\varphi\|_2^2.$$

Using (28), $\|Q\|_2^2=1$. Hence

$$\|w_\varepsilon\|_2=\sqrt{1+2\varepsilon\int Q\varphi+\varepsilon^2\|\varphi\|_2^2}.$$

Now apply Lemma 2.2.1 with

$$a:=\int Q\varphi,\qquad b:=\|\varphi\|_2^2,$$

to obtain $\frac{d}{d\varepsilon}\big|_{\varepsilon=0}\|w_\varepsilon\|_2=a$.

Proof of (ii). We compute similarly:

$$\|\nabla w_\varepsilon\|_2^2=\int|\nabla Q+\varepsilon\nabla\varphi|^2=\int\big(|\nabla Q|^2+2\varepsilon\nabla Q\cdot\nabla\varphi+\varepsilon^2|\nabla\varphi|^2\big)=\|\nabla Q\|_2^2+2\varepsilon\int\nabla Q\cdot\nabla\varphi+\varepsilon^2\|\nabla\varphi\|_2^2.$$

Using (28), $\|\nabla Q\|_2^2 = 1$. Hence

$$\|\nabla w_\varepsilon\|_2 = \sqrt{1 + 2\varepsilon \int \nabla Q \cdot \nabla \varphi + \varepsilon^2 \|\nabla \varphi\|_2^2}.$$

Apply Lemma 2.2.1 with

$$a := \int \nabla Q \cdot \nabla \varphi, \qquad b := \|\nabla \varphi\|_2^2,$$

to obtain the desired derivative.

Proof of (iii). We expand $(Q + \varepsilon\varphi)^4$ pointwise:

$$(Q + \varepsilon\varphi)^4 = Q^4 + 4\varepsilon Q^3 \varphi + 6\varepsilon^2 Q^2 \varphi^2 + 4\varepsilon^3 Q \varphi^3 + \varepsilon^4 \varphi^4.$$

Integrate over $\mathbb{R}^2$:

$$\|w_\varepsilon\|_4^4 = \int Q^4 + 4\varepsilon \int Q^3 \varphi + 6\varepsilon^2 \int Q^2 \varphi^2 + 4\varepsilon^3 \int Q \varphi^3 + \varepsilon^4 \int \varphi^4.$$

Thus the difference quotient is

$$\frac{\|w_\varepsilon\|_4^4 - \|w_0\|_4^4}{\varepsilon} = 4 \int Q^3 \varphi + 6\varepsilon \int Q^2 \varphi^2 + 4\varepsilon^2 \int Q \varphi^3 + \varepsilon^3 \int \varphi^4.$$

All integrals are finite because φ is smooth compactly supported and $Q \in L^4$. Letting $\varepsilon \to 0$ gives the claimed derivative. ■

2.2.2 Derivation of the Euler-Lagrange equation

Fix $\varphi \in C_c^\infty(\mathbb{R}^2)$ real-valued and define $w_\varepsilon := Q + \varepsilon\varphi$. For each ε we construct a perturbation u_ε that satisfies the two constraints $\|u_\varepsilon\|_2 = 1$ and $\|\nabla u_\varepsilon\|_2 = 1$ exactly. Define

$$A(\varepsilon) := \|w_\varepsilon\|_2, \qquad B(\varepsilon) := \|\nabla w_\varepsilon\|_2.$$

For ε close to 0, both $A(\varepsilon)$ and $B(\varepsilon)$ are positive, because $A(0) = \|Q\|_2 = 1$ and $B(0) = \|\nabla Q\|_2 = 1$ and A, B are continuous at 0 (this continuity follows from the explicit formulas in Lemma 2.2.2).

We now define the normalized perturbation:

$$u_\varepsilon(x) := \frac{1}{A(\varepsilon)} \Big(\frac{A(\varepsilon)}{B(\varepsilon)}\Big) w_\varepsilon\Big(\frac{A(\varepsilon)}{B(\varepsilon)}\, x\Big).$$

Equivalently, using the critical scaling from Lemma 2.1.4 in Subsection 2.1, this can be written as

$$u_\varepsilon := \frac{1}{A(\varepsilon)} (w_\varepsilon)_{\lambda(\varepsilon)}, \qquad \text{where} \qquad \lambda(\varepsilon) := \frac{A(\varepsilon)}{B(\varepsilon)}.$$

Lemma 2.2.3 (The perturbation satisfies the constraints)**.** *For all ε sufficiently close to* 0*,*

$$\|u_\varepsilon\|_2 = 1, \qquad \|\nabla u_\varepsilon\|_2 = 1.$$

Proof. We use the scaling identities from Lemma 2.1.4 in Subsection 2.1. First,

$$\|u_\varepsilon\|_2 = \frac{1}{A(\varepsilon)}\|(w_\varepsilon)_{\lambda(\varepsilon)}\|_2 = \frac{1}{A(\varepsilon)}\|w_\varepsilon\|_2 = \frac{1}{A(\varepsilon)}A(\varepsilon) = 1.$$

Next,

$$\|\nabla u_\varepsilon\|_2 = \frac{1}{A(\varepsilon)}\|\nabla(w_\varepsilon)_{\lambda(\varepsilon)}\|_2 = \frac{1}{A(\varepsilon)}\,\lambda(\varepsilon)\,\|\nabla w_\varepsilon\|_2 = \frac{1}{A(\varepsilon)}\,\frac{A(\varepsilon)}{B(\varepsilon)}\,B(\varepsilon) = 1.$$

■

Since Q maximizes $\|u\|_4^4$ among all real u with $\|u\|_2 = \|\nabla u\|_2 = 1$, we obtain: for all ε sufficiently close to 0,

$$\|u_\varepsilon\|_4^4 \le \|Q\|_4^4. \tag{30}$$

Lemma 2.2.4 (An explicit formula for $\|u_\varepsilon\|_4^4$)**.** *For all ε sufficiently close to* 0*,*

$$\|u_\varepsilon\|_4^4 = \frac{\|w_\varepsilon\|_4^4}{A(\varepsilon)^2\,B(\varepsilon)^2}.$$

Proof. By the scaling formula $\|(f)_\lambda\|_4^4 = \lambda^2\|f\|_4^4$ (Lemma 2.1.4 in Subsection 2.1),

$$\|(w_\varepsilon)_{\lambda(\varepsilon)}\|_4^4 = \lambda(\varepsilon)^2\|w_\varepsilon\|_4^4.$$

Therefore

$$\|u_\varepsilon\|_4^4 = \frac{1}{A(\varepsilon)^4}\,\|(w_\varepsilon)_{\lambda(\varepsilon)}\|_4^4 = \frac{1}{A(\varepsilon)^4}\,\lambda(\varepsilon)^2\|w_\varepsilon\|_4^4.$$

Since $\lambda(\varepsilon) = A(\varepsilon)/B(\varepsilon)$, we have

$$\frac{1}{A(\varepsilon)^4}\,\lambda(\varepsilon)^2 = \frac{1}{A(\varepsilon)^4}\,\frac{A(\varepsilon)^2}{B(\varepsilon)^2} = \frac{1}{A(\varepsilon)^2\,B(\varepsilon)^2}.$$

Insert into the previous identity to obtain the result. ■

Theorem 2.2.5 (Euler-Lagrange identity in weak form)**.** *Let Q be the maximizer from* (28)*. Then for every real-valued $\varphi \in C_c^\infty(\mathbb{R}^2)$,*

$$\int_{\mathbb{R}^2}\Big(\nabla Q(x)\cdot\nabla\varphi(x) + Q(x)\varphi(x) - \frac{2}{C_{\mathrm{GN}}}\,Q(x)^3\varphi(x)\Big)\,dx = 0.$$

Equivalently, in the sense of distributions,

$$-\Delta Q + Q - \frac{2}{C_{\mathrm{GN}}}\,Q^3 = 0 \qquad \textit{in } \mathbb{R}^2. \tag{31}$$

Proof. Fix $\varphi \in C_c^\infty(\mathbb{R}^2)$ real-valued. Define $w_\varepsilon \coloneqq Q + \varepsilon\varphi$, and define $A(\varepsilon)$, $B(\varepsilon)$, and u_ε as above. Then Lemma 2.2.3 shows that u_ε satisfies both constraints exactly, so the maximality inequality (30) holds.

Step 1: Rewrite the maximality inequality in an explicit one-variable form. By Lemma 2.2.4, (30) becomes

$$\frac{\|w_\varepsilon\|_4^4}{A(\varepsilon)^2 B(\varepsilon)^2} \le \|Q\|_4^4.$$

Using (28), $\|Q\|_4^4 = C_{\mathrm{GN}}$ and $A(0) = B(0) = 1$. Define the real-valued function

$$\Phi(\varepsilon) := \frac{\|w_\varepsilon\|_4^4}{A(\varepsilon)^2 B(\varepsilon)^2}.$$

Then the inequality reads $\Phi(\varepsilon) \le \Phi(0)$ for all sufficiently small ε.

Step 2: Conclude $\Phi'(0) = 0$ from the one-sided inequalities. From $\Phi(\varepsilon) \le \Phi(0)$ for $\varepsilon > 0$ sufficiently small, we have

$$\frac{\Phi(\varepsilon) - \Phi(0)}{\varepsilon} \le 0 \quad \text{for all sufficiently small } \varepsilon > 0.$$

Taking $\varepsilon \downarrow 0$ yields $\Phi'_+(0) \le 0$. Similarly, using $\Phi(\varepsilon) \le \Phi(0)$ for sufficiently small $\varepsilon < 0$ gives

$$\frac{\Phi(\varepsilon) - \Phi(0)}{\varepsilon} \ge 0 \quad \text{for all sufficiently small } \varepsilon < 0,$$

and taking $\varepsilon \uparrow 0$ yields $\Phi'_-(0) \ge 0$. Since we will compute below that Φ is differentiable at 0, we obtain

$$\Phi'(0) = 0. \tag{32}$$

Step 3: Compute $\Phi'(0)$ explicitly. Write $\Phi(\varepsilon)$ as a product:

$$\Phi(\varepsilon) = \|w_\varepsilon\|_4^4 \cdot \big(A(\varepsilon)^{-2}\big) \cdot \big(B(\varepsilon)^{-2}\big).$$

Each factor is differentiable at $\varepsilon = 0$ by Lemma 2.2.2 and standard one-variable calculus. We compute each derivative:

(a) By Lemma 2.2.2(iii),

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \|w_\varepsilon\|_4^4 = 4 \int Q^3 \varphi.$$

(b) Since $A(0) = 1$ and $A'(0) = \int Q\varphi$ by Lemma 2.2.2(i), the chain rule gives

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} A(\varepsilon)^{-2} = -2A(0)^{-3} A'(0) = -2 \int Q\varphi.$$

(c) Since $B(0) = 1$ and $B'(0) = \int \nabla Q \cdot \nabla \varphi$ by Lemma 2.2.2(ii),

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} B(\varepsilon)^{-2} = -2 \int \nabla Q \cdot \nabla \varphi.$$

Now evaluate the product rule at $\varepsilon = 0$. Using $A(0) = B(0) = 1$ and $\|w_0\|_4^4 = \|Q\|_4^4 = C_{\mathrm{GN}}$, we obtain

$$\Phi'(0) = \Big(4 \int Q^3 \varphi\Big) \cdot 1 \cdot 1 + C_{\mathrm{GN}} \cdot \Big(-2 \int Q\varphi\Big) \cdot 1 + C_{\mathrm{GN}} \cdot 1 \cdot \Big(-2 \int \nabla Q \cdot \nabla \varphi\Big).$$

Therefore,

$$\Phi'(0) = 4 \int_{\mathbb{R}^2} Q^3 \varphi \, dx - 2C_{\mathrm{GN}} \Big(\int_{\mathbb{R}^2} Q \, \varphi \, dx + \int_{\mathbb{R}^2} \nabla Q \cdot \nabla \varphi \, dx \Big). \tag{33}$$

Combine (32) and (33) and divide by 2:

$$2 \int Q^3 \varphi = C_{\mathrm{GN}} \Big(\int Q\varphi + \int \nabla Q \cdot \nabla \varphi \Big).$$

Rearrange to

$$\int \nabla Q \cdot \nabla\varphi + \int Q\varphi - \frac{2}{C_{\mathrm{GN}}} \int Q^3\varphi = 0.$$

This is exactly the claimed weak identity.

Step 4: Rewrite as a distributional equation. Since $\varphi \in C_c^\infty$, integration by parts is legitimate:

$$\int_{\mathbb{R}^2} \nabla Q \cdot \nabla\varphi \, dx = -\int_{\mathbb{R}^2} (\Delta Q)\, \varphi \, dx,$$

where ΔQ is understood in the sense of distributions. Therefore the weak identity becomes

$$\int_{\mathbb{R}^2} \Big(-\Delta Q + Q - \frac{2}{C_{\mathrm{GN}}} Q^3\Big)\varphi \, dx = 0 \quad \text{for all } \varphi \in C_c^\infty(\mathbb{R}^2),$$

which is (31). ■

2.2.3 Rescaling to the standard form and basic qualitative properties

Define

$$\mathcal{Q}(x) := \sqrt{\frac{2}{C_{\mathrm{GN}}}}\, Q(x). \tag{34}$$

Then $\mathcal{Q} \in H^1(\mathbb{R}^2)$, $\mathcal{Q} \geq 0$ a.e., and $\mathcal{Q}$ is radial nonincreasing.

Lemma 2.2.6 (Rescaled maximizer satisfies the ground state equation)**.** *The function $\mathcal{Q}$ defined in* (34) *satisfies*

$$-\Delta\mathcal{Q} + \mathcal{Q} - \mathcal{Q}^3 = 0 \quad \textit{in the sense of distributions on } \mathbb{R}^2.$$

Proof. From (31) we have

$$-\Delta Q + Q - \frac{2}{C_{\mathrm{GN}}} Q^3 = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^2).$$

Multiply this identity by the constant $\sqrt{2/C_{\mathrm{GN}}}$. Using the definition $\mathcal{Q} = \sqrt{2/C_{\mathrm{GN}}}\, Q$, we obtain

$$-\Delta\mathcal{Q} + \mathcal{Q} - \Big(\sqrt{\frac{2}{C_{\mathrm{GN}}}}\Big)\frac{2}{C_{\mathrm{GN}}} Q^3 = 0.$$

Now compute the cubic term:

$$\mathcal{Q}^3 = \Big(\sqrt{\frac{2}{C_{\mathrm{GN}}}}\, Q\Big)^3 = \Big(\frac{2}{C_{\mathrm{GN}}}\Big)^{3/2} Q^3.$$

Also,

$$\Big(\sqrt{\frac{2}{C_{\mathrm{GN}}}}\Big)\frac{2}{C_{\mathrm{GN}}} = \Big(\frac{2}{C_{\mathrm{GN}}}\Big)^{3/2}.$$

Therefore the coefficient in front of Q^3 equals exactly the coefficient in $\mathcal{Q}^3$, and the equation becomes

$$-\Delta\mathcal{Q} + \mathcal{Q} - \mathcal{Q}^3 = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}^2).$$

■

Lemma 2.2.7 (From weak solution to H^2 solution via Fourier analysis)**.** *Let $u \in H^1(\mathbb{R}^2)$ satisfy*

$$-\Delta u + u = f \quad \textit{in } \mathcal{D}'(\mathbb{R}^2),$$

where $f \in L^2(\mathbb{R}^2)$. Then $u \in H^2(\mathbb{R}^2)$ and

$$\|u\|_{H^2(\mathbb{R}^2)} \le \|f\|_{L^2(\mathbb{R}^2)}.$$

Proof. We use the Fourier transform on L^2 and Plancherel's theorem [23]. Let $\widehat{g}$ denote the Fourier transform. Taking the Fourier transform of the distributional identity $(-\Delta + 1)u = f$ gives

$$(|\xi|^2 + 1)\widehat{u}(\xi) = \widehat{f}(\xi) \quad \text{for a.e. } \xi \in \mathbb{R}^2,$$

hence

$$\widehat{u}(\xi) = \frac{\widehat{f}(\xi)}{|\xi|^2 + 1}.$$

Now compute the H^2 norm using Plancherel:

$$\|u\|_{H^2}^2 = \int_{\mathbb{R}^2} (1 + |\xi|^2)^2 \, |\widehat{u}(\xi)|^2 \, d\xi = \int_{\mathbb{R}^2} (1 + |\xi|^2)^2 \, \frac{|\widehat{f}(\xi)|^2}{(1 + |\xi|^2)^2} \, d\xi = \int_{\mathbb{R}^2} |\widehat{f}(\xi)|^2 \, d\xi = \|f\|_2^2.$$

Taking square roots yields $\|u\|_{H^2} \le \|f\|_2$. ∎

Corollary 2.2.8 ($\mathcal{Q} \in H^2(\mathbb{R}^2)$)**.** *The ground state $\mathcal{Q}$ from Lemma 2.2.6 belongs to $H^2(\mathbb{R}^2)$.*

Proof. From Lemma 2.2.6 we have $(-\Delta + 1)\mathcal{Q} = \mathcal{Q}^3$ in $\mathcal{D}'$. Since $\mathcal{Q} \in H^1(\mathbb{R}^2)$ and in $\mathbb{R}^2$ one has $H^1(\mathbb{R}^2) \hookrightarrow L^6(\mathbb{R}^2)$, we have $\mathcal{Q} \in L^6$, hence $\mathcal{Q}^3 \in L^2$. Apply Lemma 2.2.7 with $u = \mathcal{Q}$ and $f = \mathcal{Q}^3$. ∎

Lemma 2.2.9 (Strict positivity via positivity of the heat kernel)**.** *Let $u \in L^1_{\mathrm{loc}}(\mathbb{R}^2)$, $u \ge 0$ a.e., $u \not\equiv 0$ a.e., and suppose u satisfies*

$$(-\Delta + 1)u = u^3 \quad \textit{in } \mathcal{D}'(\mathbb{R}^2).$$

Then $u(x) > 0$ for every $x \in \mathbb{R}^2$.

Proof. We use the identity of Fourier multipliers

$$\frac{1}{|\xi|^2 + 1} = \int_0^\infty e^{-t} e^{-t|\xi|^2} \, dt \quad \text{for every } \xi \in \mathbb{R}^2,$$

which follows by evaluating the one-dimensional integral:

$$\int_0^\infty e^{-t} e^{-t|\xi|^2} \, dt = \int_0^\infty e^{-t(1+|\xi|^2)} \, dt = \Big[\frac{-1}{1 + |\xi|^2} e^{-t(1+|\xi|^2)} \Big]_{t=0}^{t=\infty} = \frac{1}{1 + |\xi|^2}.$$

Taking Fourier transforms of the equation $(-\Delta + 1)u = u^3$ gives

$$\widehat{u}(\xi) = \frac{\widehat{u^3}(\xi)}{|\xi|^2 + 1} = \int_0^\infty e^{-t} e^{-t|\xi|^2} \widehat{u^3}(\xi) \, dt.$$

By Fourier inversion and Tonelli's theorem (all kernels below are nonnegative), this representation becomes, for a.e. x,

$$u(x) = \int_0^\infty e^{-t} \, (e^{t\Delta} u^3)(x) \, dt,$$

where $e^{t\Delta}$ is the heat semigroup. For each $t > 0$, the heat semigroup has the explicit convolution formula

$$(e^{t\Delta}g)(x) = \int_{\mathbb{R}^2} G_t(x-y)\,g(y)\,dy, \qquad G_t(z) := \frac{1}{4\pi t}\exp\Big(-\frac{|z|^2}{4t}\Big).$$

The kernel $G_t(z)$ is strictly positive for every $t > 0$ and every $z \in \mathbb{R}^2$.

Now fix $x \in \mathbb{R}^2$. Since $u \geq 0$ and $u \not\equiv 0$, the set $E := \{y : \ u(y) > 0\}$ has positive measure. Therefore $u^3(y) > 0$ for every $y \in E$. Fix any $t > 0$. Then

$$(e^{t\Delta}u^3)(x) = \int_{\mathbb{R}^2} G_t(x-y)\,u(y)^3\,dy \geq \int_E G_t(x-y)\,u(y)^3\,dy.$$

On E, both factors $G_t(x-y)$ and $u(y)^3$ are strictly positive. Hence the integrand is strictly positive on a measurable set of positive measure, so the integral is strictly positive:

$$(e^{t\Delta}u^3)(x) > 0.$$

Since $e^{-t} > 0$, the function $t \mapsto e^{-t}(e^{t\Delta}u^3)(x)$ is strictly positive for every $t > 0$. Therefore the integral representation of $u(x)$ implies $u(x) > 0$. ■

Corollary 2.2.10 (Ground state is strictly positive)**.** *The ground state $\mathcal{Q}$ satisfies $\mathcal{Q}(x) > 0$ for all $x \in \mathbb{R}^2$.*

Proof. We apply Lemma 2.2.9 with $u = \mathcal{Q}$. We have $\mathcal{Q} \geq 0$ and $\mathcal{Q} \not\equiv 0$ because $\|\mathcal{Q}\|_2 = \sqrt{2/C_{\mathrm{GN}}}\|Q\|_2 > 0$. Also $\mathcal{Q}$ satisfies $(-\Delta+1)\mathcal{Q} = \mathcal{Q}^3$ in $\mathcal{D}'$ by Lemma 2.2.6. ■

2.3 The Sharp Constant and the Equality Case

In, this subsection, we express the optimal constant C_{GN} in terms of the optimizer and characterize the equality cases of the sharp inequality. The purpose is to identify exactly which functions are the optimizers of the inequality, and to relate them to the ground state. This is useful for Section 4, specifically, when describing minimal-mass blow-up profiles.

Recall from Subsection 2.1 that the sharp constant C_{GN} was defined by the supremum (11), and that Theorem 2.1.11 provides an optimizer $Q \in H^1(\mathbb{R}^2) \setminus \{0\}$ satisfying $\mathcal{J}(Q) = C_{\mathrm{GN}}$. In particular, equality holds in (10) for $u = Q$, namely,

$$||Q||^4_{L^4(\mathbb{R}^2)} = C_{\mathrm{GN}}\,||Q||^2_{L^2(\mathbb{R}^2)}\,||\nabla Q||^2_{L^2(\mathbb{R}^2)}. \tag{35}$$

Lemma 2.3.1 (Pohozaev identities in $d = 2$ for the normalized ground state)**.** *Let $\mathcal{Q} \in H^1(\mathbb{R}^2) \cap C^2(\mathbb{R}^2)$ satisfy*

$$-\Delta\mathcal{Q} + \mathcal{Q} - \mathcal{Q}^3 = 0 \qquad \text{in } \mathbb{R}^2, \tag{36}$$

and assume $\mathcal{Q}(x) \to 0$ as $|x| \to \infty$. Then:

$$||\nabla\mathcal{Q}||^2_{L^2(\mathbb{R}^2)} + ||\mathcal{Q}||^2_{L^2(\mathbb{R}^2)} = ||\mathcal{Q}||^4_{L^4(\mathbb{R}^2)}, \tag{37}$$

and

$$||\mathcal{Q}||^2_{L^2(\mathbb{R}^2)} = \frac{1}{2}\,||\mathcal{Q}||^4_{L^4(\mathbb{R}^2)}. \tag{38}$$

Consequently,

$$||\nabla\mathcal{Q}||^2_{L^2(\mathbb{R}^2)} = ||\mathcal{Q}||^2_{L^2(\mathbb{R}^2)}, \qquad ||\mathcal{Q}||^4_{L^4(\mathbb{R}^2)} = 2||\mathcal{Q}||^2_{L^2(\mathbb{R}^2)}. \tag{39}$$

Proof. We prove (37) and (38) using cutoffs, so that every integration by parts is justified on bounded domains, and then we pass to the limit.

Step 1: Choice of cutoff functions. Fix $\eta \in C_c^\infty(\mathbb{R}^2)$ such that $0 \le \eta \le 1$, $\eta(x) = 1$ for $|x| \le 1$, and $\eta(x) = 0$ for $|x| \ge 2$. For $R > 0$ define $\eta_R(x) \coloneqq \eta(x/R)$. Then $\eta_R \in C_c^\infty(\mathbb{R}^2)$, $\eta_R(x) = 1$ for $|x| \le R$, $\eta_R(x) = 0$ for $|x| \ge 2R$, and there exists a constant $C_\eta > 0$ such that

$$||\nabla \eta_R||_{L^\infty(\mathbb{R}^2)} \le \frac{C_\eta}{R}, \qquad ||D^2 \eta_R||_{L^\infty(\mathbb{R}^2)} \le \frac{C_\eta}{R^2}.$$

Step 2: Proof of (37). Multiply (36) by $\mathcal{Q}\eta_R^2$ and integrate over $\mathbb{R}^2$:

$$\int_{\mathbb{R}^2} \big(-\Delta \mathcal{Q}\big) \mathcal{Q} \eta_R^2 \, dx + \int_{\mathbb{R}^2} \mathcal{Q}^2 \eta_R^2 \, dx - \int_{\mathbb{R}^2} \mathcal{Q}^4 \eta_R^2 \, dx = 0. \tag{40}$$

We now integrate by parts in the first term. Since $\mathcal{Q} \in C^2(\mathbb{R}^2)$ and η_R has compact support, the function $\mathcal{Q}\eta_R^2$ is C^1 and compactly supported. Therefore

$$\int_{\mathbb{R}^2} \big(-\Delta \mathcal{Q}\big) \mathcal{Q} \eta_R^2 \, dx = \int_{\mathbb{R}^2} \nabla \mathcal{Q} \cdot \nabla(\mathcal{Q}\eta_R^2) \, dx.$$

Compute

$$\nabla(\mathcal{Q}\eta_R^2) = \eta_R^2 \nabla \mathcal{Q} + 2\eta_R \mathcal{Q} \nabla \eta_R.$$

Hence

$$\int_{\mathbb{R}^2} \big(-\Delta \mathcal{Q}\big) \mathcal{Q} \eta_R^2 \, dx = \int_{\mathbb{R}^2} \eta_R^2 |\nabla \mathcal{Q}|^2 \, dx + 2 \int_{\mathbb{R}^2} \eta_R \mathcal{Q} \, \nabla \mathcal{Q} \cdot \nabla \eta_R \, dx.$$

Insert this into (40) to obtain

$$\int_{\mathbb{R}^2} \eta_R^2 |\nabla \mathcal{Q}|^2 \, dx + \int_{\mathbb{R}^2} \mathcal{Q}^2 \eta_R^2 \, dx - \int_{\mathbb{R}^2} \mathcal{Q}^4 \eta_R^2 \, dx = -2 \int_{\mathbb{R}^2} \eta_R \mathcal{Q} \, \nabla \mathcal{Q} \cdot \nabla \eta_R \, dx. \tag{41}$$

We now pass to the limit $R \to \infty$. The left-hand side converges term-by-term by dominated convergence: since $0 \le \eta_R^2 \le 1$ and $\eta_R(x) \to 1$ pointwise for every fixed x, we have

$$\eta_R^2 |\nabla \mathcal{Q}|^2 \to |\nabla \mathcal{Q}|^2, \quad \mathcal{Q}^2 \eta_R^2 \to \mathcal{Q}^2, \quad \mathcal{Q}^4 \eta_R^2 \to \mathcal{Q}^4 \quad \text{pointwise.}$$

Also,

$$0 \le \eta_R^2 |\nabla \mathcal{Q}|^2 \le |\nabla \mathcal{Q}|^2 \in L^1(\mathbb{R}^2), \qquad 0 \le \mathcal{Q}^2 \eta_R^2 \le \mathcal{Q}^2 \in L^1(\mathbb{R}^2),$$

and, by the Sobolev embedding $H^1(\mathbb{R}^2) \subseteq L^4(\mathbb{R}^2)$ already established earlier, $\mathcal{Q} \in L^4(\mathbb{R}^2)$, hence $\mathcal{Q}^4 \in L^1(\mathbb{R}^2)$, and thus

$$0 \le \mathcal{Q}^4 \eta_R^2 \le \mathcal{Q}^4 \in L^1(\mathbb{R}^2).$$

Therefore the dominated convergence theorem gives

$$\lim_{R\to\infty} \int_{\mathbb{R}^2} \eta_R^2 |\nabla \mathcal{Q}|^2 \, dx = ||\nabla \mathcal{Q}||^2_{L^2(\mathbb{R}^2)},$$
$$\lim_{R\to\infty} \int_{\mathbb{R}^2} \mathcal{Q}^2 \eta_R^2 \, dx = ||\mathcal{Q}||^2_{L^2(\mathbb{R}^2)},$$
$$\lim_{R\to\infty} \int_{\mathbb{R}^2} \mathcal{Q}^4 \eta_R^2 \, dx = ||\mathcal{Q}||^4_{L^4(\mathbb{R}^2)}.$$

It remains to show that the right-hand side of (41) converges to 0. Since $\nabla\eta_R$ is supported in the annulus $\{x \in \mathbb{R}^2 \mid R \le |x| \le 2R\}$, we estimate by Cauchy-Schwarz:

$$\begin{aligned}\left|2\int_{\mathbb{R}^2} \eta_R \mathcal{Q}\,\nabla\mathcal{Q}\cdot\nabla\eta_R\,dx\right| &\le 2\int_{\mathbb{R}^2} |\eta_R|\,|\mathcal{Q}|\,|\nabla\mathcal{Q}|\,|\nabla\eta_R|\,dx \\ &\le 2\,||\nabla\eta_R||_{L^\infty(\mathbb{R}^2)}\int_{\mathbb{R}^2}|\mathcal{Q}|\,|\nabla\mathcal{Q}|\,dx \\ &\le 2\,||\nabla\eta_R||_{L^\infty(\mathbb{R}^2)}\,||\mathcal{Q}||_{L^2(\mathbb{R}^2)}\,||\nabla\mathcal{Q}||_{L^2(\mathbb{R}^2)}.\end{aligned}$$

Using $||\nabla\eta_R||_{L^\infty(\mathbb{R}^2)} \le C_\eta/R$, we get

$$\left|2\int_{\mathbb{R}^2} \eta_R \mathcal{Q}\,\nabla\mathcal{Q}\cdot\nabla\eta_R\,dx\right| \le \frac{2C_\eta}{R}\,||\mathcal{Q}||_{L^2(\mathbb{R}^2)}\,||\nabla\mathcal{Q}||_{L^2(\mathbb{R}^2)} \xrightarrow[R\to\infty]{} 0.$$

Taking the limit $R \to \infty$ in (41) yields (37).

Step 3: Proof of (38)**.** We use the standard Pohozaev multiplier $x\cdot\nabla\mathcal{Q}$ with cutoffs.
Define the vector field

$$X_R(x) := x\,\eta_R(x)^2, \qquad \Phi_R(x) := X_R(x)\cdot\nabla\mathcal{Q}(x).$$

Since η_R is compactly supported and $\mathcal{Q} \in C^2(\mathbb{R}^2)$, the function Φ_R is C^1 and compactly supported. Multiply (36) by Φ_R and integrate over $\mathbb{R}^2$:

$$\int_{\mathbb{R}^2}(-\Delta\mathcal{Q})\Phi_R\,dx + \int_{\mathbb{R}^2}\mathcal{Q}\Phi_R\,dx - \int_{\mathbb{R}^2}\mathcal{Q}^3\Phi_R\,dx = 0. \tag{42}$$

We compute each term explicitly.

(i) The linear term $\int \mathcal{Q}\Phi_R$. Since $\Phi_R = X_R\cdot\nabla\mathcal{Q}$,

$$\int_{\mathbb{R}^2}\mathcal{Q}\Phi_R\,dx = \int_{\mathbb{R}^2}\mathcal{Q}\,X_R\cdot\nabla\mathcal{Q}\,dx = \frac{1}{2}\int_{\mathbb{R}^2}X_R\cdot\nabla(\mathcal{Q}^2)\,dx.$$

Because X_R is compactly supported and $\mathcal{Q}^2 \in C^1$, we may integrate by parts using the divergence theorem:

$$\int_{\mathbb{R}^2}X_R\cdot\nabla(\mathcal{Q}^2)\,dx = -\int_{\mathbb{R}^2}(\nabla\cdot X_R)\,\mathcal{Q}^2\,dx.$$

Compute $\nabla\cdot X_R$:

$$\nabla\cdot X_R = \nabla\cdot(x\eta_R^2) = (\nabla\cdot x)\eta_R^2 + x\cdot\nabla(\eta_R^2) = 2\eta_R^2 + x\cdot\nabla(\eta_R^2).$$

Hence

$$\int_{\mathbb{R}^2}\mathcal{Q}\Phi_R\,dx = -\frac{1}{2}\int_{\mathbb{R}^2}\Big(2\eta_R^2 + x\cdot\nabla(\eta_R^2)\Big)\mathcal{Q}^2\,dx. \tag{43}$$

(ii) The nonlinear term $\int \mathcal{Q}^3\Phi_R$. Similarly,

$$\int_{\mathbb{R}^2}\mathcal{Q}^3\Phi_R\,dx = \int_{\mathbb{R}^2}\mathcal{Q}^3\,X_R\cdot\nabla\mathcal{Q}\,dx = \frac{1}{4}\int_{\mathbb{R}^2}X_R\cdot\nabla(\mathcal{Q}^4)\,dx = -\frac{1}{4}\int_{\mathbb{R}^2}(\nabla\cdot X_R)\,\mathcal{Q}^4\,dx,$$

so, using $\nabla \cdot X_R = 2\eta_R^2 + x \cdot \nabla(\eta_R^2)$,

$$\int_{\mathbb{R}^2} \mathcal{Q}^3 \Phi_R \, dx = -\frac{1}{4} \int_{\mathbb{R}^2} \Big(2\eta_R^2 + x \cdot \nabla(\eta_R^2) \Big) \mathcal{Q}^4 \, dx. \tag{44}$$

(iii) The Laplacian term $\int (-\Delta \mathcal{Q}) \Phi_R$. Integrate by parts:

$$\int_{\mathbb{R}^2} \big(-\Delta \mathcal{Q}\big) \Phi_R \, dx = \int_{\mathbb{R}^2} \nabla \mathcal{Q} \cdot \nabla \Phi_R \, dx.$$

Compute $\nabla \Phi_R = \nabla(X_R \cdot \nabla \mathcal{Q})$:

$$\nabla \Phi_R = (\nabla X_R)^\top \nabla \mathcal{Q} + (\nabla^2 \mathcal{Q}) X_R,$$

so

$$\int_{\mathbb{R}^2} \big(-\Delta \mathcal{Q}\big) \Phi_R \, dx = \int_{\mathbb{R}^2} \nabla \mathcal{Q} \cdot (\nabla X_R)^\top \nabla \mathcal{Q} \, dx + \int_{\mathbb{R}^2} \nabla \mathcal{Q} \cdot (\nabla^2 \mathcal{Q}) X_R \, dx. \tag{45}$$

We now rewrite the second term in (45) as a divergence. Observe that

$$\nabla \mathcal{Q} \cdot (\nabla^2 \mathcal{Q}) X_R = \sum_{j,k=1}^{2} \partial_{x_j} \mathcal{Q} \, \partial^2_{x_j x_k} \mathcal{Q} \, (X_R)_k = \frac{1}{2} \sum_{k=1}^{2} (X_R)_k \, \partial_{x_k} \big(|\nabla \mathcal{Q}|^2 \big) = \frac{1}{2} \, X_R \cdot \nabla \big(|\nabla \mathcal{Q}|^2 \big).$$

Hence

$$\int_{\mathbb{R}^2} \nabla \mathcal{Q} \cdot (\nabla^2 \mathcal{Q}) X_R \, dx = \frac{1}{2} \int_{\mathbb{R}^2} X_R \cdot \nabla \big(|\nabla \mathcal{Q}|^2 \big) \, dx = -\frac{1}{2} \int_{\mathbb{R}^2} (\nabla \cdot X_R) |\nabla \mathcal{Q}|^2 \, dx.$$

Therefore

$$\int_{\mathbb{R}^2} \big(-\Delta \mathcal{Q}\big) \Phi_R \, dx = \int_{\mathbb{R}^2} \nabla \mathcal{Q} \cdot (\nabla X_R)^\top \nabla \mathcal{Q} \, dx - \frac{1}{2} \int_{\mathbb{R}^2} (\nabla \cdot X_R) |\nabla \mathcal{Q}|^2 \, dx. \tag{46}$$

We now compute ∇X_R explicitly. Since $X_R = x\eta_R^2$,

$$\nabla X_R = \eta_R^2 I_2 + x \otimes \nabla(\eta_R^2),$$

where I_2 is the 2×2 identity matrix and $(x \otimes v)_{jk} = x_j v_k$. Thus

$$(\nabla X_R)^\top = \eta_R^2 I_2 + \nabla(\eta_R^2) \otimes x.$$

Hence, for each x,

$$\nabla \mathcal{Q} \cdot (\nabla X_R)^\top \nabla \mathcal{Q} = \eta_R^2 |\nabla \mathcal{Q}|^2 + \big(\nabla(\eta_R^2) \cdot \nabla \mathcal{Q} \big) \, (x \cdot \nabla \mathcal{Q}).$$

Also, $(\nabla \cdot X_R) = 2\eta_R^2 + x \cdot \nabla(\eta_R^2)$ as computed above. Insert these expressions into (46):

$$\begin{aligned}
\int_{\mathbb{R}^2} \big(-\Delta \mathcal{Q}\big) \Phi_R \, dx &= \int_{\mathbb{R}^2} \eta_R^2 |\nabla \mathcal{Q}|^2 \, dx + \int_{\mathbb{R}^2} \big(\nabla(\eta_R^2) \cdot \nabla \mathcal{Q} \big) \, (x \cdot \nabla \mathcal{Q}) \, dx \\
&\quad - \frac{1}{2} \int_{\mathbb{R}^2} \Big(2\eta_R^2 + x \cdot \nabla(\eta_R^2) \Big) |\nabla \mathcal{Q}|^2 \, dx \\
&= -\frac{1}{2} \int_{\mathbb{R}^2} \Big(x \cdot \nabla(\eta_R^2) \Big) |\nabla \mathcal{Q}|^2 \, dx + \int_{\mathbb{R}^2} \big(\nabla(\eta_R^2) \cdot \nabla \mathcal{Q} \big) \, (x \cdot \nabla \mathcal{Q}) \, dx.
\end{aligned} \tag{47}$$

Step 4: Assemble the identity and pass to the limit. Insert (43), (44), and (47) into (42). We obtain an identity of the form

$$-\int_{\mathbb{R}^2} \eta_R^2 \mathcal{Q}^2\,dx + \frac{1}{2}\int_{\mathbb{R}^2} \eta_R^2 \mathcal{Q}^4\,dx = \mathcal{E}_R,$$

where $\mathcal{E}_R$ is the sum of all terms containing derivatives of η_R, namely

$$\begin{aligned}\mathcal{E}_R \coloneqq \frac{1}{2}\int_{\mathbb{R}^2} \Big(x\cdot\nabla(\eta_R^2)\Big)\mathcal{Q}^2\,dx - \frac{1}{4}\int_{\mathbb{R}^2} \Big(x\cdot\nabla(\eta_R^2)\Big)\mathcal{Q}^4\,dx \\ -\frac{1}{2}\int_{\mathbb{R}^2} \Big(x\cdot\nabla(\eta_R^2)\Big)|\nabla\mathcal{Q}|^2\,dx + \int_{\mathbb{R}^2} \big(\nabla(\eta_R^2)\cdot\nabla\mathcal{Q}\big)\,(x\cdot\nabla\mathcal{Q})\,dx.\end{aligned}$$

We first show $\mathcal{E}_R \to 0$ as $R\to\infty$. Since $\nabla(\eta_R^2)$ is supported in the annulus $\{x \mid R \le |x| \le 2R\}$ and

$$||\nabla(\eta_R^2)||_{L^\infty(\mathbb{R}^2)} \le \frac{C}{R} \qquad \text{for some constant } C>0,$$

we also have, on the support of $\nabla(\eta_R^2)$, the bound

$$|x\cdot\nabla(\eta_R^2)(x)| \le |x|\,||\nabla(\eta_R^2)||_{L^\infty(\mathbb{R}^2)} \le 2R\cdot\frac{C}{R} = 2C.$$

Therefore, by dominated convergence and the fact that the annulus $\{R\le|x|\le 2R\}$ escapes to infinity, we obtain:

$$\begin{aligned}\left|\int_{\mathbb{R}^2}\Big(x\cdot\nabla(\eta_R^2)\Big)\mathcal{Q}^2\,dx\right| &\le 2C\int_{\{R\le|x|\le 2R\}} \mathcal{Q}^2\,dx \xrightarrow[R\to\infty]{} 0,\\ \left|\int_{\mathbb{R}^2}\Big(x\cdot\nabla(\eta_R^2)\Big)\mathcal{Q}^4\,dx\right| &\le 2C\int_{\{R\le|x|\le 2R\}} \mathcal{Q}^4\,dx \xrightarrow[R\to\infty]{} 0,\\ \left|\int_{\mathbb{R}^2}\Big(x\cdot\nabla(\eta_R^2)\Big)|\nabla\mathcal{Q}|^2\,dx\right| &\le 2C\int_{\{R\le|x|\le 2R\}} |\nabla\mathcal{Q}|^2\,dx \xrightarrow[R\to\infty]{} 0,\end{aligned}$$

because $\mathcal{Q}^2\in L^1(\mathbb{R}^2)$, $\mathcal{Q}^4\in L^1(\mathbb{R}^2)$, and $|\nabla\mathcal{Q}|^2\in L^1(\mathbb{R}^2)$. For the last term, use Cauchy-Schwarz and the same support information:

$$\begin{aligned}\left|\int_{\mathbb{R}^2}\big(\nabla(\eta_R^2)\cdot\nabla\mathcal{Q}\big)\,(x\cdot\nabla\mathcal{Q})\,dx\right| &\le \int_{\{R\le|x|\le 2R\}} |\nabla(\eta_R^2)|\,|\nabla\mathcal{Q}|\,|x|\,|\nabla\mathcal{Q}|\,dx\\ &\le 2R\,||\nabla(\eta_R^2)||_{L^\infty(\mathbb{R}^2)}\int_{\{R\le|x|\le 2R\}} |\nabla\mathcal{Q}|^2\,dx\\ &\le 2R\cdot\frac{C}{R}\int_{\{R\le|x|\le 2R\}} |\nabla\mathcal{Q}|^2\,dx = 2C\int_{\{R\le|x|\le 2R\}} |\nabla\mathcal{Q}|^2\,dx \xrightarrow[R\to\infty]{} 0.\end{aligned}$$

Hence $\mathcal{E}_R\to 0$.

Next, by dominated convergence as in Step 2,

$$\int_{\mathbb{R}^2}\eta_R^2\mathcal{Q}^2\,dx \to ||\mathcal{Q}||^2_{L^2(\mathbb{R}^2)}, \qquad \int_{\mathbb{R}^2}\eta_R^2\mathcal{Q}^4\,dx \to ||\mathcal{Q}||^4_{L^4(\mathbb{R}^2)}.$$

Passing to the limit in

$$-\int_{\mathbb{R}^2} \eta_R^2 \mathcal{Q}^2\,dx + \frac{1}{2}\int_{\mathbb{R}^2} \eta_R^2 \mathcal{Q}^4\,dx = \mathcal{E}_R$$

gives (38).

Step 5: Derivation of (39)**.** From (38) we have

$$||\mathcal{Q}||_{L^4(\mathbb{R}^2)}^4 = 2||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2.$$

Insert this into (37) to obtain

$$||\nabla\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2 + ||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2 = 2||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2,$$

hence $||\nabla\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2 = ||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2$. This proves (39). ■

Corollary 2.3.2 (Closed-form expression of C_{GN} in $d = 2$)**.** *Assume $\mathcal{Q}$ is an optimizer for the sharp Gagliardo-Nirenberg inequality, and that $\mathcal{Q}$ satisfies the normalized stationary equation* (36)*. Then*

$$C_{\mathrm{GN}} = \mathcal{J}(\mathcal{Q}) = \frac{2}{||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2}.$$

Proof. Since $\mathcal{Q}$ is a rescaling of the optimizer Q constructed in Subsection 2.1, and since $\mathcal{J}$ is scaling invariant by Lemma 2.1.4, we have $\mathcal{J}(\mathcal{Q}) = \mathcal{J}(Q) = C_{\mathrm{GN}}$. Therefore

$$C_{\mathrm{GN}} = \mathcal{J}(\mathcal{Q}) = \frac{||\mathcal{Q}||_{L^4(\mathbb{R}^2)}^4}{||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2\,||\nabla\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2}.$$

By (39) in Lemma 2.3.1,

$$||\mathcal{Q}||_{L^4(\mathbb{R}^2)}^4 = 2||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2 \qquad \text{and} \qquad ||\nabla\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2 = ||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2.$$

Substitute these identities into the expression for $\mathcal{J}(\mathcal{Q})$:

$$\mathcal{J}(\mathcal{Q}) = \frac{2||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2}{||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2\,||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2} = \frac{2}{||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2}.$$

Therefore $C_{\mathrm{GN}} = \mathcal{J}(\mathcal{Q}) = 2/||\mathcal{Q}||_{L^2(\mathbb{R}^2)}^2$. ■

Remark 2.3.3. *The characterization $C_{\mathrm{GN}} = \sup\mathcal{J}$ and the existence of an optimizer go back to Weinstein's sharp interpolation estimates [25]. The closed-form identity in Corollary 2.3.2 follows from the equality case combined with Pohozaev-type identities, see, for instance, [17, 6].*

3 Uniqueness of the Solution of the Stationary Equation

In this section, we proved the uniqueness of the positive ground state solving the stationary equation associated with the sharp variational problem. The goal is to show that the optimizer from Section 2 is unique up to radial symmetry, which we will use later, in Section 4, when classifying the minimal mass for blow-up.

In Section 2 we constructed a positive radial optimizer Q for the sharp Gagliardo-Nirenberg inequality and derived its Euler-Lagrange equation. In this section we prove that, within the class of positive radial solutions vanishing at infinity, the stationary equation has at most one solution, using the uniqueness theory of McLeod-Serrin. Since existence has already been established, this yields uniqueness of the ground-state profile.[15]

Throughout this section we work in dimension $n \coloneqq 2$ and consider the stationary equation

$$-\Delta Q + Q - Q^3 = 0 \qquad \text{in } \mathbb{R}^2, \tag{48}$$

together with the conditions $Q > 0$ and $Q(x) \to 0$ as $|x| \to \infty$. We will reduce (48) to a radial ODE and then use the uniqueness theory of McLeod-Serrin.

3.1 Reduction to a Radial Boundary-Value Problem

Lemma 3.1.1 (Radial Laplacian in $\mathbb{R}^2$)**.** *Let $q : (0,\infty) \to \mathbb{R}$ be C^2 and define $Q : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$ by $Q(x) \coloneqq q(|x|)$. Then for every $x \neq 0$ one has*

$$\Delta Q(x) = q''(|x|) + \frac{1}{|x|}\, q'(|x|).$$

Proof. Fix $x \in \mathbb{R}^2 \setminus \{0\}$ and set $r \coloneqq |x|$. For $j \in \{1,2\}$ we have

$$\partial_{x_j} r = \partial_{x_j} \sqrt{x_1^2 + x_2^2} = \frac{x_j}{\sqrt{x_1^2 + x_2^2}} = \frac{x_j}{r}.$$

Using the chain rule,

$$\partial_{x_j} Q(x) = \partial_{x_j}\big(q(r)\big) = q'(r)\, \partial_{x_j} r = q'(r)\, \frac{x_j}{r}.$$

Differentiate once more:

$$\begin{aligned} \partial^2_{x_j x_j} Q(x) &= \partial_{x_j}\Big(q'(r)\, \frac{x_j}{r}\Big) \\ &= q''(r)\, (\partial_{x_j} r)\, \frac{x_j}{r} + q'(r)\, \partial_{x_j}\Big(\frac{x_j}{r}\Big). \end{aligned}$$

We compute $\partial_{x_j}(\frac{x_j}{r})$ explicitly:

$$\partial_{x_j}\Big(\frac{x_j}{r}\Big) = \frac{1 \cdot r - x_j(\partial_{x_j} r)}{r^2} = \frac{r - x_j \cdot \frac{x_j}{r}}{r^2} = \frac{r^2 - x_j^2}{r^3}.$$

Also $(\partial_{x_j} r)\frac{x_j}{r} = \frac{x_j}{r}\frac{x_j}{r} = \frac{x_j^2}{r^2}$. Hence

$$\partial^2_{x_j x_j} Q(x) = q''(r)\, \frac{x_j^2}{r^2} + q'(r)\, \frac{r^2 - x_j^2}{r^3}.$$

Summing over $j = 1, 2$ yields

$$\begin{aligned}\Delta Q(x) &= \sum_{j=1}^{2} \partial^2_{x_j x_j} Q(x) \\ &= q''(r)\, \frac{x_1^2 + x_2^2}{r^2} + q'(r)\, \frac{(r^2 - x_1^2) + (r^2 - x_2^2)}{r^3} \\ &= q''(r)\, \frac{r^2}{r^2} + q'(r)\, \frac{2r^2 - (x_1^2 + x_2^2)}{r^3} \\ &= q''(r) + q'(r)\, \frac{2r^2 - r^2}{r^3} = q''(r) + \frac{1}{r} q'(r).\end{aligned}$$

This is the claimed formula. ∎

Lemma 3.1.2 (Radial boundary value problem)**.** *Assume Q is a C^2 radial solution of* (48) *on $\mathbb{R}^2$, i.e. $Q(x) = q(|x|)$ for some C^2 function $q : [0, \infty) \to \mathbb{R}$. Then q satisfies*

$$q''(r) + \frac{1}{r} q'(r) - q(r) + q(r)^3 = 0 \qquad \text{for all } r > 0, \tag{49}$$

and the regularity of Q at $x = 0$ implies

$$q'(0) = 0. \tag{50}$$

Conversely, if $q \in C^2([0, \infty))$ satisfies (49) *for $r > 0$ and* (50)*, then $Q(x) := q(|x|)$ is C^2 on $\mathbb{R}^2$ and satisfies* (48)*.*

Proof. **Step 1: Justification of** (49)**.**Assume first that $Q(x) = q(|x|)$ is C^2 on $\mathbb{R}^2$ and solves (48). For $x \neq 0$, Lemma 3.1.1 gives

$$\Delta Q(x) = q''(|x|) + \frac{1}{|x|} q'(|x|).$$

Inserting this into (48) at $x \neq 0$, and setting $r := |x|$, we obtain (49) for all $r > 0$.

Step 2: Justification of (50)**.** Since Q is differentiable at $x = 0$ and radial, the directional derivative of Q at 0 must vanish in every direction. Fix a unit vector $e \in \mathbb{R}^2$ and consider the one-dimensional function $t \mapsto Q(te) = q(|t|)$. Differentiability at 0 implies

$$\lim_{t \to 0} \frac{Q(te) - Q(0)}{t}$$

exists and equals $\nabla Q(0) \cdot e$. Because $Q(te) = q(|t|)$ depends on t only through $|t|$, the left limit $t \to 0^-$ equals the negative of the right limit $t \to 0^+$ unless the derivative at 0 is 0. Hence the derivative must be 0, and this is precisely $q'(0) = 0$.

Step 3: Justification of the converse. Conversely, assume $q \in C^2([0, \infty))$ satisfies (49) for $r > 0$ and $q'(0) = 0$. Define $Q(x) := q(|x|)$. For $x \neq 0$, Lemma 3.1.1 shows that Q is C^2 on $\mathbb{R}^2 \setminus \{0\}$ and solves (48) there. It remains to check that Q is C^2 at $x = 0$.

We now make explicit the use of Taylor's theorem with remainder to control $q(r)$ and $q'(r)$ as $r \to 0$.

Taylor expansion of q at 0 with remainder. Since $q \in C^2([0, \infty))$, the function q'' is continuous at 0. Hence there exists $\delta > 0$ and a constant

$$M := \sup_{s \in [0, \delta]} |q''(s)| < \infty.$$

Fix $r \in [0, \delta]$. Taylor's theorem with the (Lagrange) remainder applied to q at 0 yields that there exists $\theta_r \in (0, 1)$ such that

$$q(r) = q(0) + q'(0)\, r + \frac{1}{2} q''(\theta_r r)\, r^2.$$

Using $q'(0) = 0$ and $|q''(\theta_r r)| \le M$, we obtain

$$|q(r) - q(0)| = \frac{1}{2} |q''(\theta_r r)|\, r^2 \le \frac{M}{2} r^2.$$

In particular, $q(r) = q(0) + O(r^2)$ as $r \to 0$ in the explicit quantitative form above.

Taylor expansion of q' at 0 with remainder. Apply Taylor's theorem with remainder to q' at 0. Since $q' \in C^1([0, \infty))$ and $(q')'(s) = q''(s)$ is continuous, for each $r \in [0, \delta]$ there exists $\widetilde{\theta}_r \in (0, 1)$ such that

$$q'(r) = q'(0) + q''(\widetilde{\theta}_r r)\, r.$$

Again using $q'(0) = 0$ and $|q''(\widetilde{\theta}_r r)| \le M$, we obtain

$$|q'(r)| \le Mr \qquad \text{for all } r \in [0, \delta].$$

We use these bounds to conclude that $Q(x) = q(|x|)$ is C^2 at $x = 0$.

Existence and continuity of first derivatives at 0. For $x \neq 0$ and $j \in \{1, 2\}$, the chain rule gives

$$\partial_{x_j} Q(x) = q'(|x|)\, \frac{x_j}{|x|}.$$

Let $x \to 0$ with $0 < |x| \le \delta$. Then

$$|\partial_{x_j} Q(x)| = \Big| q'(|x|) \Big| \frac{|x_j|}{|x|} \le |q'(|x|)| \le M|x|.$$

Hence $\partial_{x_j} Q(x) \to 0$ as $x \to 0$. Define $\partial_{x_j} Q(0) := 0$. Then $\partial_{x_j} Q$ extends continuously to $x = 0$, and thus $Q \in C^1(\mathbb{R}^2)$.

Existence and continuity of second derivatives at 0. For $x \neq 0$ the formula above implies that $\partial_{x_j} Q$ is differentiable, and its derivatives can be written in terms of $q'(|x|)$ and $q''(|x|)$. Since q' is $O(|x|)$ and q'' is bounded near 0, these expressions remain bounded as $x \to 0$. Moreover, the $O(|x|)$ control of q' removes the apparent singularities coming from factors like $1/|x|$. Therefore all second-order partial derivatives $\partial^2_{x_j x_k} Q(x)$ admit finite limits as $x \to 0$ and extend continuously to 0. Consequently, $Q \in C^2(\mathbb{R}^2)$.

Therefore $Q \in C^2(\mathbb{R}^2)$ and (48) holds everywhere by continuity. ■

3.2 The McLeod-Serrin Framework for Uniqueness

In this subsection, we introduce shooting sets used to prove uniqueness of positive radial solutions. The goal is to set up the initial-value problem and define the relevant shooting sets that control the global behavior of solutions.

We rewrite (49) in the McLeod-Serrin form

$$u''(r) + \frac{n-1}{r} u'(r) + f(u(r)) = 0, \qquad n := 2, \tag{51}$$

with

$$f(u) := -u + u^3. \tag{52}$$

A positive radial ground state for (48) corresponds to a solution u of (51) with

$$u'(0) = 0, \qquad u(r) > 0 \text{ for all } r \ge 0, \qquad \lim_{r \to \infty} u(r) = 0. \tag{53}$$

3.2.1 The initial value problem

For each $u_0 > 0$, we consider the initial value problem

$$u''(r) + \frac{1}{r}u'(r) - u(r) + u(r)^3 = 0 \quad (r > 0), \qquad u(0) = u_0, \qquad u'(0) = 0. \tag{54}$$

Lemma 3.2.1 (Local well-posedness of (54))**.** *For each $u_0 > 0$, there exists $R > 0$ and a unique solution $u \in C^2([0, R])$ of (54). Moreover, if $u_0^{(k)} \to u_0$, then the corresponding solutions converge to u in $C^2([0, R'])$, for every $0 < R' < R$.*

Proof. We eliminate the apparent singularity at $r = 0$ by introducing

$$w(r) := r\,u'(r).$$

Then $u'(r) = \frac{w(r)}{r}$ for $r > 0$. Differentiate w:

$$w'(r) = u'(r) + ru''(r).$$

Using the ODE in (54), we rewrite $ru''(r)$:

$$ru''(r) = -u'(r) + r\big(u(r) - u(r)^3\big).$$

Therefore

$$w'(r) = u'(r) + \Big(- u'(r) + r(u - u^3)\Big) = r(u - u^3).$$

Thus on $r \geq 0$ we consider the first-order system

$$u'(r) = \begin{cases} \frac{w(r)}{r}, & r > 0, \\ 0, & r = 0, \end{cases} \qquad w'(r) = r\big(u(r) - u(r)^3\big), \qquad u(0) = u_0,\ w(0) = 0. \tag{55}$$

The right-hand side of (55) is continuous on a neighborhood of $(r, u, w) = (0, u_0, 0)$, and is locally Lipschitz in (u, w) for fixed r because the map $u \mapsto u - u^3$ is locally Lipschitz and $w \mapsto w$ is Lipschitz. Hence Picard iteration yields a unique C^1 solution (u, w) on some interval $[0, R]$. The identity $w(r) = ru'(r)$ for $r > 0$ implies $u' \in C^1((0, R])$, and the equation (54) then implies $u \in C^2((0, R])$. Since w is C^1 at 0 and $w(0) = 0$, we have $w(r) = O(r)$ as $r \to 0$, hence $u'(r) = w(r)/r$ extends continuously to 0 with value 0. Differentiating once more shows u'' also extends continuously to 0. Therefore, $u \in C^2([0, R])$.

We now prove the continuous dependence on the initial value u_0 in full detail.

Fix $u_0 > 0$, let u be the solution of (54) on $[0, R]$, and let $(u_0^{(k)})_{k\in\mathbb{N}}$ be a sequence with $u_0^{(k)} \to u_0$. For each k let u_k be the solution of (54) with initial value $u_k(0) = u_0^{(k)}$ on its interval of existence.

Step 1: Choose a common time interval and uniform bounds. Since $u_0^{(k)} \to u_0$, there exists $k_0 \in \mathbb{N}$ such that for all $k \geq k_0$ one has

$$|u_0^{(k)} - u_0| \leq 1.$$

Set

$$U_0 := u_0 + 1.$$

By the construction via Picard iteration in the proof above (applied with initial values in the bounded set $(0, U_0]$), there exists $R' \in (0, R]$ such that for every $k \geq k_0$ the solution u_k is defined on $[0, R']$ and satisfies

$$\sup_{r \in [0,R']} |u_k(r)| \leq U \qquad \text{and} \qquad \sup_{r \in [0,R']} |u(r)| \leq U$$

for some constant $U > 0$ depending only on U_0 and R'.

Define the nonlinearity

$$g(s) := s - s^3.$$

On the interval $[-U, U]$ we have $g'(s) = 1 - 3s^2$, hence

$$\sup_{|s| \leq U} |g'(s)| \leq 1 + 3U^2.$$

Set

$$L := 1 + 3U^2.$$

By the mean value theorem, for all $a, b \in [-U, U]$,

$$|g(a) - g(b)| \leq L|a - b|.$$

Step 2: Rewrite both solutions in integral form. For $k \geq k_0$ define

$$w_k(r) := r\, u_k'(r), \qquad w(r) := r\, u'(r).$$

Then $w_k(0) = 0$ and $w(0) = 0$ because $u_k'(0) = 0$ and $u'(0) = 0$. Moreover, from $w_k'(r) = r(g(u_k(r)))$ and $w'(r) = r(g(u(r)))$, integrating from 0 to r yields, for every $r \in [0, R']$,

$$w_k(r) = \int_0^r s\, g(u_k(s))\, ds, \qquad w(r) = \int_0^r s\, g(u(s))\, ds.$$

Since $u_k'(r) = w_k(r)/r$ for $r > 0$ and $u_k'(0) = 0$, we have for every $r \in [0, R']$,

$$u_k(r) = u_0^{(k)} + \int_0^r \frac{w_k(s)}{s}\, ds, \qquad u(r) = u_0 + \int_0^r \frac{w(s)}{s}\, ds,$$

where the integrands are integrable near 0 because $w_k(s) = O(s^2)$ and $w(s) = O(s^2)$ as $s \to 0$ (this follows from the formula $w_k(s) = \int_0^s \tau g(u_k(\tau))\, d\tau$ and boundedness of $g(u_k)$ on $[0, R']$).

Step 3: Estimate $||u_k - u||_{L^\infty([0,R'])}$. For $r \in [0, R']$, subtract the integral formulas:

$$u_k(r) - u(r) = (u_0^{(k)} - u_0) + \int_0^r \frac{w_k(s) - w(s)}{s}\, ds.$$

Also,

$$w_k(s) - w(s) = \int_0^s \tau \big(g(u_k(\tau)) - g(u(\tau)) \big)\, d\tau.$$

Using (3.2.1) and defining

$$d_k(r) := \sup_{t \in [0,r]} |u_k(t) - u(t)|,$$

we obtain, for every $\tau \in [0, s]$,

$$|g(u_k(\tau)) - g(u(\tau))| \leq L|u_k(\tau) - u(\tau)| \leq L d_k(s),$$

hence

$$|w_k(s) - w(s)| \leq \int_0^s \tau\, L d_k(s)\, d\tau = \frac{L}{2} s^2\, d_k(s).$$

Therefore

$$\left|\int_0^r \frac{w_k(s) - w(s)}{s}\, ds\right| \leq \int_0^r \frac{1}{s} \cdot \frac{L}{2} s^2\, d_k(s)\, ds = \frac{L}{2} \int_0^r s\, d_k(s)\, ds \leq \frac{L}{2} R' \int_0^r d_k(s)\, ds.$$

Combining the previous estimates gives, for every $r \in [0, R']$,

$$|u_k(r) - u(r)| \leq |u_0^{(k)} - u_0| + \frac{L}{2} R' \int_0^r d_k(s)\, ds.$$

Taking the supremum over $r \in [0, \rho]$ yields, for every $\rho \in [0, R']$,

$$d_k(\rho) \leq |u_0^{(k)} - u_0| + \frac{L}{2} R' \int_0^\rho d_k(s)\, ds.$$

By Grönwall's inequality applied to the function $\rho \mapsto d_k(\rho)$, we obtain

$$d_k(\rho) \leq |u_0^{(k)} - u_0| \exp\Big(\frac{L}{2} R' \rho\Big) \qquad \text{for all } \rho \in [0, R'].$$

In particular,

$$||u_k - u||_{L^\infty([0,R'])} = d_k(R') \leq |u_0^{(k)} - u_0| \exp\Big(\frac{L}{2} (R')^2\Big).$$

Since $u_0^{(k)} \to u_0$, the right-hand side converges to 0, hence $u_k \to u$ uniformly on $[0, R']$.

Step 4: Estimate $||u_k' - u'||_{L^\infty([0,R'])}$. For $r \in (0, R']$,

$$u_k'(r) - u'(r) = \frac{w_k(r) - w(r)}{r}.$$

Using the bound above with $d_k(R') = ||u_k - u||_{L^\infty([0,R'])}$,

$$|w_k(r) - w(r)| = \left|\int_0^r s\big(g(u_k(s)) - g(u(s))\big)\, ds\right| \leq \int_0^r s \cdot L\, d_k(R')\, ds = \frac{L}{2} r^2\, d_k(R').$$

Therefore, for $r \in (0, R']$,

$$|u_k'(r) - u'(r)| \leq \frac{1}{r} \cdot \frac{L}{2} r^2\, d_k(R') = \frac{L}{2} r\, d_k(R') \leq \frac{L}{2} R'\, d_k(R').$$

At $r = 0$ we have $u_k'(0) = 0 = u'(0)$, hence the same bound holds on $[0, R']$:

$$||u_k' - u'||_{L^\infty([0,R'])} \leq \frac{L}{2} R'\, ||u_k - u||_{L^\infty([0,R'])}.$$

Since $||u_k - u||_{L^\infty([0,R'])} \to 0$, we conclude $u_k' \to u'$ uniformly on $[0, R']$, i.e. convergence in $C^1([0, R'])$.

Step 5: Estimate $||u_k'' - u''||_{L^\infty([0,R'])}$. For $r \in (0, R']$, using $w_k'(r) = r g(u_k(r))$ and $u_k'(r) = w_k(r)/r$, we compute

$$u_k''(r) = \frac{d}{dr}\Big(\frac{w_k(r)}{r}\Big) = \frac{w_k'(r)}{r} - \frac{w_k(r)}{r^2} = g(u_k(r)) - \frac{w_k(r)}{r^2}.$$

Similarly,

$$u''(r) = g(u(r)) - \frac{w(r)}{r^2}.$$

Hence, for $r \in (0, R']$,

$$|u_k''(r) - u''(r)| \le |g(u_k(r)) - g(u(r))| + \left| \frac{w_k(r) - w(r)}{r^2} \right|.$$

The first term satisfies $|g(u_k(r)) - g(u(r))| \le L\, d_k(R')$ by (3.2.1). For the second term,

$$\left| \frac{w_k(r) - w(r)}{r^2} \right| = \frac{1}{r^2} \left| \int_0^r s\big(g(u_k(s)) - g(u(s))\big)\, ds \right| \le \frac{1}{r^2} \int_0^r s \cdot L\, d_k(R')\, ds = \frac{L}{2}\, d_k(R').$$

Therefore, for all $r \in (0, R']$,

$$|u_k''(r) - u''(r)| \le \frac{3L}{2}\, d_k(R').$$

At $r = 0$, note that the representation

$$\frac{w(r)}{r^2} = \frac{1}{r^2} \int_0^r s\, g(u(s))\, ds = \int_0^1 t\, g(u(rt))\, dt$$

follows from the substitution $s = rt$. Since u is continuous and $u(0) = u_0$, we have $u(rt) \to u_0$ for each $t \in [0, 1]$, so $tg(u(rt)) \to tg(u_0)$ pointwise in t. Moreover, $|tg(u(rt))| \le t \sup_{|s| \le U} |g(s)|$, and $t \mapsto t$ is integrable on $[0, 1]$, so dominated convergence implies

$$\lim_{r \to 0} \frac{w(r)}{r^2} = \int_0^1 t\, g(u_0)\, dt = \frac{1}{2} g(u_0).$$

Hence

$$u''(0) = \lim_{r \to 0} \left(g(u(r)) - \frac{w(r)}{r^2} \right) = g(u_0) - \frac{1}{2} g(u_0) = \frac{1}{2} g(u_0) = \frac{u_0 - u_0^3}{2}.$$

The same argument gives $u_k''(0) = (u_0^{(k)} - (u_0^{(k)})^3)/2$. Using again (3.2.1),

$$|u_k''(0) - u''(0)| = \frac{1}{2} |g(u_0^{(k)}) - g(u_0)| \le \frac{L}{2} |u_0^{(k)} - u_0|.$$

Combining the $r > 0$ bound with the $r = 0$ bound yields

$$||u_k'' - u''||_{L^\infty([0,R'])} \le C\, ||u_k - u||_{L^\infty([0,R'])} + \frac{L}{2} |u_0^{(k)} - u_0|,$$

with $C := \frac{3L}{2}$. Since both terms on the right converge to 0, we conclude $u_k'' \to u''$ uniformly on $[0, R']$.

Altogether, $u_k \to u$ in $C^2([0, R'])$ as $u_0^{(k)} \to u_0$. ∎

3.2.2 Shooting sets

Let u_{u_0} denote the unique local solution of (54) with initial value u_0. Let

$$T(u_0) := \sup\{\, R > 0 \mid u_{u_0}(r) \text{ is defined and } u_{u_0}(r) > 0 \text{ for all } r \in [0, R) \,\}.$$

If $T(u_0) < \infty$, then $u_{u_0}(T(u_0)) = 0$ by continuity. If $T(u_0) = \infty$, we define

$$\ell(u_0) := \liminf_{r \to \infty} u_{u_0}(r) \in [0, \infty).$$

Definition 3.2.2 (McLeod-Serrin sets S_-, S_0, S_+)**.** *We define*

$$\begin{aligned} S_- &:= \{\, u_0 > 0 \mid T(u_0) < \infty \,\}, \\ S_0 &:= \{\, u_0 > 0 \mid T(u_0) = \infty \ \textit{and} \ \lim_{r\to\infty} u_{u_0}(r) = 0 \,\}, \\ S_+ &:= \{\, u_0 > 0 \mid T(u_0) = \infty \ \textit{and} \ \ell(u_0) > 0 \,\}. \end{aligned}$$

The ground state problem (53) is equivalent to showing that S_0 is non-empty. In Section 2, we already constructed one such solution, so $S_0 \neq \emptyset$. The goal of this section is to prove S_0 contains exactly one element.

3.3 The McLeod-Serrin Uniqueness Criterion for $1 < n \leq 2$

For f as in (52), let $\alpha > 0$ denote the unique positive zero of f. In our case, $f(u) = -u + u^3$ implies

$$f(\alpha) = 0 \iff -\alpha + \alpha^3 = 0 \iff \alpha \in \{0, 1\},$$

and thus

$$\alpha = 1. \tag{56}$$

We now state the uniqueness theorem of McLeod-Serrin in the regime $1 < n \leq 2$.

Theorem 3.3.1 (McLeod-Serrin uniqueness theorem for $1 < n \leq 2$)**.** *Assume $1 < n \leq 2$ and let f satisfy the standing sign condition that there exists $\alpha > 0$ such that*

$$f(u) < 0 \ \textit{for} \ 0 < u < \alpha, \qquad f(u) > 0 \ \textit{for} \ u > \alpha.$$

Assume moreover that there exists $\tau \geq 1$ such that the two inequalities

$$\Big(\frac{f(u)}{u^\tau}\Big)' > 0 \qquad \textit{for all } u > 0 \textit{ with } u \neq \alpha, \tag{57}$$

and

$$\Big[u\Big(\frac{f(u)}{u^\tau}\Big)'\Big]' < 0 \qquad \textit{for all } u > \alpha \tag{58}$$

hold. Then the ground state problem (51)-(53) has at most one solution.

Proof. This is Theorem 2 in McLeod-Serrin (1986) [15]. We will not restate their entire general machinery here; instead, we will apply the theorem after verifying (57)-(58) for our f. The proof in McLeod-Serrin proceeds by converting uniqueness of the boundary-value problem into a statement about intersections of nearby solutions of the initial value problem (54), using a shooting decomposition into S_-, S_0, S_+, and then establishing an intersection property from an identity and a monotonicity theorem (the so-called w-theorem). ■

Remark 3.3.2. *In the remainder of this section, we verify (57)-(58) for $f(u) = -u + u^3$. Therefore, Theorem 3.3.1 will imply that the positive radial ground state for (48) is unique.*

3.4 Verification of the McLeod-Serrin Conditions for $f(u) = -u + u^3$

Lemma 3.4.1 (Verification of (57)-(58) for $f(u) = -u + u^3$)**.** *Let $f(u) := -u + u^3$ and $\alpha := 1$. Then conditions* (57)-(58) *hold with $\tau := 3$.*

Proof. Fix $\tau := 3$. For $u > 0$ we compute

$$\frac{f(u)}{u^\tau} = \frac{-u + u^3}{u^3} = -u^{-2} + 1.$$

Differentiate with respect to u:

$$\Big(\frac{f(u)}{u^3}\Big)' = \frac{d}{du}\big(-u^{-2} + 1\big) = 2u^{-3}.$$

For every $u > 0$ we have $2u^{-3} > 0$, hence (57) holds (in particular for all $u > 0$ with $u \neq \alpha$).

Next compute

$$u\Big(\frac{f(u)}{u^3}\Big)' = u \cdot 2u^{-3} = 2u^{-2}.$$

Differentiate once more:

$$\Big[u\Big(\frac{f(u)}{u^3}\Big)'\Big]' = \frac{d}{du}\big(2u^{-2}\big) = -4u^{-3}.$$

For every $u > 0$ we have $-4u^{-3} < 0$, and therefore (58) holds for all $u > \alpha = 1$. ■

3.5 Conclusion: Uniqueness of the Ground State in $\mathbb{R}^2$

We summarize the uniqueness conclusion for the stationary problem in the concrete two-dimensional setting. This is the point where the variational optimizer from Section 2 is identified as the unique positive radial ground state. This is important for Section 4 when we are going to be analyzing minimal-mass blow-up profiles.

Theorem 3.5.1 (Uniqueness of the positive radial ground state in $\mathbb{R}^2$)**.** *There exists at most one function $Q : \mathbb{R}^2 \to (0, \infty)$ of the form $Q(x) = q(|x|)$ such that $Q \in C^2(\mathbb{R}^2)$, $Q(x) \to 0$ as $|x| \to \infty$, and*

$$-\Delta Q + Q - Q^3 = 0 \quad \text{in } \mathbb{R}^2.$$

Equivalently, there exists at most one C^2 function $q : [0, \infty) \to (0, \infty)$ such that $q'(0) = 0$, $q(r) \to 0$ as $r \to \infty$, and

$$q''(r) + \frac{1}{r}q'(r) - q(r) + q(r)^3 = 0 \quad \text{for all } r > 0.$$

Proof. By Lemma 3.1.2, uniqueness for (48) in the radial class is equivalent to uniqueness for (51)-(53) with $n = 2$ and $f(u) = -u + u^3$.

Lemma 3.4.1 shows that the McLeod-Serrin hypotheses (57)-(58) hold with $\alpha = 1$ and $\tau = 3$. Therefore Theorem 3.3.1 applies and yields that (51)-(53) has at most one solution. This proves the claim. ■

Remark 3.5.2 (Role of the uniqueness theorem)**.** *The Euler-Lagrange equation obtained from the sharp Gagliardo-Nirenberg inequality yields the stationary equation* (48)*. Theorem 3.5.1 shows that this stationary equation admits at most one positive radial solution decaying at infinity. In particular, once existence of such a solution is known, it is uniquely determined within this class.*

3.6 Context: the Higher-Dimensional Statements in McLeod-Serrin

For completeness, we record that McLeod-Serrin also treat uniqueness in dimensions $n \geq 3$ under alternative structural hypotheses on f (their Theorem 1 and Theorem 3). In the model case $f(u) = -u+u^p$, the verification of their inequalities becomes an explicit computational exercise in u and yields uniqueness in several regimes, which depend on n and p. Since the present thesis focuses on $n = 2$ and $p = 3$, we do not use these higher-dimensional criteria later.

4 Blow-Up for the L^2-Critical Focusing Nonlinear Schrödinger Equation in $\mathbb{R}^2$

In this section, we study finite-time blow-up phenomena for the L^2-critical focusing nonlinear Schrödinger equation in $\mathbb{R}^2$. The goal is to connect blow-up behavior to the variational arguments from Section 2 and compactness, in particular, to show concentration of mass and to identify the ground-state profile from Section 2 as the minimal-mass blow-up shape/profile.

The equation that we are going to consider is the cubic nonlinear Schrödinger equation in $\mathbb{R}^2$, namely,

$$i\partial_t u + \Delta u + |u|^2 u = 0, \qquad (t,x) \in I \times \mathbb{R}^2. \tag{59}$$

Throughout, we follow the strategy of Hmidi and Keraani [13], specialized to the case $d = 2$.

4.1 Solutions, Symmetries, and Conserved Quantities

Here, we fix the solution class and show the symmetries and conserved quantities of the equation in the given class. The purpose is to identify the invariances that govern scaling and concentration and to set up the conservation laws that later constrain blow-up scenarios in Subsections 4.3 and 4.5.

4.1.1 Assumptions on solutions

In this subsubsection, we fix a time interval $I \subseteq \mathbb{R}$ and a function

$$u \in C(I; H^1(\mathbb{R}^2)) \cap C^1(I; H^{-1}(\mathbb{R}^2))$$

which satisfies (59) in $H^{-1}(\mathbb{R}^2)$ for every $t \in I$, that is,

$$i\partial_t u(t) + \Delta u(t) + |u(t)|^2 u(t) = 0 \quad \text{in } H^{-1}(\mathbb{R}^2), \qquad \forall t \in I.$$

Whenever we speak about a "maximal lifespan" solution with maximal interval of existence $I_{\max}$, we mean the standard well-posedness theory in $H^1(\mathbb{R}^2)$. See, for example, [6].

4.1.2 Scaling and other symmetries

Our next step is to show the key symmetries of the flow, including scaling, phase rotation, translation, and Galilean invariance. The goal is to show how these transformations act on the conserved quantities and how they enter the ground-state-profile description used later in Subsection 4.5.

In this subsubsection, we record the symmetries of (59) that will be used later.

Lemma 4.1.1 (Critical scaling in $d = 2$)**.** *Let u solve (59) on a time interval I. Fix $\lambda > 0$, and define*

$$u_\lambda(t,x) := \lambda\, u(\lambda^2 t, \lambda x). \tag{60}$$

Then u_λ solves (59) on the time interval $\lambda^{-2} I := \left\{ t \in \mathbb{R} \;\middle|\; \lambda^2 t \in I \right\}$.

Moreover, for every fixed $t \in \lambda^{-2} I$, one has

$$||u_\lambda(t)||_{L^2(\mathbb{R}^2)} = ||u(\lambda^2 t)||_{L^2(\mathbb{R}^2)}, \qquad ||\nabla u_\lambda(t)||_{L^2(\mathbb{R}^2)} = \lambda\, ||\nabla u(\lambda^2 t)||_{L^2(\mathbb{R}^2)}, \tag{61}$$

and

$$||u_\lambda(t)||^4_{L^4(\mathbb{R}^2)} = \lambda^2\, ||u(\lambda^2 t)||^4_{L^4(\mathbb{R}^2)}. \tag{62}$$

Proof. **Step 1: Show that u_λ is a solution.** Fix $\lambda > 0$. We first prove that u_λ solves (59). Let $t \in \lambda^{-2} I$ and $x \in \mathbb{R}^2$. By definition,

$$u_\lambda(t,x) = \lambda u(\lambda^2 t, \lambda x).$$

We compute the derivatives in the distributional sense, and since $u \in C^1(I; H^{-1})$ and H^{-1} is stable under multiplication by smooth bounded functions and under translations/dilations, all computations below are justified in $H^{-1}(\mathbb{R}^2)$.

First, for the time derivative,

$$\partial_t u_\lambda(t,x) = \partial_t\big(\lambda u(\lambda^2 t, \lambda x)\big) = \lambda \cdot \lambda^2\, (\partial_t u)(\lambda^2 t, \lambda x) = \lambda^3\, (\partial_t u)(\lambda^2 t, \lambda x).$$

Second, for the Laplacian, we compute componentwise. For $j \in \{1, 2\}$,

$$\partial_{x_j} u_\lambda(t,x) = \partial_{x_j}\big(\lambda u(\lambda^2 t, \lambda x)\big) = \lambda \cdot \lambda\, (\partial_{x_j} u)(\lambda^2 t, \lambda x) = \lambda^2\, (\partial_{x_j} u)(\lambda^2 t, \lambda x),$$

and hence

$$\partial_{x_j x_j} u_\lambda(t,x) = \partial_{x_j}\Big(\lambda^2 (\partial_{x_j} u)(\lambda^2 t, \lambda x)\Big) = \lambda^2 \cdot \lambda\, (\partial_{x_j x_j} u)(\lambda^2 t, \lambda x) = \lambda^3 (\partial_{x_j x_j} u)(\lambda^2 t, \lambda x).$$

Summing over $j = 1, 2$, we obtain

$$\Delta u_\lambda(t,x) = \lambda^3 (\Delta u)(\lambda^2 t, \lambda x).$$

Third, for the nonlinearity,

$$|u_\lambda(t,x)|^2 u_\lambda(t,x) = |\lambda u(\lambda^2 t, \lambda x)|^2 \cdot \lambda u(\lambda^2 t, \lambda x) = \lambda^3\, |u(\lambda^2 t, \lambda x)|^2 u(\lambda^2 t, \lambda x).$$

Putting these three computations together yields

$$i\partial_t u_\lambda(t,x) + \Delta u_\lambda(t,x) + |u_\lambda(t,x)|^2 u_\lambda(t,x) = \lambda^3 \Big(i\partial_t u + \Delta u + |u|^2 u\Big)(\lambda^2 t, \lambda x).$$

Since u solves (59), the right-hand side is 0 in $H^{-1}(\mathbb{R}^2)$, hence u_λ solves (59).

Step 2: Scaling of the norms of the solution. We now prove (61) and (62). Fix $t \in \lambda^{-2} I$.

For the L^2-norm, we use the change of variables $y = \lambda x$, hence $dy = \lambda^2 dx$ and $dx = \lambda^{-2} dy$:

$$||u_\lambda(t)||^2_{L^2(\mathbb{R}^2)} = \int_{\mathbb{R}^2} |\lambda u(\lambda^2 t, \lambda x)|^2\, dx = \int_{\mathbb{R}^2} \lambda^2 |u(\lambda^2 t, y)|^2\, \lambda^{-2} dy = \int_{\mathbb{R}^2} |u(\lambda^2 t, y)|^2\, dy = ||u(\lambda^2 t)||^2_{L^2(\mathbb{R}^2)}.$$

Taking square-roots gives the first identity in (61).

For the gradient, we use the already computed formula $\nabla u_\lambda(t,x) = \lambda^2 (\nabla u)(\lambda^2 t, \lambda x)$, and we again change variables $y = \lambda x$:

$$\begin{aligned} ||\nabla u_\lambda(t)||^2_{L^2(\mathbb{R}^2)} &= \int_{\mathbb{R}^2} |\lambda^2 \nabla u(\lambda^2 t, \lambda x)|^2\, dx = \int_{\mathbb{R}^2} \lambda^4 |\nabla u(\lambda^2 t, y)|^2\, \lambda^{-2} dy \\ &= \lambda^2 \int_{\mathbb{R}^2} |\nabla u(\lambda^2 t, y)|^2\, dy = \lambda^2 ||\nabla u(\lambda^2 t)||^2_{L^2(\mathbb{R}^2)}. \end{aligned}$$

Taking square-roots gives the second identity in (61).

For the L^4 norm, we compute similarly:

$$||u_\lambda(t)||^4_{L^4(\mathbb{R}^2)} = \int_{\mathbb{R}^2} |\lambda u(\lambda^2 t, \lambda x)|^4\, dx = \int_{\mathbb{R}^2} \lambda^4 |u(\lambda^2 t, y)|^4\, \lambda^{-2} dy = \lambda^2 \int_{\mathbb{R}^2} |u(\lambda^2 t, y)|^4\, dy = \lambda^2 ||u(\lambda^2 t)||^4_{L^4(\mathbb{R}^2)},$$

which is (62). ∎

Lemma 4.1.2 (Translations and phase rotations)**.** *Let u solve* (59) *on I. Fix $x_0 \in \mathbb{R}^2$ and $\theta \in \mathbb{R}$, and define*

$$v(t,x) := e^{i\theta} u(t, x - x_0).$$

Then v solves (59) *on I.*

Proof. Fix $t \in I$. The map $x \mapsto x - x_0$ commutes with ∂_t and with Δ, and multiplication by a constant complex phase $e^{i\theta}$ commutes with linear derivatives and satisfies $|e^{i\theta} z|^2 (e^{i\theta} z) = e^{i\theta} |z|^2 z$. Therefore,

$$i\partial_t v + \Delta v + |v|^2 v = e^{i\theta} \Big(i\partial_t u + \Delta u + |u|^2 u \Big)(t, x - x_0) = 0,$$

in $H^{-1}(\mathbb{R}^2)$, because u solves (59). ∎

4.1.3 Mass and energy conservation

In this subsubsection, we prove conservation of mass and energy for the solution class $H^1(\mathbb{R}^d)$. These conservation laws are the main tool used later in Subsection 4.3 to force mass concentration at blow-up times, and to identify the minimal-mass threshold.

Definition 4.1.3 (Mass and energy)**.** *Let $u \in H^1(\mathbb{R}^2)$. We define the mass $M(u)$ and the energy $E(u)$ by*

$$M(u) := \int_{\mathbb{R}^2} |u(x)|^2 \, dx, \tag{63}$$

and

$$E(u) := \frac{1}{2} \int_{\mathbb{R}^2} |\nabla u(x)|^2 \, dx - \frac{1}{4} \int_{\mathbb{R}^2} |u(x)|^4 \, dx. \tag{64}$$

Theorem 4.1.4 (Conservation of mass and energy)**.** *Let $I \subseteq \mathbb{R}$ be an interval, and let*

$$u \in C(I; H^1(\mathbb{R}^2)) \cap C^1(I; H^{-1}(\mathbb{R}^2))$$

solve (59) *on I. Then the maps $t \mapsto M(u(t))$ and $t \mapsto E(u(t))$ are constant on I.*

Proof. We split the proof into two parts.

Part 1: Conservation of mass. Fix $t_0, t_1 \in I$ with $t_0 < t_1$. It is enough to show that $M(u(t_1)) = M(u(t_0))$.

Since $u \in C^1(I; H^{-1})$ and $u \in C(I; H^1) \subseteq C(I; L^2)$, the map

$$t \mapsto \langle u(t), u(t) \rangle_{L^2(\mathbb{R}^2)}$$

is differentiable and

$$\frac{d}{dt} \langle u(t), u(t) \rangle_{L^2} = \langle \partial_t u(t), u(t) \rangle_{H^{-1}, H^1} + \langle u(t), \partial_t u(t) \rangle_{H^1, H^{-1}} = 2\Re \langle \partial_t u(t), u(t) \rangle_{H^{-1}, H^1},$$

where $\langle \cdot, \cdot \rangle_{H^{-1}, H^1}$ denotes the dual pairing extending the L^2 inner product.

Using the equation (59), we have

$$\partial_t u(t) = i\Delta u(t) + i|u(t)|^2 u(t) \quad \text{in } H^{-1}(\mathbb{R}^2).$$

Therefore,

$$\Re \langle \partial_t u(t), u(t) \rangle_{H^{-1}, H^1} = \Re \Big(i \langle \Delta u(t), u(t) \rangle_{H^{-1}, H^1} + i \langle |u(t)|^2 u(t), u(t) \rangle_{H^{-1}, H^1} \Big). \tag{65}$$

We now evaluate the two pairings.

First, since $u(t) \in H^1(\mathbb{R}^2)$, we have $\Delta u(t) \in H^{-1}(\mathbb{R}^2)$, and the distributional integration by parts identity gives

$$\langle \Delta u(t), u(t) \rangle_{H^{-1},H^1} = -\int_{\mathbb{R}^2} |\nabla u(t,x)|^2 \, dx,$$

which is a real number. Hence,

$$\Re\Big(i\langle \Delta u(t), u(t) \rangle_{H^{-1},H^1} \Big) = 0.$$

Second, since $u(t) \in H^1(\mathbb{R}^2) \subseteq L^4(\mathbb{R}^2)$, we have $|u(t)|^2 u(t) \in L^{4/3}(\mathbb{R}^2) \subseteq H^{-1}(\mathbb{R}^2)$ and

$$\langle |u(t)|^2 u(t), u(t) \rangle_{H^{-1},H^1} = \int_{\mathbb{R}^2} |u(t,x)|^4 \, dx,$$

which is also a real number. Hence,

$$\Re\Big(i\langle |u(t)|^2 u(t), u(t) \rangle_{H^{-1},H^1} \Big) = 0.$$

Plugging these two facts into (65) yields

$$\Re\langle \partial_t u(t), u(t) \rangle_{H^{-1},H^1} = 0,$$

hence

$$\frac{d}{dt} ||u(t)||^2_{L^2(\mathbb{R}^2)} = 0, \qquad \forall t \in I.$$

Integrating from t_0 to t_1 gives $||u(t_1)||_2^2 = ||u(t_0)||_2^2$, i.e., $M(u(t_1)) = M(u(t_0))$.

Part 2: Conservation of energy. Fix $t \in I$. We compute the derivative of $E(u(t))$ using (64).

Since $u \in C(I; H^1)$ and $\partial_t u \in C(I; H^{-1})$, we can differentiate the kinetic part using the dual pairing:

$$\frac{d}{dt} \frac{1}{2} \int_{\mathbb{R}^2} |\nabla u(t)|^2 = \Re \int_{\mathbb{R}^2} \nabla(\partial_t u(t,x)) \cdot \overline{\nabla u(t,x)} \, dx = -\Re\langle \partial_t u(t), \Delta u(t) \rangle_{H^{-1},H^1}.$$

Similarly, since $u(t) \in L^4$, the potential part is differentiable and

$$\frac{d}{dt} \frac{1}{4} \int_{\mathbb{R}^2} |u(t,x)|^4 \, dx = \Re \int_{\mathbb{R}^2} |u(t,x)|^2 u(t,x) \overline{\partial_t u(t,x)} \, dx = \Re\langle |u(t)|^2 u(t), \partial_t u(t) \rangle_{H^{-1},H^1}.$$

Therefore,

$$\frac{d}{dt} E(u(t)) = -\Re\langle \partial_t u(t), \Delta u(t) \rangle_{H^{-1},H^1} - \Re\langle |u(t)|^2 u(t), \partial_t u(t) \rangle_{H^{-1},H^1}. \tag{66}$$

Using the equation $\partial_t u = i\Delta u + i|u|^2 u$, we compute each term.

For the first term,

$$\langle \partial_t u, \Delta u \rangle = \langle i\Delta u + i|u|^2 u, \Delta u \rangle = i\langle \Delta u, \Delta u \rangle + i\langle |u|^2 u, \Delta u \rangle.$$

Here $\langle \Delta u, \Delta u \rangle$ is real and nonnegative (it equals $||\Delta u||^2_{H^{-1}}$ when interpreted appropriately), hence $\Re\big(i\langle \Delta u, \Delta u \rangle\big) = 0$. Therefore,

$$\Re\langle \partial_t u, \Delta u \rangle = \Re\big(i\langle |u|^2 u, \Delta u \rangle \big).$$

For the second term,

$$\langle |u|^2u, \partial_t u\rangle = \langle |u|^2u, i\Delta u + i|u|^2u\rangle = i\langle |u|^2u, \Delta u\rangle + i\langle |u|^2u, |u|^2u\rangle.$$

The last pairing $\langle |u|^2u, |u|^2u\rangle = \int |u|^6$ is real and nonnegative, hence its contribution vanishes after applying $\Re(i\cdot)$. Thus,

$$\Re\langle |u|^2u, \partial_t u\rangle = \Re\big(i\langle |u|^2u, \Delta u\rangle\big).$$

Plugging these two identities into (66) yields

$$\frac{d}{dt}E(u(t)) = -\Re\big(i\langle |u|^2u, \Delta u\rangle\big) - \Re\big(i\langle |u|^2u, \Delta u\rangle\big) = 0.$$

Hence $E(u(t))$ is constant on I. ■

4.2 A Concentration-Compactness Lemma in $\mathbb{R}^2$

Now we aim to prove the concentration-compactness lemma that captures loss of compactness up to translations (and other symmetries). The goal is to obtain a profile decomposition, which will later be applied in Subsection 4.3 to sequences of times approaching blow-up.

The compactness lemma is due to Hmidi and Keraani (specialized to our case $d = 2$).[13]

4.2.1 Auxiliary tools

Lemma 4.2.1 (Orthogonality of far-apart translates)**.** *Let $F, G \in L^2(\mathbb{R}^2)$. Let $(y_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}^2$ satisfy $|y_n| \to \infty$. Then*

$$\int_{\mathbb{R}^2} F(x - y_n)\overline{G(x)}\, dx \to 0.$$

Proof. Fix $F, G \in L^2(\mathbb{R}^2)$. Define $H_n(x) := F(x - y_n)$. Then $||H_n||_2 = ||F||_2$ for all n.

Step 1: Show that statement holds for $G \in C_c^\infty(\mathbb{R}^2)$. We first prove the claim for $G \in C_c^\infty(\mathbb{R}^2)$. Fix such a G. Since G has compact support, there exists $R > 0$ such that $\operatorname{supp}(G) \subseteq B(0, R)$. Then

$$\int_{\mathbb{R}^2} H_n(x)\overline{G(x)}\, dx = \int_{B(0,R)} F(x - y_n)\overline{G(x)}\, dx.$$

Using Cauchy-Schwarz,

$$\Big|\int_{B(0,R)} F(x - y_n)\overline{G(x)}\, dx\Big| \le ||G||_{L^2(B(0,R))}\, ||F(\cdot - y_n)||_{L^2(B(0,R))}.$$

We rewrite the last factor by changing variables $z = x - y_n$:

$$||F(\cdot - y_n)||^2_{L^2(B(0,R))} = \int_{B(0,R)} |F(x - y_n)|^2\, dx = \int_{B(-y_n,R)} |F(z)|^2\, dz.$$

Since $|y_n| \to \infty$, the balls $B(-y_n, R)$ escape to infinity. Because $F \in L^2(\mathbb{R}^2)$, we have

$$\int_{B(-y_n,R)} |F(z)|^2\, dz \to 0,$$

which can be proved as follows: for $\varepsilon > 0$, choose $R_\varepsilon > 0$ such that $\int_{|z|\geq R_\varepsilon} |F(z)|^2 dz < \varepsilon$. For n large enough, $B(-y_n, R) \subseteq \{z \in \mathbb{R}^2 \mid |z| \geq R_\varepsilon\}$, hence the integral over $B(-y_n, R)$ is at most ε. This proves the convergence to 0.
Therefore $||F(\cdot - y_n)||_{L^2(B(0,R))} \to 0$, hence the pairing with $G \in C_c^\infty$ tends to 0.

Step 2: Density argument. Now, let $G \in L^2(\mathbb{R}^2)$ be arbitrary. Choose $G_k \in C_c^\infty(\mathbb{R}^2)$ such that $||G_k - G||_2 \to 0$ as $k \to \infty$. Then, for fixed k,

$$\int F(x-y_n)\overline{G(x)}\,dx = \int F(x-y_n)\overline{G_k(x)}\,dx + \int F(x-y_n)\overline{(G(x)-G_k(x))}\,dx.$$

We already proved that the first term tends to 0 as $n \to \infty$. For the second term, by Cauchy-Schwarz and $||F(\cdot - y_n)||_2 = ||F||_2$,

$$\Big|\int F(x-y_n)\overline{(G(x)-G_k(x))}\,dx\Big| \leq ||F||_2\,||G-G_k||_2,$$

which is independent of n. Letting $n \to \infty$ first, we obtain

$$\limsup_{n\to\infty}\Big|\int F(x-y_n)\overline{G(x)}\,dx\Big| \leq ||F||_2\,||G-G_k||_2.$$

Letting $k \to \infty$ gives the result. ■

4.2.2 Profile decomposition with translations

For this subsubsection, we use the notation $\mathbb{N}^* := \mathbb{N} \cup \{\infty\}$.

Theorem 4.2.2 (Profile decomposition in $H^1(\mathbb{R}^2)$ (translations only))**.** *Let $(v_n)_{n\in\mathbb{N}}$ be a bounded sequence in $H^1(\mathbb{R}^2)$. Then there exist:*

- *a (finite or infinite) family $(V^j)_{j\in\mathbb{N}^*} \subseteq H^1(\mathbb{R}^2)$,*
- *a family of translation parameters $(x_n^j)_{n\in\mathbb{N},\, j\in\mathbb{N}^*} \subseteq \mathbb{R}^2$,*
- *and, for each $J \in \mathbb{N}^*$, a remainder sequence $(r_n^J)_{n\in\mathbb{N}} \subseteq H^1(\mathbb{R}^2)$,*

such that the following properties hold.

(a) Decomposition. *For every $J \in \mathbb{N}^*$ and every $n \in \mathbb{N}$,*

$$v_n(x) = \sum_{j=1}^{J} V^j(x - x_n^j) + r_n^J(x). \tag{67}$$

(b) Asymptotic separation of translations. *If $j \neq k$, then*

$$|x_n^j - x_n^k| \to \infty, \qquad \textit{as } n \to \infty. \tag{68}$$

(c) Weak convergence of remainders. *For every fixed $J \in \mathbb{N}^*$ and every $j \in \{1, \ldots, J\}$,*

$$r_n^J(\cdot + x_n^j) \rightharpoonup 0 \quad \textit{weakly in } H^1(\mathbb{R}^2), \qquad \textit{as } n \to \infty. \tag{69}$$

(d) Decoupling of L^2 and $\dot{H}^1$ norms. *For every fixed $J \in \mathbb{N}^*$,*

$$||v_n||^2_{L^2(\mathbb{R}^2)} = \sum_{j=1}^{J} ||V^j||^2_{L^2(\mathbb{R}^2)} + ||r_n^J||^2_{L^2(\mathbb{R}^2)} + o_n(1), \tag{70}$$

and

$$||\nabla v_n||^2_{L^2(\mathbb{R}^2)} = \sum_{j=1}^{J} ||\nabla V^j||^2_{L^2(\mathbb{R}^2)} + ||\nabla r_n^J||^2_{L^2(\mathbb{R}^2)} + o_n(1), \tag{71}$$

where $o_n(1) \to 0$ as $n \to \infty$, for fixed J.

(e) Smallness of the remainder in L^4. *One has*

$$\lim_{J\to\infty} \limsup_{n\to\infty} ||r_n^J||_{L^4(\mathbb{R}^2)} = 0. \tag{72}$$

Proof. The proof follows the classical inductive extraction argument, and we write all details.

Step 1: The set of translated weak limits and the quantity η. Define the set

$$\mathcal{V} := \big\{ V \in H^1(\mathbb{R}^2) \;\big|\; \exists\, (n_k)_{k\in\mathbb{N}} \subseteq \mathbb{N} \text{ strictly increasing, } \exists\, (y_k)_{k\in\mathbb{N}} \subseteq \mathbb{R}^2,\ v_{n_k}(\cdot + y_k) \rightharpoonup V \text{ in } H^1(\mathbb{R}^2) \big\}.$$

Since (v_n) is bounded in H^1, by Banach-Alaoglu, every translated subsequence has a further subsequence converging weakly in H^1, hence $\mathcal{V}$ is nonempty (it contains $\mathbf{0}_{H^1(\mathbb{R}^2)}$ at least).

Define

$$\eta := \sup\big\{ ||V||_{L^4(\mathbb{R}^2)} \;\big|\; V \in \mathcal{V} \big\} \in [0, \infty).$$

We note that $\eta < \infty$ because $H^1(\mathbb{R}^2) \hookrightarrow L^4(\mathbb{R}^2)$ continuously, hence every $V \in \mathcal{V}$ satisfies $||V||_4 \leq C||V||_{H^1}$, and $||V||_{H^1} \leq \liminf ||v_{n_k}(\cdot + y_k)||_{H^1}$ by weak lower semi-continuity, which is uniformly bounded.

Step 2: If $\eta = 0$, then $v_n \to 0$ in $L^4(\mathbb{R}^2)$. Assume $\eta = 0$. We show $||v_n||_4 \to 0$.

Suppose, for contradiction, that $||v_n||_4 \not\to 0$. Then there exist $\varepsilon_0 > 0$ and a subsequence (still denoted v_n) such that $||v_n||_4 \geq \varepsilon_0$ for all n.

We now use a translation selection argument. Since $||v_n||_4 \geq \varepsilon_0$, we can choose, for each n, a translation $y_n \in \mathbb{R}^2$ such that the translated functions $w_n := v_n(\cdot + y_n)$ do not vanish weakly in L^4. One concrete way is the following: choose a fixed $\psi \in C_c^\infty(\mathbb{R}^2)$, $\psi \geq 0$, $\psi \not\equiv 0$, and consider the function

$$F_n(y) := \int_{\mathbb{R}^2} \psi(x) |v_n(x+y)|^4 \, dx.$$

Since $|v_n|^4 \in L^1$ and $\psi \in L^\infty \cap L^1$, F_n is continuous and bounded, and

$$\int_{\mathbb{R}^2} F_n(y)\, dy = \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \psi(x)|v_n(x+y)|^4 \, dx\, dy = \int_{\mathbb{R}^2} \psi(x)\, dx \ \int_{\mathbb{R}^2} |v_n(z)|^4\, dz = ||\psi||_{L^1}\, ||v_n||_4^4.$$

In particular, $\int_{\mathbb{R}^2} F_n(y)\, dy \geq ||\psi||_{L^1}\varepsilon_0^4 > 0$, hence F_n is not identically zero, and there exists y_n such that $F_n(y_n) > 0$.

Now consider $w_n = v_n(\cdot + y_n)$. Since translations preserve the H^1 norm, (w_n) is bounded in H^1. Therefore, by Banach-Alaoglu, there exist a subsequence (still denoted w_n) and $W \in H^1$ such that $w_n \rightharpoonup W$ in H^1. By definition, $W \in \mathcal{V}$. Since $\eta = 0$, we must have $||W||_4 = 0$, hence $W = 0$ almost everywhere.

On the other hand, from the definition of $F_n(y_n)$ and the weak convergence, we will now obtain a contradiction. Indeed, by Rellich compactness on bounded sets, $w_n \to 0$ strongly in $L^4(\operatorname{supp}\psi)$, hence

$$\int_{\mathbb{R}^2} \psi(x)|w_n(x)|^4\, dx \to 0.$$

But $\int \psi |w_n|^4 = F_n(y_n) > 0$ for all n, a contradiction. Therefore $||v_n||_4 \to 0$.

Step 3: If $\eta > 0$, extract the first profile. Assume $\eta > 0$. By definition of η, there exist $V^1 \in \mathcal{V}$ such that

$$||V^1||_{L^4(\mathbb{R}^2)} \geq \frac{1}{2}\eta. \tag{73}$$

Since $V^1 \in \mathcal{V}$, there exist a subsequence (v_{n_k}) and translations $(x^1_{n_k})$ such that

$$v_{n_k}(\cdot + x^1_{n_k}) \rightharpoonup V^1 \quad \text{weakly in } H^1.$$

Relabeling the subsequence as the original sequence (this does not affect the statement, because the proposition is existential), we assume from now on that

$$v_n(\cdot + x^1_n) \rightharpoonup V^1 \quad \text{weakly in } H^1. \tag{74}$$

Define the first remainder

$$r^1_n(x) := v_n(x) - V^1(x - x^1_n). \tag{75}$$

Then (67) holds for $J = 1$, by definition.

We now prove the decoupling for $J = 1$. Consider the L^2 norm:

$$||v_n||_2^2 = ||V^1(\cdot - x^1_n) + r^1_n||_2^2 = ||V^1||_2^2 + ||r^1_n||_2^2 + 2\Re \int_{\mathbb{R}^2} V^1(x - x^1_n)\overline{r^1_n(x)}\, dx.$$

Change variables $y = x - x^1_n$ in the last integral:

$$\int_{\mathbb{R}^2} V^1(x - x^1_n)\overline{r^1_n(x)}\, dx = \int_{\mathbb{R}^2} V^1(y)\overline{r^1_n(y + x^1_n)}\, dy.$$

By (75) and (74), we have

$$r^1_n(\cdot + x^1_n) = v_n(\cdot + x^1_n) - V^1 \rightharpoonup 0 \quad \text{weakly in } H^1,$$

hence also weakly in L^2. Therefore,

$$\int_{\mathbb{R}^2} V^1(y)\overline{r^1_n(y + x^1_n)}\, dy \to 0.$$

This proves (70) for $J = 1$.

The proof of (71) for $J = 1$ is identical, replacing V^1 by ∇V^1 and r^1_n by ∇r^1_n, using the weak convergence in H^1.

Step 4: Inductive step. Assume we have constructed profiles $V^1, \ldots, V^J$, translations $x^1_n, \ldots, x^J_n$, and remainders r^J_n such that (67) holds and such that (68), (69), (70), and (71) hold up to level J.

We define the next profile by repeating Step 1-3 with the sequence $(r^J_n)_{n\in\mathbb{N}}$ in place of $(v_n)_{n\in\mathbb{N}}$.

Define

$$\mathcal{V}_J := \left\{ W \in H^1(\mathbb{R}^2) \;\middle|\; \exists\, (n_k), \exists\, (y_k),\ r^J_{n_k}(\cdot + y_k) \rightharpoonup W \text{ in } H^1 \right\}, \qquad \eta_J := \sup\{||W||_{L^4} \mid W \in \mathcal{V}_J\}.$$

If $\eta_J = 0$, we stop the extraction process and set $V^{J+1} = 0$ and $r^{J'}_n = r^J_n$ for all $J' \geq J$. If $\eta_J > 0$, then we choose $V^{J+1} \in \mathcal{V}_J$ such that $||V^{J+1}||_4 \geq \eta_J/2$, choose translations (x^{J+1}_n) such that

$$r^J_n(\cdot + x^{J+1}_n) \rightharpoonup V^{J+1} \quad \text{in } H^1,$$

and define

$$r_n^{J+1}(x) := r_n^J(x) - V^{J+1}(x - x_n^{J+1}).$$

Then (67) holds for $J+1$.

We now verify the separation of translations. Fix $k \in \{1, \dots, J\}$. Suppose, for contradiction, that $|x_n^{J+1} - x_n^k|$ does not tend to infinity. Then there exists a subsequence (not relabeled) such that $(x_n^{J+1} - x_n^k)$ converges in $\mathbb{R}^2$, hence is bounded. Consider

$$r_n^J(\cdot + x_n^k) = v_n(\cdot + x_n^k) - \sum_{j=1}^{J} V^j(\cdot + x_n^k - x_n^j).$$

By the induction hypothesis, $r_n^J(\cdot + x_n^k) \rightharpoonup 0$ in H^1 as $n \to \infty$. On the other hand, since $x_n^{J+1} - x_n^k$ is bounded, after extracting a subsequence we may assume $x_n^{J+1} - x_n^k \to \bar{x} \in \mathbb{R}^2$. Then

$$r_n^J(\cdot + x_n^{J+1}) = r_n^J(\cdot + x_n^k + (x_n^{J+1} - x_n^k)).$$

Because translations act continuously on H^1 and weak limits are preserved under fixed translations, this implies that if $r_n^J(\cdot + x_n^k) \rightharpoonup 0$, then also $r_n^J(\cdot + x_n^{J+1}) \rightharpoonup 0$. This contradicts the fact that $r_n^J(\cdot + x_n^{J+1}) \rightharpoonup V^{J+1}$ with $||V^{J+1}||_4 \geq \eta_J/2 > 0$, hence $V^{J+1} \neq 0$. Therefore $|x_n^{J+1} - x_n^k| \to \infty$, proving (68) at level $J+1$.

The decoupling (70) and (71) at level $J+1$ follows by expanding squares as in Step 3 and using Lemma 4.2.1 together with (68) to show that cross terms vanish.

Step 5: Smallness of the remainder in L^4. We prove (72). Fix J. By construction, η_J is the supremum of L^4 norms of translated weak limits of (r_n^J). If $\eta_J = 0$, then, by repeating Step 2 with r_n^J instead of v_n, we obtain $||r_n^J||_4 \to 0$. If $\eta_J > 0$, then we extract another profile.

The key point is that the decoupling (70) implies

$$\sum_{j=1}^{\infty} ||V^j||_2^2 \leq \limsup_{n\to\infty} ||v_n||_2^2 < \infty,$$

hence $||V^j||_2 \to 0$ as $j \to \infty$. Since $H^1 \hookrightarrow L^4$, also $||V^j||_4 \to 0$.

By construction, $\eta_J \leq 2||V^{J+1}||_4$. Therefore $\eta_J \to 0$ as $J \to \infty$. Applying Step 2 to the sequence (r_n^J) for each fixed J gives

$$\limsup_{n\to\infty} ||r_n^J||_4 \leq C\,\eta_J,$$

for some absolute constant C coming from the argument in Step 2, hence taking $J \to \infty$ yields (72). ■

4.2.3 The compactness lemma and its $d = 2$ consequence

In this subsubsection, we state and prove the compactness lemma in the precise form used for blow-up.

Theorem 4.2.3 (Compactness lemma in $d = 2$)**.** *Let $(v_n)_{n\in\mathbb{N}}$ be a bounded sequence in $H^1(\mathbb{R}^2)$ such that*

$$\limsup_{n\to\infty} ||\nabla v_n||_{L^2(\mathbb{R}^2)} \leq M \qquad \text{and} \qquad \limsup_{n\to\infty} ||v_n||_{L^4(\mathbb{R}^2)} \geq m, \tag{76}$$

for some $M \in (0,\infty)$ and $m \in (0,\infty)$.

Then, there exist a subsequence (not relabeled), a sequence of translations $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}^2$, *and a function* $V \in H^1(\mathbb{R}^2)$ *such that*

$$v_n(\cdot + x_n) \rightharpoonup V \quad \textit{weakly in } H^1(\mathbb{R}^2), \tag{77}$$

and

$$||V||^2_{L^2(\mathbb{R}^2)} \geq \frac{m^4}{C_{\mathrm{GN}}\, M^2}. \tag{78}$$

In particular, using the identity $C_{\mathrm{GN}} = 2/||Q||^2_{L^2(\mathbb{R}^2)}$ *from the sharp Gagliardo-Nirenberg theory, one obtains*

$$||V||_{L^2(\mathbb{R}^2)} \geq \frac{1}{\sqrt{2}} \frac{m^2}{M}\, ||Q||_{L^2(\mathbb{R}^2)}. \tag{79}$$

Proof. We apply Proposition 4.2.2 to (v_n).

By (67), for each fixed J,

$$v_n = \sum_{j=1}^{J} V^j(\cdot - x_n^j) + r_n^J.$$

By (72), we can choose J large enough so that

$$\limsup_{n\to\infty} ||r_n^J||_{L^4(\mathbb{R}^2)} \leq \frac{m}{2}. \tag{80}$$

We now relate the L^4 norms of v_n and of the profiles. Since translations preserve the L^4 norm, and because of (68), the profiles become asymptotically disjoint in space. This implies a Brezis-Lieb type decoupling for the L^4 norm, and we now justify it.

Fix the above J. For each n, define

$$S_n(x) \coloneqq \sum_{j=1}^{J} V^j(x - x_n^j).$$

Then $v_n = S_n + r_n^J$. Passing to a subsequence, we may assume $v_n(x) \to v(x)$ almost everywhere and $r_n^J(x) \to r(x)$ almost everywhere (this is standard: boundedness in H^1 implies boundedness in L^4, hence after extraction one has almost everywhere convergence; one can use a diagonal argument on balls and Rellich compactness). Consequently, $S_n = v_n - r_n^J$ converges almost everywhere as well.

Apply Lemma 1.9.1 with $p = 4$, $f_n = v_n$ and $f = r_n^J$. We obtain

$$||v_n||_4^4 - ||S_n||_4^4 \to ||r_n^J||_4^4, \qquad \text{as } n \to \infty. \tag{81}$$

In particular,

$$\limsup_{n\to\infty} ||S_n||_4^4 \geq \limsup_{n\to\infty} ||v_n||_4^4 - \limsup_{n\to\infty} ||r_n^J||_4^4. \tag{82}$$

Using the assumptions (76) and (80), we get

$$\limsup_{n\to\infty} ||S_n||_4^4 \geq m^4 - \left(\frac{m}{2}\right)^4 \geq \frac{15}{16} m^4.$$

Now we decouple $||S_n||_4^4$ among the profiles. By (68) and another application of Lemma 1.9.1 (iterated J times), one proves

$$||S_n||_4^4 \to \sum_{j=1}^{J} ||V^j||_4^4, \qquad \text{as } n \to \infty. \tag{83}$$

Hence,
$$\sum_{j=1}^{J} ||V^j||_4^4 \geq \frac{15}{16} m^4.$$
Therefore, there exists at least one index $j_* \in \{1, \dots, J\}$ such that
$$||V^{j_*}||_4^4 \geq \frac{15}{16} \frac{m^4}{J}. \tag{84}$$
We now choose $V := V^{j_*}$ and $x_n := x_n^{j_*}$. Then (77) follows from the definition of profiles.

It remains to prove the L^2 lower bound. We use the sharp Gagliardo-Nirenberg inequality in $d = 2$:
$$||f||_{L^4(\mathbb{R}^2)}^4 \leq C_{\mathrm{GN}} \, ||f||_{L^2(\mathbb{R}^2)}^2 \, ||\nabla f||_{L^2(\mathbb{R}^2)}^2, \qquad \forall f \in H^1(\mathbb{R}^2). \tag{85}$$
Applying (85) to $f = V$ gives
$$||V||_4^4 \leq C_{\mathrm{GN}} \, ||V||_2^2 \, ||\nabla V||_2^2. \tag{86}$$
We next bound $||\nabla V||_2$ in terms of M. By (71) with J fixed, we have
$$||\nabla v_n||_2^2 = \sum_{j=1}^{J} ||\nabla V^j||_2^2 + ||\nabla r_n^J||_2^2 + o_n(1).$$
Taking $\limsup_{n\to\infty}$ and using $||\nabla r_n^J||_2^2 \geq 0$ yields
$$\limsup_{n\to\infty} ||\nabla v_n||_2^2 \geq \sum_{j=1}^{J} ||\nabla V^j||_2^2 \geq ||\nabla V||_2^2.$$
By the assumption $\limsup ||\nabla v_n||_2 \leq M$, we obtain $||\nabla V||_2 \leq M$. Insert this into (86) to get
$$||V||_4^4 \leq C_{\mathrm{GN}} \, ||V||_2^2 \, M^2,$$
hence
$$||V||_2^2 \geq \frac{||V||_4^4}{C_{\mathrm{GN}} \, M^2}.$$
Finally, since $||V||_4^4 \geq 0$, we may drop the non-optimal constants coming from (84) by observing that we only need a lower bound in terms of m, and by repeating the extraction with J chosen minimal so that (80) holds, we can ensure $||V||_4 \geq m$ (this is exactly the choice made in [13]). In that case, we get (78):
$$||V||_2^2 \geq \frac{m^4}{C_{\mathrm{GN}} \, M^2}.$$
Using $C_{\mathrm{GN}} = 2/||Q||_2^2$ gives (79). ■

4.3 Mass Concentration at Blow-Up

We prove that approaching a finite blow-up time forces a nontrivial amount of L^2-mass to concentrate in balls whose radius shrinks at an appropriate rate. The goal is to link blow-up to the variational threshold of the ground state from Section 2.

Theorem 4.3.1 (Mass concentration at blow-up)**.** *Let u be a solution of the L^2-critical focusing nonlinear Schrödinger equation in $\mathbb{R}^2$,*

$$i\partial_t u + \Delta u + |u|^2 u = 0, \tag{87}$$

defined on a maximal forward time interval $[0, T)$ with $T \in (0, \infty)$, and assume

$$u \in C([0, T); H^1(\mathbb{R}^2)).$$

Assume, moreover, that u blows up at T, in the sense that

$$\lim_{t \uparrow T} ||\nabla u(t)||_{L^2(\mathbb{R}^2)} = +\infty.$$

Let $\lambda : [0, T) \to (0, \infty)$ satisfy

$$\lim_{t \uparrow T} \lambda(t) \, ||\nabla u(t)||_{L^2(\mathbb{R}^2)} = +\infty. \tag{88}$$

Let Q denote the positive radial ground state solving

$$-\Delta Q + Q - Q^3 = 0 \qquad \text{in } \mathbb{R}^2. \tag{89}$$

Then there exists a map $x : [0, T) \to \mathbb{R}^2$ such that

$$\liminf_{t \uparrow T} \int_{|x - x(t)| \le \lambda(t)} |u(t, x)|^2 \, dx \ge ||Q||^2_{L^2(\mathbb{R}^2)}. \tag{90}$$

Proof. We first recall the conservation laws: the mass

$$M(u(t)) := \int_{\mathbb{R}^2} |u(t, x)|^2 \, dx$$

and the energy

$$E(u(t)) := \frac{1}{2} \int_{\mathbb{R}^2} |\nabla u(t, x)|^2 \, dx - \frac{1}{4} \int_{\mathbb{R}^2} |u(t, x)|^4 \, dx$$

are constant on $[0, T)$, in particular,

$$M(u(t)) = M(u_0), \qquad E(u(t)) = E(u_0), \qquad \forall \, t \in [0, T).$$

Step 1: Reduction to a normalized sequence at times $t_n \uparrow T$. Let $(t_n)_{n \in \mathbb{N}}$ be any sequence with $t_n \uparrow T$. Set

$$\alpha_n := \frac{1}{||\nabla u(t_n)||_{L^2(\mathbb{R}^2)}} \in (0, \infty).$$

Then $\alpha_n \to 0$ because $||\nabla u(t_n)||_{L^2} \to \infty$.

Define the rescaled functions

$$v_n(x) := \alpha_n \, u(t_n, \alpha_n x), \qquad x \in \mathbb{R}^2.$$

We verify the scaling identities carefully.

<ins>Mass.</ins> Using the change of variables $y := \alpha_n x$, hence $dx = \alpha_n^{-2} dy$,

$$||v_n||^2_{L^2(\mathbb{R}^2)} = \int_{\mathbb{R}^2} \alpha_n^2 |u(t_n, \alpha_n x)|^2 \, dx = \int_{\mathbb{R}^2} |u(t_n, y)|^2 \, dy = ||u(t_n)||^2_{L^2(\mathbb{R}^2)}. \tag{91}$$

Thus $||v_n||_{L^2} = ||u_0||_{L^2}$ for all n.

Gradient. We compute $\nabla v_n(x) = \alpha_n^2 (\nabla u)(t_n, \alpha_n x)$. Hence, again by $y = \alpha_n x$,

$$||\nabla v_n||^2_{L^2(\mathbb{R}^2)} = \int_{\mathbb{R}^2} \alpha_n^4 |\nabla u(t_n, \alpha_n x)|^2 \, dx = \alpha_n^2 \int_{\mathbb{R}^2} |\nabla u(t_n, y)|^2 \, dy = \alpha_n^2 ||\nabla u(t_n)||^2_{L^2(\mathbb{R}^2)} = 1. \tag{92}$$

Thus, (v_n) is bounded in $H^1(\mathbb{R}^2)$, and its gradient norm is exactly normalized.

L^4-norm and energy scaling. Similarly,

$$||v_n||^4_{L^4(\mathbb{R}^2)} = \int_{\mathbb{R}^2} \alpha_n^4 |u(t_n, \alpha_n x)|^4 \, dx = \alpha_n^2 \int_{\mathbb{R}^2} |u(t_n, y)|^4 \, dy = \alpha_n^2 ||u(t_n)||^4_{L^4(\mathbb{R}^2)}. \tag{93}$$

Using (92) and (93), we compute the energy of v_n:

$$\begin{aligned} E(v_n) = \frac{1}{2}||\nabla v_n||^2_{L^2(\mathbb{R}^2)} - \frac{1}{4}||v_n||^4_{L^4(\mathbb{R}^2)} = \\ = \alpha_n^2 \left(\frac{1}{2}||\nabla u(t_n)||^2_{L^2(\mathbb{R}^2)} - \frac{1}{4}||u(t_n)||^4_{L^4(\mathbb{R}^2)} \right) = \alpha_n^2 E(u(t_n)). \end{aligned} \tag{94}$$

Since $E(u(t_n)) = E(u_0)$ is constant and $\alpha_n \to 0$, it follows that

$$\lim_{n\to\infty} E(v_n) = 0. \tag{95}$$

Combining (95) with $||\nabla v_n||^2_{L^2} = 1$ in (94) yields

$$\frac{1}{2} - \frac{1}{4}||v_n||^4_{L^4(\mathbb{R}^2)} = E(v_n) \to 0, \qquad \text{hence} \qquad ||v_n||^4_{L^4(\mathbb{R}^2)} \to 2. \tag{96}$$

In particular,

$$\liminf_{n\to\infty} ||v_n||_{L^4(\mathbb{R}^2)} > 0. \tag{97}$$

Step 2: Extraction of a concentrating profile. We now apply the compactness lemma 4.2.3. By (91), (92), and (97), the assumptions of 4.2.3 hold for the bounded sequence $(v_n) \subseteq H^1(\mathbb{R}^2)$.

Hence there exist a sequence of translations $(x_n)_{n\in\mathbb{N}} \subseteq \mathbb{R}^2$ and a function

$$V \in H^1(\mathbb{R}^2), \qquad V \not\equiv 0,$$

such that, up to extraction of a subsequence (not relabeled),

$$v_n(\cdot + x_n) \rightharpoonup V \quad \text{weakly in } H^1(\mathbb{R}^2),$$

and such that

$$||V||_{L^2(\mathbb{R}^2)} \geq ||Q||_{L^2(\mathbb{R}^2)}. \tag{98}$$

Step 3: From the weak limit to a quantitative local L^2 lower bound. Fix $\varepsilon \in (0, 1)$. Since $V \in L^2(\mathbb{R}^2)$, there exists $R_\varepsilon \in (0, \infty)$ such that

$$\int_{|x|\leq R_\varepsilon} |V(x)|^2 \, dx \geq ||V||^2_{L^2(\mathbb{R}^2)} - \varepsilon. \tag{99}$$

We now show that

$$\liminf_{n\to\infty} \int_{|x|\le R_\varepsilon} |v_n(x+x_n)|^2\,dx \ge \int_{|x|\le R_\varepsilon} |V(x)|^2\,dx. \tag{100}$$

To prove (100), we use only weak convergence in L^2. Indeed, the map

$$H^1(\mathbb{R}^2) \to \mathbb{R}, \qquad f \mapsto \int_{|x|\le R_\varepsilon} |f(x)|^2\,dx,$$

is weakly lower semicontinuous on $L^2(\mathbb{R}^2)$, hence also on $H^1(\mathbb{R}^2)$. A fully explicit proof is as follows.

Let $\mathbf{1}_{R_\varepsilon} \coloneqq \mathbf{1}_{\{x\in\mathbb{R}^2 \mid |x|\le R_\varepsilon\}}$. Since $0 \le \mathbf{1}_{R_\varepsilon} \le 1$ and $\mathbf{1}_{R_\varepsilon} \in L^\infty(\mathbb{R}^2)$, the multiplication operator

$$T_{R_\varepsilon} : L^2(\mathbb{R}^2) \to L^2(\mathbb{R}^2), \qquad T_{R_\varepsilon}(f) \coloneqq \mathbf{1}_{R_\varepsilon} f,$$

is bounded and linear. Therefore, since $v_n(\cdot + x_n) \rightharpoonup V$ weakly in $L^2(\mathbb{R}^2)$, it follows that $T_{R_\varepsilon}(v_n(\cdot + x_n)) \rightharpoonup T_{R_\varepsilon}(V)$ weakly in $L^2(\mathbb{R}^2)$.

Now we use the standard fact: if $w_n \rightharpoonup w$ weakly in a Hilbert space, then

$$||w||^2 \le \liminf_{n\to\infty} ||w_n||^2.$$

Applying this with $w_n \coloneqq T_{R_\varepsilon}(v_n(\cdot + x_n))$ and $w \coloneqq T_{R_\varepsilon}(V)$ yields

$$\int_{|x|\le R_\varepsilon} |V(x)|^2\,dx = ||T_{R_\varepsilon}(V)||^2_{L^2(\mathbb{R}^2)} \le \liminf_{n\to\infty} ||T_{R_\varepsilon}(v_n(\cdot+x_n))||^2_{L^2(\mathbb{R}^2)} = \liminf_{n\to\infty} \int_{|x|\le R_\varepsilon} |v_n(x+x_n)|^2\,dx,$$

which is exactly (100).

Combining (100) with (99) and (98), we obtain

$$\liminf_{n\to\infty} \int_{|x|\le R_\varepsilon} |v_n(x+x_n)|^2\,dx \ge ||V||^2_{L^2(\mathbb{R}^2)} - \varepsilon \ge ||Q||^2_{L^2(\mathbb{R}^2)} - \varepsilon. \tag{101}$$

Step 4: Return to the original variables and obtain a sup-type estimate. Define $y_n \coloneqq \alpha_n x_n \in \mathbb{R}^2$. Using the change of variables $z = \alpha_n x$ and the definition of v_n, we compute, for every $R > 0$,

$$\int_{|x-y_n|\le R\alpha_n} |u(t_n,x)|^2\,dx = \int_{|z-x_n|\le R} |v_n(z)|^2\,dz = \int_{|z|\le R} |v_n(z+x_n)|^2\,dz. \tag{102}$$

Taking $R = R_\varepsilon$ in (102) and using (101) gives

$$\liminf_{n\to\infty} \int_{|x-y_n|\le R_\varepsilon\alpha_n} |u(t_n,x)|^2\,dx \ge ||Q||^2_{L^2(\mathbb{R}^2)} - \varepsilon. \tag{103}$$

We now compare the radii $R_\varepsilon\alpha_n$ and $\lambda(t_n)$. Since $\alpha_n = 1/||\nabla u(t_n)||_{L^2}$, condition (88) reads

$$\frac{\lambda(t_n)}{\alpha_n} = \lambda(t_n)\,||\nabla u(t_n)||_{L^2(\mathbb{R}^2)} \to +\infty.$$

Thus, for the fixed R_ε, there exists $n_\varepsilon \in \mathbb{N}$ such that

$$\lambda(t_n) \ge R_\varepsilon\alpha_n, \qquad \forall\, n \ge n_\varepsilon. \tag{104}$$

For $n \geq n_\varepsilon$, the inclusion of balls

$$\{x \in \mathbb{R}^2 \mid |x - y_n| \leq R_\varepsilon \alpha_n\} \subseteq \{x \in \mathbb{R}^2 \mid |x - y_n| \leq \lambda(t_n)\}$$

implies

$$\int_{|x-y_n|\leq\lambda(t_n)} |u(t_n, x)|^2 \, dx \geq \int_{|x-y_n|\leq R_\varepsilon\alpha_n} |u(t_n, x)|^2 \, dx. \tag{105}$$

Combining (103) with (105) yields

$$\liminf_{n\to\infty} \int_{|x-y_n|\leq\lambda(t_n)} |u(t_n, x)|^2 \, dx \geq ||Q||^2_{L^2(\mathbb{R}^2)} - \varepsilon. \tag{106}$$

Since $\varepsilon \in (0, 1)$ is arbitrary, we conclude that

$$\liminf_{n\to\infty} \int_{|x-y_n|\leq\lambda(t_n)} |u(t_n, x)|^2 \, dx \geq ||Q||^2_{L^2(\mathbb{R}^2)}. \tag{107}$$

Now define, for $t \in [0, T)$,

$$m(t) := \sup_{y\in\mathbb{R}^2} \int_{|x-y|\leq\lambda(t)} |u(t, x)|^2 \, dx.$$

Because the supremum dominates any particular choice of center, (107) implies

$$\liminf_{n\to\infty} m(t_n) \geq ||Q||^2_{L^2(\mathbb{R}^2)}. \tag{108}$$

Since (t_n) was an arbitrary sequence with $t_n \uparrow T$, we infer the full liminf statement

$$\liminf_{t\uparrow T} m(t) \geq ||Q||^2_{L^2(\mathbb{R}^2)}. \tag{109}$$

Indeed, if (109) were false, then there would exist $\eta > 0$ and a sequence $t_n \uparrow T$ with $m(t_n) \leq ||Q||^2_{L^2} - \eta$, contradicting (108).

Step 5: Attainment of the spatial supremum for each fixed time. For $t \in [0, T)$, we set

$$m(t) := \sup_{y\in\mathbb{R}^2} \int_{|x-y|\leq\lambda(t)} |u(t, x)|^2 \, dx.$$

From the previous part of the proof, we already know that

$$\liminf_{t\uparrow T} m(t) \geq ||Q||^2_{L^2(\mathbb{R}^2)}. \tag{110}$$

We now show that, $\forall\, t \in [0, T)$, the supremum defining $m(t)$ is attained.

Fix $t \in [0, T)$ and define

$$F_t : \mathbb{R}^2 \to [0, \infty), \qquad F_t(y) := \int_{|x-y|\leq\lambda(t)} |u(t, x)|^2 \, dx.$$

We prove that F_t is continuous on $\mathbb{R}^2$ and that $F_t(y) \to 0$ as $|y| \to \infty$.

Continuity of F_t. Let $(y_n)_{n\in\mathbb{N}} \subset \mathbb{R}^2$ satisfy $y_n \to y$ in $\mathbb{R}^2$. For each n, set

$$g_n(x) := \mathbf{1}_{\{x\in\mathbb{R}^2 \mid |x-y_n|\le\lambda(t)\}}(x)\,|u(t,x)|^2, \qquad g(x) := \mathbf{1}_{\{x\in\mathbb{R}^2 \mid |x-y|\le\lambda(t)\}}(x)\,|u(t,x)|^2.$$

Then

$$F_t(y_n) = \int_{\mathbb{R}^2} g_n(x)\,dx, \qquad F_t(y) = \int_{\mathbb{R}^2} g(x)\,dx.$$

We claim that $g_n(x) \to g(x)$ for Lebesgue-a.e. $x \in \mathbb{R}^2$.

Indeed, fix $x \in \mathbb{R}^2$ such that $|x-y| \neq \lambda(t)$.

- If $|x-y| < \lambda(t)$, then $\delta := \frac{1}{2}(\lambda(t) - |x-y|) > 0$. Since $y_n \to y$, there exists n_0 such that $|y_n - y| < \delta$ for all $n \ge n_0$. For such n,

  $$|x-y_n| \le |x-y| + |y-y_n| < |x-y| + \delta < \lambda(t),$$

  hence $\mathbf{1}_{\{|x-y_n|\le\lambda(t)\}}(x) = 1 = \mathbf{1}_{\{|x-y|\le\lambda(t)\}}(x)$ for all $n \ge n_0$.
- If $|x-y| > \lambda(t)$, then $\delta := \frac{1}{2}(|x-y| - \lambda(t)) > 0$. For n large enough so that $|y_n - y| < \delta$, we have

  $$|x-y_n| \ge |x-y| - |y-y_n| > |x-y| - \delta > \lambda(t),$$

  hence $\mathbf{1}_{\{|x-y_n|\le\lambda(t)\}}(x) = 0 = \mathbf{1}_{\{|x-y|\le\lambda(t)\}}(x)$ for all large n.

Thus, if $|x-y| \neq \lambda(t)$, we have $\mathbf{1}_{\{|x-y_n|\le\lambda(t)\}}(x) \to \mathbf{1}_{\{|x-y|\le\lambda(t)\}}(x)$, and therefore $g_n(x) \to g(x)$.

It remains to justify that the set

$$\big\{x \in \mathbb{R}^2 \;\big|\; |x-y| = \lambda(t)\big\}$$

has Lebesgue measure 0. This set is the circle of radius $\lambda(t)$ centered at y. For $\varepsilon > 0$, it is contained in the annulus

$$A_\varepsilon := \big\{x \in \mathbb{R}^2 \;\big|\; \lambda(t) - \varepsilon < |x-y| < \lambda(t) + \varepsilon\big\}.$$

Hence, by polar coordinates around y,

$$|A_\varepsilon| = \pi(\lambda(t)+\varepsilon)^2 - \pi(\lambda(t)-\varepsilon)^2 = 4\pi\lambda(t)\,\varepsilon.$$

Since $4\pi\lambda(t)\,\varepsilon \to 0$ as $\varepsilon \downarrow 0$, it follows that the circle has Lebesgue measure 0. Consequently, $g_n(x) \to g(x)$ for Lebesgue-a.e. $x \in \mathbb{R}^2$.

Moreover, for every n and every x,

$$0 \le g_n(x) \le |u(t,x)|^2, \qquad 0 \le g(x) \le |u(t,x)|^2.$$

Since $u(t) \in L^2(\mathbb{R}^2)$, we have $|u(t)|^2 \in L^1(\mathbb{R}^2)$ and

$$\int_{\mathbb{R}^2} |u(t,x)|^2\,dx = ||u(t)||^2_{L^2(\mathbb{R}^2)} < \infty.$$

Therefore, by the dominated convergence theorem,

$$\lim_{n\to\infty} F_t(y_n) = \lim_{n\to\infty} \int_{\mathbb{R}^2} g_n(x)\,dx = \int_{\mathbb{R}^2} g(x)\,dx = F_t(y).$$

This proves that F_t is continuous on $\mathbb{R}^2$.

Vanishing of F_t at infinity. Let $\varepsilon > 0$. Since $|u(t)|^2 \in L^1(\mathbb{R}^2)$, there exists $R > 0$ such that

$$\int_{|x|>R} |u(t,x)|^2 \, dx < \varepsilon.$$

Justification of the tail estimate. Let $f \in L^1(\mathbb{R}^d)$. For $R > 0$, set $A : (0, \infty) \to [0, \infty)$,

$$A(R) := \int_{|x|>R} |f(x)| \, dx, \qquad g_R(x) := |f(x)| \, \mathbf{1}_{\{|x|>R\}}(x).$$

Then, $A(R) = \int_{\mathbb{R}^d} g_R(x) \, dx$. Moreover, for every fixed $x \in \mathbb{R}^d$, we have $\mathbf{1}_{\{|x|>R\}}(x) \to 0$ as $R \to \infty$, hence $g_R(x) \to 0$ for a.e. $x \in \mathbb{R}^d$. Also, for every $R > 0$ and every $x \in \mathbb{R}^d$, we have $0 \le g_R(x) \le |f(x)|$, and, since $f \in L^1(\mathbb{R}^d)$, we have $|f| \in L^1(\mathbb{R}^d)$. Therefore, by the dominated convergence theorem,

$$\lim_{R\to\infty} A(R) = \lim_{R\to\infty} \int_{\mathbb{R}^d} g_R(x) \, dx = \int_{\mathbb{R}^d} \lim_{R\to\infty} g_R(x) \, dx = 0.$$

Consequently, for the given $\varepsilon > 0$, there exists $R_\varepsilon > 0$ such that, for all $R \ge R_\varepsilon$,

$$\int_{|x|>R} |f(x)| \, dx = A(R) < \varepsilon.$$

Applying this with $f(x) := |u(t,x)|^2 \in L^1(\mathbb{R}^2)$ yields the claimed estimate.

If $|y| > R + \lambda(t)$ and $|x - y| \le \lambda(t)$, then

$$|x| \ge |y| - |x - y| > R + \lambda(t) - \lambda(t) = R,$$

hence $B(y, \lambda(t)) \subseteq \left\{ x \in \mathbb{R}^2 \,\middle|\, |x| > R \right\}$ and thus

$$0 \le F_t(y) = \int_{|x-y|\le\lambda(t)} |u(t,x)|^2 \, dx \le \int_{|x|>R} |u(t,x)|^2 \, dx < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, this proves $\lim_{|y|\to\infty} F_t(y) = 0$.

Existence of a maximizer $x(t)$. Let $M_t := \sup_{y\in\mathbb{R}^2} F_t(y) \in [0, \infty)$. If $M_t = 0$, choose $x(t) := 0$. Assume $M_t > 0$. Since $F_t(y) \to 0$ as $|y| \to \infty$, there exists $R_t > 0$ such that $|y| > R_t$ implies $F_t(y) < M_t/2$. Hence,

$$M_t = \sup_{y\in\mathbb{R}^2} F_t(y) = \sup_{|y|\le R_t} F_t(y).$$

The set $\left\{ y \in \mathbb{R}^2 \,\middle|\, |y| \le R_t \right\}$ is compact, and F_t is continuous, hence $\exists\, x(t) \in \mathbb{R}^2$ with $|x(t)| \le R_t$ such that

$$F_t(x(t)) = \sup_{|y|\le R_t} F_t(y) = \sup_{y\in\mathbb{R}^2} F_t(y) = m(t).$$

Therefore,

$$m(t) = \sup_{y\in\mathbb{R}^2} \int_{|x-y|\le\lambda(t)} |u(t,x)|^2 \, dx = \int_{|x-x(t)|\le\lambda(t)} |u(t,x)|^2 \, dx. \tag{111}$$

Finally, combining (110) with (111), we obtain

$$\liminf_{t\uparrow T} \int_{|x-x(t)|\le\lambda(t)} |u(t,x)|^2 \, dx = \liminf_{t\uparrow T} m(t) \ge ||Q||^2_{L^2(\mathbb{R}^2)},$$

which is exactly (90). ■

4.4 Minimal-Mass Blow-Up and the Asymptotic Ground-State Profile

We consider the minimal-mass regime and show that the concentrating profile must asymptotically approach the ground state. The goal is to identify the unique limiting shape/profile using the sharp inequality from Section 2 together with the uniqueness result from Section 3.

Theorem 4.4.1 (Minimal-mass blow-up profile, $d = 2$)**.** *Let u be a solution of* (59) *on a maximal forward lifespan interval $[0, T)$ with $0 < T < \infty$, and assume that*

$$||u_0||_{L^2(\mathbb{R}^2)} = ||Q||_{L^2(\mathbb{R}^2)} \qquad \text{and} \qquad ||\nabla u(t)||_{L^2(\mathbb{R}^2)} \to \infty \quad \text{as } t \uparrow T. \tag{112}$$

Define

$$\rho(t) := \frac{||\nabla Q||_{L^2(\mathbb{R}^2)}}{||\nabla u(t)||_{L^2(\mathbb{R}^2)}}, \qquad t \in [0, T). \tag{113}$$

Then $\rho(t) \to 0$ as $t \uparrow T$, and there exist functions $\theta : [0, T) \to \mathbb{R}$ and $x : [0, T) \to \mathbb{R}^2$ such that

$$\rho(t)\, e^{i\theta(t)} u\big(t, \rho(t)\ \cdot + x(t)\big) \to Q \quad \text{strongly in } H^1(\mathbb{R}^2), \qquad \text{as } t \uparrow T. \tag{114}$$

Proof. We prove (114) by a contradiction argument, using subsequential compactness plus the rigidity of the sharp Gagliardo-Nirenberg inequality.

Step 1: Subsequence extraction and normalization. Let (t_n) be any sequence with $t_n \uparrow T$. Define

$$\rho_n := \rho(t_n) = \frac{||\nabla Q||_2}{||\nabla u(t_n)||_2}.$$

Then $\rho_n \to 0$. Define the normalized rescalings

$$v_n(x) := \rho_n\, u(t_n, \rho_n x).$$

As in the proof of Theorem 4.3.1, one checks that

$$||v_n||_2 = ||u(t_n)||_2 = ||u_0||_2 = ||Q||_2,$$

and

$$||\nabla v_n||_2 = \rho_n ||\nabla u(t_n)||_2 = ||\nabla Q||_2.$$

Thus (v_n) is bounded in H^1.

Moreover, the energy scales as $E(v_n) = \rho_n^2 E(u(t_n)) = \rho_n^2 E(u_0) \to 0$.

Step 2: Compactness up to translations and identification of the weak limit. Apply Theorem 4.2.3 to (v_n) after dividing by $||\nabla Q||_2$ if desired, to obtain translations (x_n) and a weak limit $V \in H^1$ such that, after extraction,

$$v_n(\cdot + x_n) \rightharpoonup V \quad \text{weakly in } H^1,$$

and

$$||V||_2 \geq ||Q||_2.$$

On the other hand, since translations preserve the L^2 norm,

$$||v_n(\cdot + x_n)||_2 = ||v_n||_2 = ||Q||_2, \qquad \forall n.$$

By weak lower semi-continuity of the L^2 norm,

$$||V||_2 \leq \liminf_{n\to\infty} ||v_n(\cdot + x_n)||_2 = ||Q||_2.$$

Combining with $||V||_2 \geq ||Q||_2$ gives

$$||V||_2 = ||Q||_2. \tag{115}$$

Since $v_n(\cdot + x_n) \rightharpoonup V$ in L^2 and the norms converge, we have strong convergence in L^2:

$$v_n(\cdot + x_n) \to V \quad \text{strongly in } L^2(\mathbb{R}^2). \tag{116}$$

Indeed, expand

$$||v_n(\cdot + x_n) - V||_2^2 = ||v_n(\cdot + x_n)||_2^2 + ||V||_2^2 - 2\Re \int v_n(\cdot + x_n)\overline{V},$$

and note that the integral tends to $||V||_2^2$ by weak convergence. Using $||v_n(\cdot + x_n)||_2^2 \to ||V||_2^2$ from (115) gives the claim.

Step 3: Strong H^1 convergence and saturation of the sharp Gagliardo-Nirenberg inequality. We next show that $v_n(\cdot + x_n) \to V$ strongly in H^1 and that V is a sharp optimizer, hence V lies in the orbit of Q.

Since $E(v_n) \to 0$ and energy is translation-invariant, we have

$$E(v_n(\cdot + x_n)) = E(v_n) \to 0.$$

By weak lower semi-continuity,

$$\int |\nabla V|^2 \leq \liminf_{n\to\infty} \int |\nabla v_n(\cdot + x_n)|^2 = ||\nabla Q||_2^2.$$

We now show the reverse inequality. Using the sharp Gagliardo-Nirenberg inequality (85) with $||V||_2 = ||Q||_2$,

$$||V||_4^4 \leq C_{\text{GN}}\, ||V||_2^2\, ||\nabla V||_2^2 = C_{\text{GN}}\, ||Q||_2^2\, ||\nabla V||_2^2 = 2||\nabla V||_2^2,$$

because $C_{\text{GN}} = 2/||Q||_2^2$.

Therefore,

$$E(V) = \frac{1}{2}||\nabla V||_2^2 - \frac{1}{4}||V||_4^4 \geq \frac{1}{2}||\nabla V||_2^2 - \frac{1}{4}(2||\nabla V||_2^2) = 0.$$

On the other hand, by weak convergence and Brezis-Lieb (Lemma 1.9.1), one has

$$E(V) \leq \liminf_{n\to\infty} E(v_n(\cdot + x_n)) = 0.$$

Hence $E(V) = 0$, and the inequality above must be an equality. Thus

$$||V||_4^4 = 2||\nabla V||_2^2,$$

which is exactly saturation of the sharp Gagliardo-Nirenberg inequality at mass $||Q||_2$. By the variational characterization from the sharp inequality theory (in particular, the characterization of equality cases), it follows that there exist $\theta_* \in \mathbb{R}$, $\rho_* > 0$, and $x_* \in \mathbb{R}^2$ such that

$$V(x) = e^{i\theta_*}\rho_* Q(\rho_*(x - x_*)).$$

Since $||V||_2 = ||Q||_2$ and the L^2 norm is invariant under the critical scaling, ρ_* is not constrained by the mass condition, but it will be absorbed into the choice of $\rho(t)$ in the next step.

Finally, we show strong convergence in H^1. We already have strong L^2 convergence (116). It remains to show $||\nabla v_n(\cdot + x_n)||_2 \to ||\nabla V||_2$. This follows from energy convergence plus L^4 convergence: by Brezis-Lieb in L^4 and the already established strong L^2 convergence, one obtains $||v_n(\cdot+x_n)||_4 \to ||V||_4$, hence from

$$E(v_n(\cdot + x_n)) = \frac{1}{2}||\nabla v_n(\cdot + x_n)||_2^2 - \frac{1}{4}||v_n(\cdot + x_n)||_4^4$$

and $E(v_n(\cdot + x_n)) \to E(V)$, we deduce $||\nabla v_n(\cdot + x_n)||_2 \to ||\nabla V||_2$. Together with weak convergence in H^1, this implies strong convergence in H^1.

Step 4: Promote subsequential convergence to full convergence as $t \uparrow T$. We have shown: for every sequence $t_n \uparrow T$, there exist parameters (x_n) and phases (θ_n) such that

$$\rho(t_n)e^{i\theta_n}u\big(t_n, \rho(t_n)\cdot + x_n\big) \to Q \quad \text{in } H^1.$$

Suppose, for contradiction, that the full limit (114) fails. Then there exist $\varepsilon_0 > 0$ and a sequence $t_n \uparrow T$ such that for every choice of $\theta \in \mathbb{R}$ and $x \in \mathbb{R}^2$,

$$\Big\|\rho(t_n)e^{i\theta}u\big(t_n, \rho(t_n)\cdot + x\big) - Q\Big\|_{H^1(\mathbb{R}^2)} \geq \varepsilon_0.$$

But, by the subsequential result from Steps 1-3, after extracting a subsequence there exist (θ_n) and (x_n) such that the left-hand side tends to 0, contradicting the inequality. Hence the full convergence (114) holds for some choices $\theta(t)$ and $x(t)$. ■

4.5 Classification of Minimal-Mass Finite-Time Blow-Up Data

Lastly, we classify all initial data that lead to finite-time blow-up at the minimal-mass threshold. The goal is to show that such solutions are, up to the equation's symmetries, generated by the ground-state profile. This completes the connection between sharp inequalities and critical blow-up dynamics. The argument is due to Banica.[2]

Definition 4.5.1 (Ground-state orbit)**.** *Define*

$$\mathcal{A} := \Big\{\rho\, e^{i\theta}Q(\rho x + x_0) \;\Big|\; \rho > 0,\ \theta \in \mathbb{R},\ x_0 \in \mathbb{R}^2\Big\}.$$

Theorem 4.5.2 (Classification of minimal-mass blow-up data in $d = 2$)**.** *Let u be a solution of* (59) *on a maximal forward lifespan interval $[0, T)$ with $0 < T < \infty$, and assume*

$$||u_0||_{L^2(\mathbb{R}^2)} = ||Q||_{L^2(\mathbb{R}^2)}.$$

Assume that u blows up at time T, in the sense that $||\nabla u(t)||_2 \to \infty$ as $t \uparrow T$.

Then there exists $x_0 \in \mathbb{R}^2$ such that

$$e^{i\frac{|x-x_0|^2}{4T}}u_0 \in \mathcal{A}. \tag{117}$$

Proof. Let (t_n) be an arbitrary sequence with $t_n \uparrow T$. By Theorem 4.4.1, there exist $\rho_n \to 0$, phases $\theta_n \in \mathbb{R}$, and translations $x_n \in \mathbb{R}^2$ such that

$$\rho_n e^{i\theta_n}u(t_n, \rho_n x + x_n) \to Q \quad \text{strongly in } H^1(\mathbb{R}^2). \tag{118}$$

In particular, by strong L^2 convergence, the measures $|u(t_n,x)|^2dx$ concentrate at scale ρ_n around x_n. More precisely, for every $\varphi \in C_c(\mathbb{R}^2)$,

$$\int_{\mathbb{R}^2} \varphi(x)|u(t_n,x)|^2\,dx = \int_{\mathbb{R}^2} \varphi(x_n+\rho_n y)\,|\rho_n u(t_n,x_n+\rho_n y)|^2\,dy \to \varphi(x_\infty)\,||Q||_2^2,$$

provided $x_n \to x_\infty$. Up to extracting a subsequence, we may assume either $x_n \to x_0 \in \mathbb{R}^2$ or $|x_n| \to \infty$. We will show that the latter is impossible.

Step 1: Truncated variance and its derivative estimate. Choose $\phi \in C^\infty(\mathbb{R}^2)$ such that:

- ϕ is radial and $\phi \geq 0$,
- $\phi(x) = |x|^2$ for $|x| \leq 1$,
- ϕ is constant for $|x| \geq 2$,
- and $|\nabla\phi(x)|^2 \leq C\,\phi(x)$ for all x, for some constant C.

Such a function can be built by smoothing a suitable radial cutoff.

For $p \in \mathbb{N}^*$, define the rescaled cutoff

$$\phi_p(x) := p^2\phi\Big(\frac{x}{p}\Big).$$

Define the truncated variance

$$g_p(t) := \int_{\mathbb{R}^2} \phi_p(x)|u(t,x)|^2\,dx. \tag{119}$$

We now compute $\dot{g}_p(t)$ and prove an estimate.

Differentiate (119) using that $u \in C^1([0,T);H^{-1})$ and $\phi_p \in W^{1,\infty}$:

$$\dot{g}_p(t) = 2\Re\int_{\mathbb{R}^2} \phi_p(x)\,\partial_t u(t,x)\,\overline{u(t,x)}\,dx.$$

Using $\partial_t u = i\Delta u + i|u|^2u$ and $\Re(i|u|^2u\overline{u}) = \Re(i|u|^4) = 0$, we get

$$\dot{g}_p(t) = 2\Re\int_{\mathbb{R}^2} \phi_p(x)\,i\Delta u(t,x)\,\overline{u(t,x)}\,dx.$$

Integrate by parts in the distributional sense:

$$\int \phi_p\,\Delta u\,\overline{u} = -\int \nabla u\cdot\nabla(\phi_p\overline{u}) = -\int \nabla u\cdot\big((\nabla\phi_p)\overline{u} + \phi_p\nabla\overline{u}\big).$$

Therefore,

$$\dot{g}_p(t) = -2\Re\Big(i\int_{\mathbb{R}^2} \nabla u(t,x)\cdot(\nabla\phi_p(x))\,\overline{u(t,x)}\,dx\Big) - 2\Re\Big(i\int_{\mathbb{R}^2} \phi_p(x)\nabla u(t,x)\cdot\nabla\overline{u(t,x)}\,dx\Big).$$

The second integral is real, because $\nabla u\cdot\nabla\overline{u} = |\nabla u|^2$, hence its contribution vanishes after applying $\Re(i\cdot)$. Thus,

$$\dot{g}_p(t) = -2\Re\Big(i\int_{\mathbb{R}^2} \overline{u(t,x)}\,\nabla u(t,x)\cdot\nabla\phi_p(x)\,dx\Big). \tag{120}$$

Using $|\Re(iz)| \leq |z|$, we obtain

$$|\dot{g}_p(t)| \leq 2\Big|\int_{\mathbb{R}^2} \overline{u(t,x)}\,\nabla u(t,x)\cdot\nabla\phi_p(x)\,dx\Big|. \tag{121}$$

Step 2: A discriminant argument giving a Cauchy-Schwarz type estimate. Fix $t \in [0,T)$. Define, for $s \in \mathbb{R}$, the gauge-modulated function

$$u_s(x) := e^{is\phi_p(x)} u(t,x).$$

We note that $||u_s||_2 = ||u(t)||_2 = ||Q||_2$.

By the sharp Gagliardo-Nirenberg inequality at critical mass (as in the proof of Theorem 4.4.1), we have, for every $f \in H^1(\mathbb{R}^2)$ with $||f||_2 = ||Q||_2$,

$$E(f) \geq 0.$$

Therefore,

$$E(u_s) \geq 0, \qquad \forall s \in \mathbb{R}. \tag{122}$$

We now compute $E(u_s)$ as a quadratic polynomial in s.

First compute the gradient:

$$\nabla u_s = \nabla\big(e^{is\phi_p} u\big) = e^{is\phi_p}\big(is(\nabla\phi_p)u + \nabla u\big).$$

Hence

$$|\nabla u_s|^2 = |is(\nabla\phi_p)u + \nabla u|^2 = s^2|u|^2|\nabla\phi_p|^2 + |\nabla u|^2 + 2s\,\Im(\overline{u}\,\nabla u \cdot \nabla\phi_p),$$

where we used the identity $2\Re(iz) = -2\Im(z)$ and the standard expansion of $|A+B|^2$.

Also, the nonlinearity is invariant under multiplication by unimodular phases:

$$|u_s|^4 = |u|^4.$$

Therefore,

$$\begin{aligned}
E(u_s) &= \frac{1}{2}\int |\nabla u_s|^2 - \frac{1}{4}\int |u_s|^4 \\
&= \frac{1}{2}\int \Big(s^2|u|^2|\nabla\phi_p|^2 + |\nabla u|^2 + 2s\,\Im(\overline{u}\,\nabla u\cdot\nabla\phi_p)\Big) - \frac{1}{4}\int |u|^4 \\
&= \Big(\frac{1}{2}\int |u|^2|\nabla\phi_p|^2\Big)s^2 + \Big(\int \Im(\overline{u}\,\nabla u\cdot\nabla\phi_p)\Big)s + E(u(t)).
\end{aligned}$$

Define the coefficients

$$A_p(t) := \frac{1}{2}\int_{\mathbb{R}^2} |u(t,x)|^2|\nabla\phi_p(x)|^2\,dx, \qquad B_p(t) := \int_{\mathbb{R}^2} \Im(\overline{u(t,x)}\,\nabla u(t,x)\cdot\nabla\phi_p(x))\,dx,$$

$$C(t) := E(u(t)) = E(u_0).$$

Then

$$E(u_s) = A_p(t)s^2 + B_p(t)s + C(t). \tag{123}$$

By (122), the quadratic polynomial in (123) is nonnegative for all $s \in \mathbb{R}$. A necessary and sufficient condition for a real quadratic As^2+Bs+C with $A \geq 0$ to be nonnegative for all s is that its discriminant is nonpositive:

$$B^2 - 4AC \leq 0.$$

We justify this: if $A = 0$, then the function is affine and nonnegative for all s forces $B = 0$ and $C \geq 0$, hence $B^2 - 4AC = 0$. If $A > 0$, then the polynomial attains its minimum at $s_* = -B/(2A)$, with minimal value $C - B^2/(4A)$. Nonnegativity of the polynomial is equivalent to $C - B^2/(4A) \geq 0$, i.e., $B^2 - 4AC \leq 0$.

Applying this to (123), we obtain

$$B_p(t)^2 \leq 4A_p(t)C(t) = 4A_p(t)\, E(u_0). \tag{124}$$

Finally, we relate $B_p(t)$ to $\dot{g}_p(t)$. Comparing (120) with the definition of $B_p(t)$ and using $\Re(iz) = -\Im(z)$, we get

$$\dot{g}_p(t) = 2B_p(t).$$

Therefore, (124) gives

$$|\dot{g}_p(t)|^2 \leq 16A_p(t)\, E(u_0). \tag{125}$$

Step 3: Close the estimate using $|\nabla\phi_p|^2 \leq C\phi_p$. By construction of ϕ, we have $|\nabla\phi|^2 \leq C\phi$. Therefore,

$$|\nabla\phi_p(x)|^2 = p^2 \Big|\nabla\phi\Big(\frac{x}{p}\Big)\Big|^2 \leq p^2 C\, \phi\Big(\frac{x}{p}\Big) = C\, \phi_p(x).$$

Hence,

$$A_p(t) = \frac{1}{2}\int |u(t,x)|^2 |\nabla\phi_p(x)|^2\, dx \leq \frac{C}{2}\int \phi_p(x)|u(t,x)|^2\, dx = \frac{C}{2}\, g_p(t).$$

Insert this into (125):

$$|\dot{g}_p(t)|^2 \leq 16\Big(\frac{C}{2} g_p(t)\Big) E(u_0) = 8C\, E(u_0)\, g_p(t).$$

Taking square-roots yields

$$|\dot{g}_p(t)| \leq \sqrt{8C\, E(u_0)}\, \sqrt{g_p(t)}, \qquad \forall t \in [0, T). \tag{126}$$

Step 4: Integrate the differential inequality. Fix $t \in [0, T)$. Consider the function $h_p(t) := \sqrt{g_p(t)}$. Since $g_p(t) \geq 0$, h_p is well-defined. Whenever $g_p(t) > 0$, we can differentiate:

$$\dot{h}_p(t) = \frac{1}{2\sqrt{g_p(t)}} \dot{g}_p(t).$$

Using (126), we obtain, for all such t,

$$|\dot{h}_p(t)| \leq \frac{1}{2\sqrt{g_p(t)}} |\dot{g}_p(t)| \leq \frac{1}{2\sqrt{g_p(t)}} \cdot \sqrt{8C\, E(u_0)}\, \sqrt{g_p(t)} = \sqrt{2C\, E(u_0)}.$$

Thus h_p is Lipschitz with Lipschitz constant $\sqrt{2C\, E(u_0)}$, hence for all $t \in [0, T)$,

$$\sqrt{g_p(t)} \leq \sqrt{g_p(0)} + \sqrt{2C\, E(u_0)}\, t. \tag{127}$$

A symmetric estimate also holds with t replaced by $T - t$, after translating time; in particular, the estimate in [13] yields a bound of the form

$$g_p(t) \leq C(u_0)\, (T - t)^2, \qquad \forall t \in [0, T), \tag{128}$$

for a constant $C(u_0)$ depending only on u_0.

Step 5: Let $p \to \infty$ and conclude. Since $\phi_p(x) \uparrow |x|^2$ pointwise as $p \to \infty$ and $\phi_p(x) \le |x|^2$ for all p and x, we can pass to the limit in (128) and obtain that $u_0 \in \Sigma \coloneqq \{ f \in H^1(\mathbb{R}^2) \mid \int |x|^2 |f(x)|^2 dx < \infty \}$, and that

$$\int_{\mathbb{R}^2} |x|^2 |u(t,x)|^2 dx \le C(u_0)(T-t)^2.$$

In particular, letting $t \uparrow T$ gives that the variance tends to 0 at rate $(T-t)^2$. Combining this with the strong profile convergence (118) forces the concentration centers (x_n) to remain bounded, hence (after extraction) $x_n \to x_0$.

Finally, using the pseudo-conformal identity (equivalently, the virial identity rewritten as in [13]), one shows that

$$e^{i \frac{|x-x_0|^2}{4T}} u_0$$

must lie in the orbit $\mathcal{A}$, which is exactly (117). ■

5 Conclusion & Summary

In this master thesis, we studied the focusing L^2-critical cubic nonlinear Schrödinger equation on $\mathbb{R}^2$,

$$i\,\partial_t u + \Delta u + |u|^2 u = 0, \qquad u(0,\cdot) = u_0 \in H^1(\mathbb{R}^2),$$

with a particular emphasis on the mechanism and structure of finite-time blow-up. The guiding principle is that, in the mass-critical regime, blow-up and concentration are controlled by a sharp functional inequality and its optimizer, and, as a consequence, by the associated stationary, ground-state equation

$$\Delta Q - Q + Q^3 = 0, \quad \text{on } \mathbb{R}^2$$

too.

In Section 1, we assembled the functional-analytic tools used throughout the thesis, including the basic properties of Banach and Hilbert spaces, L^p and Sobolev spaces, and the rearrangement formalism needed for variational arguments. These preliminaries were chosen so that the later sections can be read with no gaps. The greatest amount of attention was paid to Subsection 1.7, as it contains the most amount of results that were necessary for the calculations that succeeded it in Section 2.

In Section 2, we proved the sharp Gagliardo-Nirenberg inequality on $\mathbb{R}^2$, identified the optimal constant C_{GN}, and established the existence of an optimizer via a variational approach in Subsection 2.1. From the Euler-Lagrange equation, we derived, in Subsection 2.2, the stationary equation

$$-\Delta Q + Q - Q^3 = 0 \qquad \text{on } \mathbb{R}^2,$$

and we explained how the optimizer controls the sharp constant and the equality case. This variational framework goes back to Weinstein, and, in particular, it is the origin of the mass threshold $||Q||_{L^2(\mathbb{R}^2)}$ that separates global behavior from the onset of blow-up scenarios.

In Section 3, we proved that the stationary equation admits a unique positive radial solution, up to the natural symmetries, by specializing the theory of McLeod and Serrin to our parameters. This uniqueness result has important consequences for the blow-up regime, since it ensures that, once concentration forces a limiting profile to be a ground state, that given profile is determined uniquely, up to symmetries.

Finally, in Section 4, following Hmidi and Keraani, we developed the blow-up analysis for the L^2-mass-critical regime in $\mathbb{R}^2$. Starting from the conserved quantities and the symmetries of the equation, we then proved a concentration-compactness lemma adapted to translations. This, in turn, yielded mass concentration at blow-up and, in the minimal-mass regime, the asymptotic ground-state profile. In particular, the minimal-mass blow-up dynamics are described by the pseudo-conformal structure generated from the ground state. As a consequence, minimal-mass finite-time blow-up initial data admit a complete classification in terms of the symmetries of the time-dependent equation.

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